

Analysis of Porous Media and Objects of Cultural Heritage by Mobile NMR

Von der Fakultät für Mathematik, Informatik und Naturwissenschaften der
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Berichte aus der Physik

Agnes Haber

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To my family...

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Chapter 1

Introduction

The development of unilateral mobile NMR sensors started in 1980 by Jasper Jackson, 35 years after the NMR signal was first successfully detected by Felix Bloch (Blümich 05a). Already 15 years after Jackson, the unilateral inside-out magnets were commercialized by the oil industry as well-logging tools (Jackson 80). The use of permanent magnets which generate an inhomogeneous magnetic field was important also for many other research areas, soft-matter science, environmental science, chemical engineering, etc.. The research in the mobile NMR field is still ongoing, with different applications (Casanova 11).

Why is this thesis focusing on porous media and cultural heritage? Because I think these research areas are vast, complex, interesting and influencing our everyday life. Porous media are in the walls of our house, in the streets on which we are walking, in the food that we are eating and in the cultural heritage objects that we inherited from our ancestors. NMR proved to be a good tool to probe the dynamics of fluids inside these porous systems from a microscopic to a macroscopic range over a long period of time. Moreover, single-sided NMR demonstrated not only once its efficiency in analyzing cultural heritage objects. One aim of this work is to contribute to the understanding of fluid dynamics in porous systems through 1D and 2D NMR relaxation measurements. In the case of the cultural heritage objects the aim is to help the conservators to identify suitable conservation strategies and to detect undocumented restoration efforts of the past.

After a short overview of the instrumentation used for obtaining the results presented in this thesis and the pulse sequences used in the experiments, the

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chapters following focus on the dynamics of water in porous media. Chapter 3 presents the theory of multi-site exchange NMR with focus on a simple two site exchange problem. This method correlates the relaxation time distribution at two different time intervals, observing in this way how the proton spins migrate during this time period from one environment to the other. Based on this theory, numerical simulations were done and compared to experimental results, which evidenced the exchange between the two sites. Moreover, the exchange rate was determined which provides information on the pore connectivity. Furthermore, another system was tested for the same reason but with more than two sites.

Chapter 4 focuses on water dynamics in natural porous systems, which are one fine sand sample and two silt loams samples with different compositions. The functions of each soil type are directly related to the structure of the pore system, which is characterized by the pore size distribution, pore connectivity, water content, and water dynamics. For this reason, the water dynamics is analyzed in both saturated and unsaturated systems. For these systems the focus was to evidence how the different structural compositions are influencing the dynamics and to investigate what can be learned from the relaxation time distributions in both one and two dimensions.

Chapter 5 presents the experimental results obtained for two different concrete types with different composition that were tested on-site as well as in the laboratory. Because of their different composition, especially the aggregate composition, the water ingress in the saturation process is behaving totally different. This study was relevant for the industrial partner to compare the MRE and NMR techniques based on different analytical methods, and from our academic point of view to gain deeper understanding of complex porous systems. The use of the NMR-MOUSE showed to be appropriate, especially for the on-site measurements, where the measurements could be performed fast and non-invasively.

The last chapter focuses on cultural heritage objects. The first part deals with wall paintings and mosaics, located in the ancient Roman city of Herculaneum. Depth profiles were recorded from the Mosaic of Neptune and Amphitrite, and from the wall paintings located in the House with the Black Room and in the Villa of the Papyri. These measurements were made to identify the type of information available from unilateral NMR with the NMR-MOUSE and their significance to

the conservation and restoration of mosaics and wall paintings. In the second part two oil paintings (a Rembrandt self portrait and an anonymous Cologne painting) were examined with the aim to determine the stratigraphy of them and to find clues about the age of the paintings through the NMR measurements.

Mobile NMR turns out to be a simple but reliable and powerful tool to investigate moisture distributions and pore structures in porous media as well as the conservation state and history of objects of cultural heritage.

1. INTRODUCTION

Chapter 2

Nuclear Magnetic Resonance

2.1 Introduction

Nuclear magnetic resonance (NMR) is a physical resonance phenomenon, which occurs when atomic nuclei, that have a $I \neq 0$ nuclear spin, are placed in a static magnetic field B_0 and exposed to a B_1 oscillating magnetic field (Abragam 61, Ernst 87). Since the phenomenon's observation in 1945 by Purcell and Bloch, the variety and number of applications is still increasing. It is used for the analysis of matter with various applications in medicine, chemistry, materials science, geophysics, and chemical engineering (Ernst 87, Blümich 05a). In conventional NMR, massive, superconducting magnets with field strengths of up to 22 T are used to obtain high sensitivity. Among the drawbacks for these high-field magnets are: they cannot be transported easily, they need to be cooled with liquid helium and nitrogen, large samples do not fit the given magnet bore. This is why mobile permanent magnets were introduced to materials testing where the object of interest is positioned in the stray field outside the magnet. First applications in this mobile low-field were made in the early 1950s by the oil companies (Cooper 80, Burnett 80, Jackson 80).

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2.2 Relaxation

In the absence of an external magnetic field, the spins are uniformly distributed in space. When an static magnetic field is applied a bulk magnetization is built up along the field direction. The magnetization will precess along the field direction with the Larmor frequency ω_0 defined as:

$$\omega_0 = -\gamma B_0 \quad (2.1)$$

where γ is the gyromagnetic ratio and B_0 the applied static magnetic field strength. The stronger the magnetic field is, the faster the spins will precess.

When a radio-frequency field, B_1 , is applied orthogonal to the main magnetic field, B_0 , with the frequency ω_0 , the magnetization vector will flip conventionally from z -direction to the xy plane, but still precessing around the z axis. So for example if the B_1 field is induced on the x axis the B_0 field will flip from z -direction to the y direction. This precession will induce a current in the coil, the same coil which generated before the B_1 field. The current signal will decay with the time constant T_2 , the transverse relaxation time, because of the loss of the net magnetization in the xy plane. The initial amplitude of the signal is proportional to the number of the excited spins. This signal is called free induction decay (FID).

The relaxation process refers to the process in which the spins return to the thermodynamic equilibrium (the spins are relaxing back into their lowest energy state) and no transverse magnetization is present. This implies also that all the coherences disappear (Levitt 08). In NMR two major relaxation processes can occur: longitudinal relaxation (T_1) and transverse relaxation (T_2).

1. The longitudinal relaxation time is the time needed by the spins to realign with the axis of the B_0 magnetic field (lowest energy state) and give up all their excess energy to the lattice in order to restore the thermal equilibrium. The longitudinal relaxation time, T_1 , called also spin-lattice relaxation time characterizes the rate at which the z component of the net magnetization, M_z , recovers its initial value M_0 . It can be described by an exponential function:

$$M_z(t) = M_0(t)(1 - \exp(-t/T_1)) \quad (2.2)$$

2. The transverse relaxation time is the time in which the net magnetization in the transverse plane decays. In this time the spins get out of phase with respect to each other because they are experiencing slightly different precession frequencies, due to molecular interactions or magnetic field inhomogeneities. The transverse relaxation time, T_2 , called also spin-spin relaxation time, in a perfectly homogeneous B_0 magnetic field is described by an exponential function:

$$M_{xy}(t) = M_{xy,0}(t)\exp(-t/T_2) \quad (2.3)$$

2.2.1 Relaxation in porous media

The different relaxation times measured in porous media give important information about parameters that characterize the pore space, such as porosity, surface-to-volume ratio, permeability, formation factor, tortuosity, pore size distribution and pore connectivity (Barrie 00). This is possible because the relaxation times are sensitive to molecular motion and pore geometry. For a porous medium the dimensions of the pores are related to relaxation times by the Brownstein - Tarr equation (Brownstein 77, Brownstein 79). The T_1 and T_2 relaxation times can be written as:

$$\frac{1}{T_1} = \frac{1}{T_{1\text{bulk}}} + \rho_1 \frac{S}{V} \quad (2.4)$$

$$\frac{1}{T_2} = \frac{1}{T_{2\text{bulk}}} + \rho_2 \frac{S}{V} + \frac{D}{12}(\gamma G t_E)^2 \quad (2.5)$$

where T_1 and T_2 are the apparent relaxation times measured in seconds (s), ρ_1 and ρ_2 are the surface relaxivities measured in $\mu\text{ m/s}$, S/V is the surface-to-volume ratio measured in m^{-1} , D is the diffusion coefficient measured in m^2/s , γ is the gyromagnetic ratio measured in MHz/T , G is the magnetic field gradient strength measured in T/m and t_E is the echo time measured in s.

If the echo time, is chosen small enough, then the 3rd element in equation 2.5 can be neglected and the relaxation is dominated just by the surface relaxation. In this case, the relaxation rate of a fluid in a pore, with surface-to-volume S/V is given by (Casanova 11):

$$\frac{1}{T_{1,2}} = \frac{1}{T_{1,2\text{bulk}}} + \rho_{1,2} \frac{S}{V} \quad (2.6)$$

2. NUCLEAR MAGNETIC RESONANCE

2.3 Magnet designs

In the following section the two magnet geometries will be presented, which were used in the experiments presented in the following chapters of this thesis. Both geometries used are made of permanent magnets. The magnetic field strength is in both cases low, which is preferable for characterizing porous media. The magnets can be divided into open and closed geometries (Blümich 11). In case of the open magnets, the investigated object is exposed to the stray field of the magnet and the NMR signal is acquired from a sensitive volume outside the magnet and inside the object. This principle was first used in the well-logging industry and become popular for materials testing with the invention of the NMR-MOUSE[®] (Eidmann 96, Blümich 08). Stray-field NMR does not limit the size of the object, but has low sensitivity. A large sensitive volume is obtained with homogeneous and weakly inhomogeneous fields inside closed magnets, where the most popular design is due to Halbach (Raich 04).

2.3.1 Profile NMR-MOUSE[®]

The original sensor consists of two permanent magnets mounted on a iron yoke, with opposite polarizations. In the gap between the two magnets an rf surface coil is placed as shown in figure 2.1. This coil is used for both, excitation and detection. This arrangement of magnet and coil creates a sensitive volume outside the sensor, from which the signal is received. In order to improve the sensitivity, newer geometries were made of four magnets with the rf coil placed in the gap between them. In this way a highly flat sensitive volume is achieved (Profile NMR-MOUSE) (Perlo 05). For the two Profile NMR-MOUSE devices used in this work, these sensitive volumes are a flat slice of about 1 cm respectively 5 cm diameter parallel to the surface of the sensor and allowing a penetration depth of 5 mm respectively 25 mm. For measurements, the NMR-MOUSE is mounted on a precision lift and its position is controlled by a computer. Then the assembly is positioned close to the object in such a way that the sensitive slice is parallel to the object. The measurement starts with the smallest distance between sensor and object at the largest measurement depth, and then the sensor is retracted

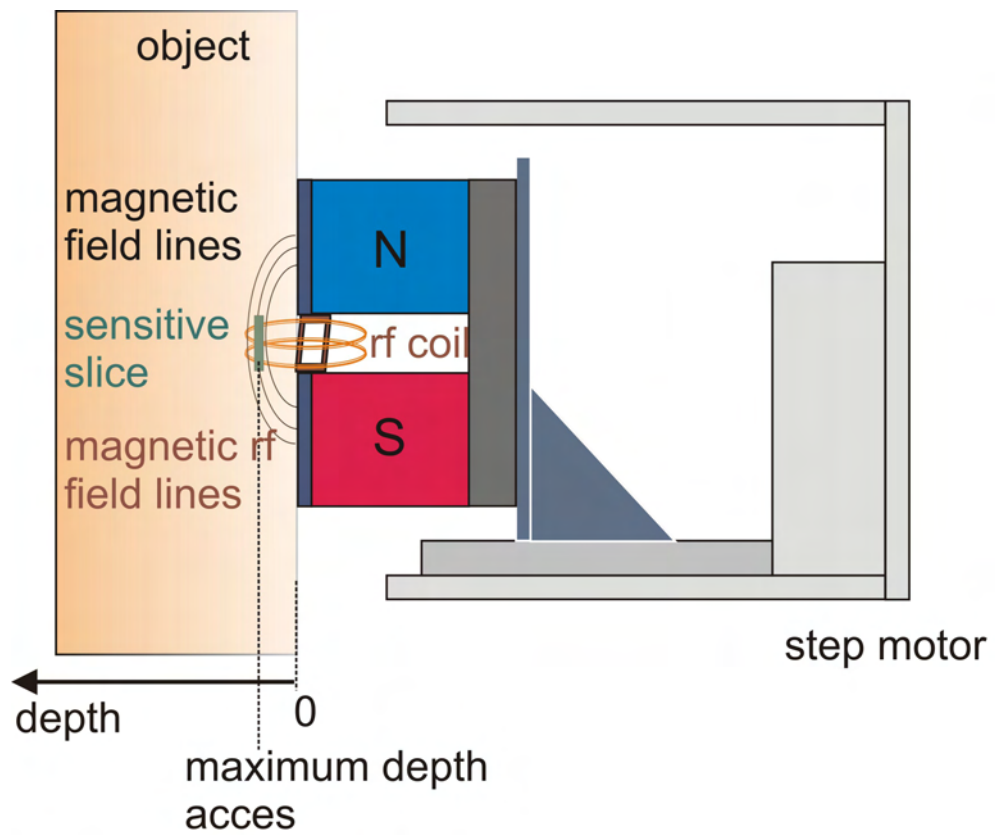


Figure 2.1: The NMR-MOUSE[®] for measuring depth profiles. Components and principle of operation. The magnet is positioned on a high precision lift, in order to record the signal at different depths in the sample.

2. NUCLEAR MAGNETIC RESONANCE

step by step. At each depth an NMR signal is received, from which a number is computed which determines the amplitude in a depth profile.

2.3.2 Halbach sensor

Following Klaus Halbach, a closed magnet with zero stray field outside and a comparatively homogeneous field inside transverse to the cylinder axis can be constructed from individual blocks of permanent magnets (Raich 04, Blümich 09). Such a Halbach magnet is composed of several rings with discrete magnet elements. The distance between the rings is adjusted to achieve axial field homogeneity. The magnet used for our experiments consists of two rings with 12 hexagonal bar magnets each (Fig. 2.2). The rings are separated by a 2.5 cm gap, which leads to a more homogeneous field along the magnet axis. The field strength is $B_0 = 0.51$ T and defines a proton resonance frequency of 21.85 MHz. The radio frequency coil was wound on a hollow Teflon cylinder with an inner diameter of 11 mm using five windings of copper sheet wire across a length of 5 cm. This allowed us to measure 10 mm thick samples.

2.4 Pulse sequences

In this section the pulse sequences are presented, by which the results presented in this thesis were obtained.

2.4.1 CPMG sequence

The most important and most frequently used pulse sequence is the CPMG named after its discoverers Carr, Purcell, Meiboom, Gill (Carr 54, Meiboom 58). It is used to measure the transverse relaxation time T_2 . A sketch of the sequence is presented in Fig. 2.3. It consists of a 90° rf pulse followed by a train of 180° pulses. The 180° pulse is double in amplitude but has the same length as the 90° pulse. In this way the frequency bandwidth is the same and the uniform slice selection is maintained. The time between two consecutive 180° is called echo time, t_E . With the help of the 180° pulses the magnetization, that started before to dephase due to B_0 inhomogeneities, is being refocused and an echo is formed. The envelope of

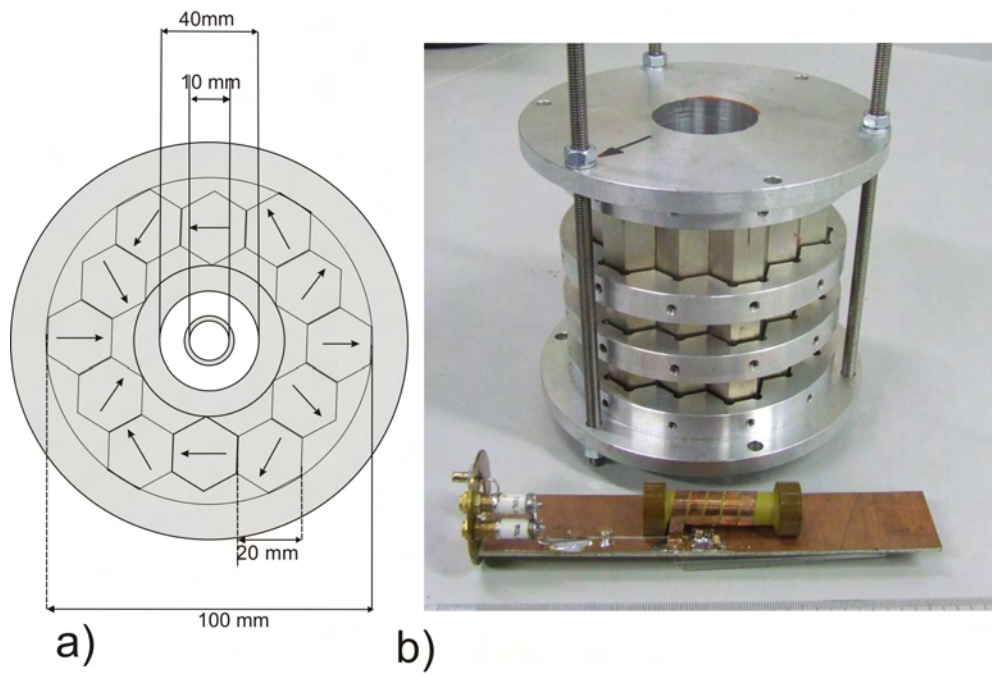


Figure 2.2: Halbach magnet geometry and setup. a) Sketch of one Halbach ring made of hexagonal magnet blocks with the positions of the blocks and the dimensions of the components. b) The magnet and the coil used for the measurements.

2. NUCLEAR MAGNETIC RESONANCE

the CPMG echo train is recorded (Blümich 05a). The echo amplitude decays with time. Its initial amplitude provides the proton density, which is proportional to

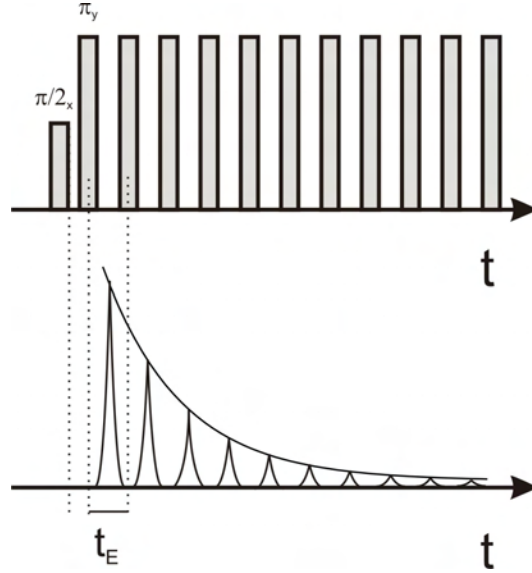


Figure 2.3: CPMG pulse sequence. Echoes can be generated with 180° impulses, the peaks of which stroboscopically sample the homogeneous-field decay. The echo envelope decays with time constant T_2 .

the content of hydrogen atoms in the sample. The decay-time constant T_2 varies with the mobility of the protons. It is proved that this is a robust method that can be used even in porous media with strong internal gradients (Song 02a).

The experimental CPMG signal $s(t)$ can be processed in different ways. A good approximation of the signal is a sum of exponential functions with relaxation times $T_{2eff,i}$

$$s(t) = \sum_i A_i \exp\left\{\frac{-t}{T_{2eff,i}}\right\} \quad (2.7)$$

where A_i is the relative spin density or mole fraction of components with the relaxation times $T_{2eff,i}$ in the sensitive volume if $s(t)$ is normalized to one. Another type of exponential function is the stretched function, where b is the shape parameter.

$$s(t) = A \exp\left\{-\frac{1}{b} \left\{\frac{-t}{T_{2eff,i}}\right\}^b\right\} \quad (2.8)$$

Often used is also the weighting parameter w . It is defined by the ratio of partial integrals over $s(t)$,

$$w = \frac{\int_{t_1}^{t_2} s(t) dt}{\int_{t_0}^{t_1} s(t) dt}. \quad (2.9)$$

2.4.2 T_1 saturation recovery

This sequence is used to measure the longitudinal relaxation time T_1 . It consist of a series of 90° pulses which will bring the magnetization in the xy plane and destroy it by totally dephasing it. Actually this is a 2D experiment ad it is used also for the T_1 - T_2 correlation experiments. During the recovery time τ_{re} the magnetization is restored on the z axis. The signal is read out with a CPMG sequence. So the longer the recovery time the more signal amplitude we obtain, until the magnetization is completely recovered and the maximum signal intensity is recorded. A sketch of the pulse sequence is shown in Fig. 2.4 For single-

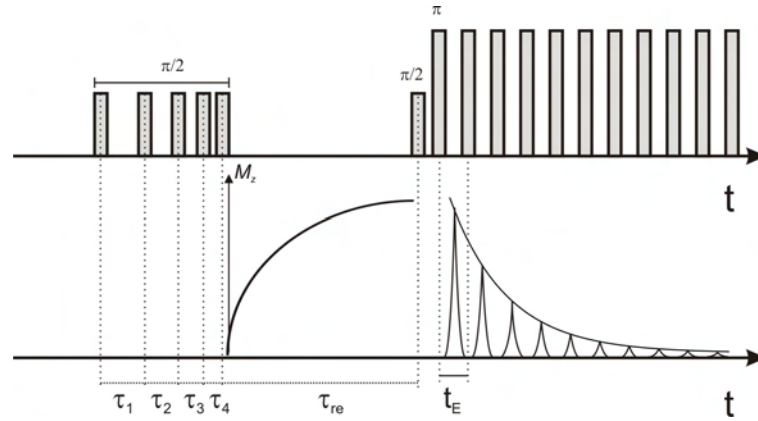


Figure 2.4: T_1 saturation recovery pulse sequence. In the first part of the sequence the net magnetization is destroyed. During the recovery time the magnetization is restored and read out with a CPMG sequence.

exponential relaxation the experimental signal is described as a function of the recovery time:

$$s(t) = A \left\{ 1 - \exp \left\{ \frac{-\tau_{re}}{T_1} \right\} \right\} \quad (2.10)$$

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The spin-lattice relaxation time give us information about the molecular mobility. Knowing the T_1 value, we can determine the recycle delay that we need to wait between two consecutive scans. Theoretically this is $5T_1$ and corresponds to the time in which 99 % of the magnetization recovers to the equilibrium state.

2.4.3 2D T_2 - T_2 exchange

The pulse sequence of the two-dimensional T_2 - T_2 relaxation exchange experiment (Washburn 06) is presented in Fig. 2.5. It has three main time periods: a T_2 encoding period for the indirectly detected relaxation dimension, a mixing time t_m , and an encoding period for read-out of the signal decay times after the mixing time. During the encoding periods, transverse relaxation is detected by the CPMG sequence. The time between the echoes, t_E , is kept at a constant value. The numbers of echoes used in the first encoding period is increased logarithmically so that the maximum first encoding time covers the same range as the second encoding time. The mixing times used between the two T_2 encoding times is varied. The second CPMG sequence is used for signal read out. This section and

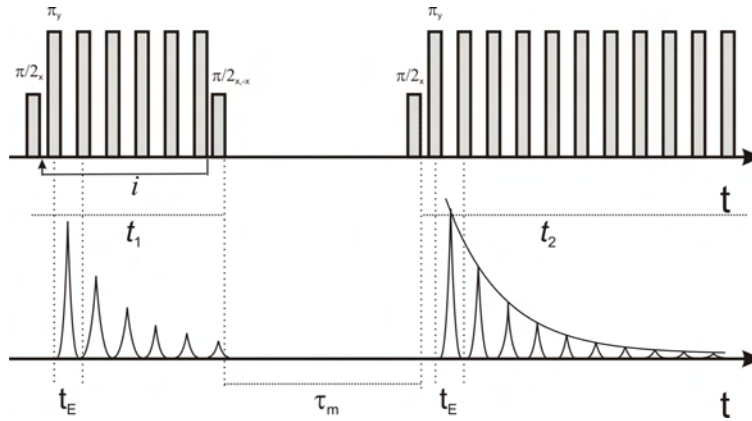


Figure 2.5: 2D T_2 - T_2 exchange pulse sequence. t_E is the echo time, which is kept fixed, and t_m is the mixing time, which extends the exchange time. The index i counts the 180° pulses in the indirect dimension. It is varied logarithmically in such a way that for maximum i , the evolution time in the indirect dimension is equal to the detection time in the direct dimension.

the theory to it will be discussed in detail in the next chapter.

2.4.4 Stimulated echo sequence

This sequence is used to measure the self-diffusion coefficient of the sample. The sketch of the pulse sequence is presented in Fig. 2.6. This sequence helps us to measure diffusion, D , as a function of the evolution time, Δ . It can be even measured at longer evolution times because in this time the magnetization is stored in the z direction and it decays with T_1 . The information obtained from

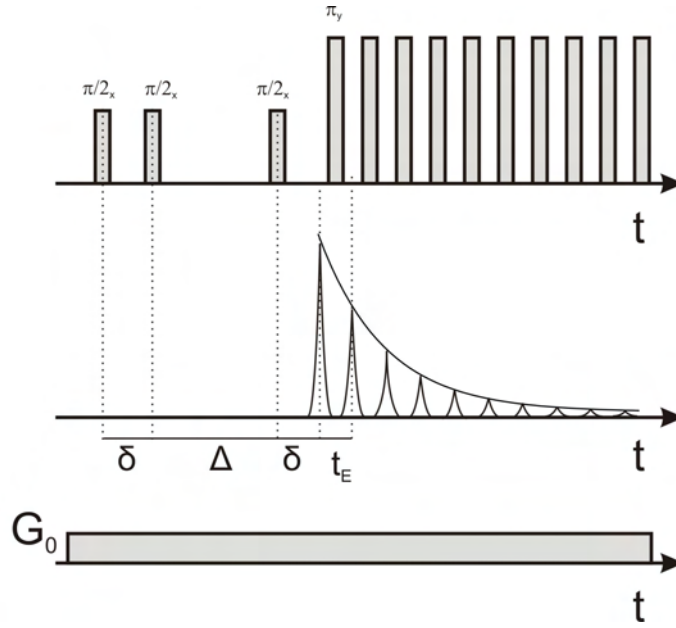


Figure 2.6: Stimulated echo sequence. In the first part the magnetization is stored in the z direction and the spins are allowed to diffuse. The signal is read out by a CPMG sequence. The more time the spins are allowed to diffuse the more the signal intensity will be attenuated.

the time depended diffusion coefficient is used in porous media to probe length scales. The measured signal is described by:

$$\ln \frac{I}{I_0} = -D\gamma^2 G^2 \delta^2 \left\{ \Delta + \frac{2}{3} \delta \right\} \quad (2.11)$$

where δ is the encoding time. This sequence is used also for the 2D D - T_2 correlation experiments, in which δ is varied.

2. NUCLEAR MAGNETIC RESONANCE

Chapter 3

Simple porous systems

Porous media have great influence in everyday life. They are present in different areas, starting from soil science and geology, continuing in medicine and biology and ending in civil engineering. The main idea is to understand the dynamics of the fluids captured inside the pores or moving from one pore environment to another. Once the dynamics is understood valuable informations are gained about the internal structure and surface properties. Moreover, with the help of the simulations and comparison with the experimental results, further helpful quantitative information can be gained for such porous systems. In order to gain deep understanding about the fluid dynamics in porous systems, first simple porous systems were tested and afterwards gradually more complex systems.

NMR can offer various solutions to solve the fluid dynamics problem inside a porous system. As an example local transport phenomena can be probed with different flow imaging techniques (Paciok 11). The well logging industry is using nowadays 2D relaxation correlation experiments to probe different fluid properties inside rocks down hole (Casanova 11). This shows that NMR techniques can probe microscopic as well as macroscopic phenomena and the changes inside the porous systems can be monitored from milliseconds up to days (Barrie 00). In this chapter fluid dynamics inside water saturated silica particles and water saturated aluminium oxide particles were tested using 2D relaxation exchange NMR experiments. Some of the results were published in the doctoral thesis of Maxime van Landeghem, as in this common project he was mainly focusing on the simulations and myself more on the experiments.

3. SIMPLE POROUS SYSTEMS

3.1 Introduction

2D relaxation exchange NMR is a form of 2D NMR where molecular motion and transport are mapped by relaxation exchange. This NMR technique has a lot of applications, especially in porous media studies (Hürlimann 02, Washburn 06, Mitchell 07). This experiment is analogous to 2D Fourier Transform frequency exchange NMR but appears more appropriate for the study of exchange of liquids between pores since relaxivity is expected to depend on pore size while the chemical shift does not (McDonald 05).

2D FT frequency exchange NMR spectroscopy is able to discriminate different sites in terms of their chemical shifts in an NMR spectrum (Bain 03). Site in this case is a given magnetic environment. But, in order to be able to measure a chemical shift spectrum a highly homogeneous magnetic field is needed and the nuclear magnetic response of a molecule to a particular environment is not characterized only by a chemical shift but also by a rate of decay. Thus, the chemical shift scale can be replaced by a relaxation or a diffusion scale, so that the occupancy of the different sites is obtained not by the structure of the frequency distribution but instead by the distribution of relaxation rates or diffusion coefficients. This information can easily be obtained with simple magnets and inhomogeneous fields. This concept of relaxation exchange NMR is not new (Kroeker 86, Lee 93) but gained recent popularity with the development of a stable 2D inverse Laplace transformation (ILT) algorithm (Venkataramanan 02).

However, the interpretation of 2D exchange maps is not so simple. The reason is that the evolution times must be long enough to allow relaxation encoding. To simplify the problem, one would ideally want the exchange rate to be short enough so that exchange would be negligible during the relaxation encoding periods. This limit is called the slow exchange limit. The mixing time on the other hand must be long enough so that exchange occurs significantly. The applicability of the slow exchange limit during the evolution period requires that the system relaxes faster than it exchanges. This means, that relaxation can only be significant during the mixing period and thus that relaxation and exchange must be considered simultaneously at least during the mixing period. On the other hand, if the exchange rate is fast enough, the mixing period must be kept short enough

to avoid relaxation implying that significant exchange would occur during the relaxation encoding periods. The fact that it is relaxation which is probed during the encoding periods implies that exchange can never be restricted to the mixing period and must also be considered during the encoding periods. Consequently, contrary to the case of 2D Fourier exchange spectra, 2D relaxation exchange maps can never be interpreted as pure exchange maps that is measurements of the exchange probability densities. In the next sections the theory of multi-site exchange NMR is presented followed by the results obtained for two and multiple site exchange (Landeghem 10).

3.2 Multi-site exchange NMR

Time evolution

Generalizing the problem and considering a porous system with N sites, which means N different environments responding in different ways to a magnetic environment, $s(t)$ would be the NMR signal coming from the superposition of all the magnetization components.

$$s(t) = \sum_{i=1}^N M_i(t) \quad (3.1)$$

Considering the magnetization evolution just for one site $M_i(t)$ in the absence of exchange, it will follow the differential equation

$$\frac{d}{dt}[M_i(t) - M_i^{\text{eq}}] = -R_i[M_i(t) - M_i^{\text{eq}}], \quad (3.2)$$

where the relaxation rate, R , is $R_i = 1/T_1$ and the magnetization at equilibrium, M^{eq} , is $M_i^{\text{eq}} = M_i^0$ for longitudinal magnetization and $R_i = 1/T_2$ and $M_i^{\text{eq}} = 0$ for transverse magnetization. If we consider N magnetization components they will be brought together in the magnetization vector $\mathbf{M}(t)$ and equation 3.2 becomes:

$$\frac{d}{dt}[\mathbf{M}(t) - \mathbf{M}^{\text{eq}}] = -\mathbf{R}[\mathbf{M}(t) - \mathbf{M}^{\text{eq}}], \quad (3.3)$$

where $\mathbf{R}$ is a diagonal matrix with the corresponding relaxation rates.

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If exchange occur between the different sites an off-diagonal exchange matrix $\mathbf{K}$, that will mix the magnetization components, needs to be introduced in equation 3.3.

$$\frac{d}{dt}[\mathbf{M}(t) - \mathbf{M}^{\text{eq}}] = -(\mathbf{R} + \mathbf{K})[\mathbf{M}(t) - \mathbf{M}^{\text{eq}}] \quad (3.4)$$

This equation will have the familiar solution

$$\mathbf{M}(t) - \mathbf{M}^{\text{eq}} = \exp\{-(\mathbf{R} + \mathbf{K})(t - t_0)\}[\mathbf{M}(t_0) - \mathbf{M}^{\text{eq}}] \quad (3.5)$$

The exchange matrix

The generalized exchange matrix $\mathbf{K}$ can be written as

$$\mathbf{K} = \begin{bmatrix} k_{11} & -k_{12} & \cdots & -k_{1n} \\ -k_{21} & k_{22} & \cdots & -k_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ -k_{n1} & -k_{n2} & \cdots & k_{nn} \end{bmatrix},$$

where k_{ij} is the exchange rate between sites i and site j . As $\mathbf{K}$ only mixes magnetization components and preserves the total magnetization, detailed mass balance requires that whatever amount of magnetization is lost in one component has to be recovered from other components:

$$\mathbf{K}\mathbf{M}^{\text{eq}} = 0 \quad (3.6)$$

For example, for magnetization component M_1 can be written

$$k_{11}M_1^{\text{eq}} - \sum_{j=2}^n k_{1j}M_j^{\text{eq}} = 0 \quad (3.7)$$

Likewise, whatever total amount of magnetization is received by one of the magnetization components from other magnetization components has to be restored to them. For example, for magnetization component M_1 ,

$$\sum_{j=2}^n k_{1j}M_j^{\text{eq}} = \sum_{i=2}^n k_{i1}M_1^{\text{eq}} \quad (3.8)$$

This will lead to

$$k_{11} - \sum_{i=2}^n k_{i1} = 0 \quad (3.9)$$

so that the column sums of $\mathbf{K}$ vanish. The kinetic matrix is determined by $(n-1)^2$ independent parameters. Note that the rates k_{ij} apply to the forward kinetics from state i to state j , so that the rates k_{ji} apply to the reverse kinetics from state j to state i . The associated equilibrium constant K_{ij} which is defined by the ratio $[M_j]/[M_i]$ of concentrations is given by

$$K_{ij} = \frac{[M_j]}{[M_i]} = \frac{k_{ij}}{k_{ji}} \quad (3.10)$$

Magnetization evolution in T_2 - T_2 exchange NMR experiments

The time evolution of the magnetization is examined for a T_2 - T_2 exchange experiment with a mixing period during which the magnetization experiences T_1 relaxation. The pulse sequence is presented in Fig. 2.5 in section 2.4.3. The magnetization after the mixing time t_m follows from Eqn. 3.5:

$$\mathbf{M}(t_m + t_1 + t_0) - \mathbf{M}^0 = \exp\{-[\mathbf{R}^{(1)} + \mathbf{K}](t_m)\}(\mathbf{M}(t_1 + t_0) - \mathbf{M}^0) \quad (3.11)$$

During the evolution period t_1 and the detection period t_2 the magnetization is given by

$$\mathbf{M}(t_1 + t_0) = \exp\{-[\mathbf{R}^{(2)} + \mathbf{K}](t_1 + t_0)\}\mathbf{M}(t_0) \quad (3.12)$$

and

$$\mathbf{M}(t_2 + t_m + t_1 + t_0) = \exp\{-[\mathbf{R}^{(2)} + \mathbf{K}](t_2)\}\mathbf{M}(t_m + t_1 + t_0) \quad (3.13)$$

The last three equations combine into

$$\begin{aligned} \mathbf{M}(t_2 + t_m + t_1 + t_0) = & \exp\{-[\mathbf{R}^{(2)} + \mathbf{K}](t_2)\}[\exp\{-[\mathbf{R}^{(1)} + \mathbf{K}](t_m)\} \\ & \times (\exp\{-[\mathbf{R}^{(2)} + \mathbf{K}](t_1 + t_0)\}\mathbf{M}(t_0) - \mathbf{M}^0) + \mathbf{M}^0] \end{aligned} \quad (3.14)$$

Statistical interpretation of peak intensities in the slow exchange limit

In 2D Fourier exchange NMR experiments, the fast exchange limit may apply during the mixing time so that relaxation can be neglected during t_m , while

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Table 3.1: Evolution operators for the three cases that can occur in 2D relaxation exchange NMR

	$U(t_1)$	$U(t_m)$	$U(t_2)$
Case 1	$\exp\{-[\mathbf{R}](t_1)\}$	$\exp\{-[\mathbf{R} + \mathbf{K}](t_m)\}$	$\exp\{-[\mathbf{R}](t_2)\}$
Case 2	$\exp\{-[\mathbf{R} + \mathbf{K}](t_1)\}$	$\exp\{-[\mathbf{K}](t_m)\}$	$\exp\{-[\mathbf{R} + \mathbf{K}](t_2)\}$
Case 3	$\exp\{-[\mathbf{R} + \mathbf{K}](t_1)\}$	$\exp\{-[\mathbf{R} + \mathbf{K}](t_m)\}$	$\exp\{-[\mathbf{R} + \mathbf{K}](t_2)\}$

the slow exchange limit applies during the evolution and detection times. This situation is never encountered in 2D relaxation exchange NMR. Only the following three cases may be encountered. The corresponding evolution operators are given in Table 3.1.

1. The exchange is slow during the evolution periods, which means that the time scale of exchange is long compared the time scale of relaxation, and pure relaxation encoding is possible during the evolution periods. Consequently, there is no fast exchange during the mixing period.
2. The exchange is fast compared to relaxation. Then the fast exchange limit is satisfied during the mixing time. In this case, the mixing period involves only exchange but not relaxation, while the evolution and detection periods encode both, relaxation and exchange.
3. The exchange rate is commensurate to the relaxation rates, and both must be considered during the evolution, mixing, and detection periods.

Only in the first case can the 2D relaxation cross peaks be analyzed in terms of exchange kinetics, although not easy. Being in the slow exchange limit during evolution, means that one can measure the relaxation rate R_i at time t and identify it with the magnetization of the corresponding site M_i . In this case, the slow exchange limit applies to the t_1 and t_2 encoding periods, and Eqn. 3.14 simplifies to

$$\begin{aligned}
 \mathbf{M}(t_2 + t_m + t_1 + t_0) = & \exp\{-[\mathbf{R}^{(2)}](t_2)\}[\exp\{-[\mathbf{R}^{(1)} + \mathbf{K}](t_m)\} \\
 & \times (\exp\{-[\mathbf{R}^{(2)} + \mathbf{K}](t_1 + t_0)\}\mathbf{M}(t_0) - \mathbf{M}^0) + \mathbf{M}^0]
 \end{aligned}
 \tag{3.15}$$

Since the relaxation matrix $\mathbf{R}^{(2)}$ is diagonal, the magnetization of a particular site i at the end of the evolution period is obtained as

$$M_i(t_1 + t_0) \approx \exp\{-R_i^{(2)}t_1\}M_i(t_0) = \exp\{-t_1/T_{2,i}\}M_i(t_0) \quad (3.16)$$

After the mixing time t_m , one obtains from Eqn. 3.5,

$$M_j(t_2 + t_m + t_1 + t_0) \approx \exp\{-R_j^{(2)}t_2\}M_j(t_m + t_1 + t_0) = \exp\{-t_2/T_{2,j}\}M_j(t_m + t_1 + t_0) \quad (3.17)$$

The resultant time domain signal is then approximated by

$$M_j(t_2 + t_m + t_1 + t_0) \approx \exp\{-t_2/T_{2,j}\} \left[\sum_i \{ [\exp\{-(\mathbf{R}^{(1)} + \mathbf{K})t_m\}]_{ji} (\exp\{-t_2/T_{2,i}\}M_i(t_0) - M_i^0) \} + M_j^0 \right] \quad (3.18)$$

After the 2D inversion over t_1 and t_2 cross peaks arise at coordinates $(T_{2,i}, T_{2,j})$ with amplitudes given by

$$M_{ij} = [\exp\{-(\mathbf{R}^{(1)} + \mathbf{K})t_m\}]_{ij} M_i(t_0) \quad (3.19)$$

Therefore, when this slow exchange limit is fulfilled for the encoding and detection periods t_1 and t_2 , the cross-peak amplitude represents the magnetization from site j that originated from site i before the mixing time t_m .

The other two cases are even more complex as magnetization is not conserved or follows multiple single relaxation paths. As a consequence, 2D relaxation exchange maps cannot be interpreted in terms of conditional probability densities, and the observed decay rates are not associated with single site relaxation rates (McDonald 05, Monteilhet 06).

3.3 The ILT problem

2D Laplace NMR did not evolve as fast as 2D Fourier NMR. This is because of the ill-defined ILT which inverts the data from the time domain into the relaxation rate domain. This problem was presented in papers by Venkataramanan and Song (Venkataramanan 02, Song 02b, Song 10) and finally became possible after Venkataramanan (Venkataramanan 02) proposed an ILT computing method

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that was not that consuming for the PC. The oldest algorithm used for ILT is the CONTIN program (Provencher 82a, Provencher 82b). The software that I used is the one developed by Prof. P.T. Callaghan but it is based on the algorithm of Venkataramanan (Venkataramanan 02, Godefroy 03).

Inside a porous medium different environments can be found which will influence the NMR signal in different ways (Casanova 11). This signal can be observed, in general terms, as a multi-exponential decay, $s(t_i)$. In case of the 1D experiments this will be:

$$s(t_i) = \sum_s A(T_s) \exp\left\{-\frac{t_i}{T_s}\right\} + \varepsilon_i, \quad (3.20)$$

where T_s is the relaxation times and $A(T_s)$ the relation time distribution of the studied system. ε_i represent here the noise of the signal.

The procedure to invert the data and to determine the distribution of relaxation times $A(T_s)$ consist in applying the non-negative least square algorithm described in (Lawson 74). This consists in minimization of:

$$\min\{\chi^2 = \sum_{i=1}^n (s(t_i) - \sum A(T_s) \exp\{-\frac{t_i}{T_s}\})^2\}. \quad (3.21)$$

However, because the real signal has some noise, too, in it, the determination of the distribution by a simple non-negative least square fit would lead to a multitude of peaks, in this way the problem is ill-posed. To avoid it, a regularization function is introduced (or the so called smoothing parameter)(Borgia 98).

$$\min\{\chi^2 = \sum_{i=1}^n (s(t_i) - \sum A(T_s) \exp\{-\frac{t_i}{T_s}\})^2 + \alpha^{-1} \sum_{s=1}^m (2A(T_s) - A(T_{s-1}) - A(T_{s+1}))^2\}, \quad (3.22)$$

where α represents the regularization parameter. This expression can be rewritten as:

$$Y = KX + E, \quad (3.23)$$

where X is the distribution that need to be determined, Y is the signal measured, K is the known matrix of the $\exp(-t_i/T_s)$. The least square function has now the form:

$$\chi^2 = \|KX + E\|^2 + \alpha^{-1} \|X\|^2. \quad (3.24)$$

In case of a 2D experiment, for example for a relaxation-relaxation correlation experiment, the signal measured follows:

$$s(t_1, t_2) = \sum A(T_1, T_2) \exp\{-\frac{t_1}{T_1}\} \exp\{-\frac{t_2}{T_2}\} + \varepsilon_{t_1, t_2}. \quad (3.25)$$

and in the matrix form it would be:

$$Y = K_1 X K_2' + E. \quad (3.26)$$

Here the K_1 and K_2 matrices are the $\exp(-t_{1i}/T_{1s})$ and $\exp(-t_{2i}/T_{2s})$. In order to get a 2D Laplace inversion of the data defined by Y , the matrix Y of size $m \times n$ needs to be reduced to a vector of length $m \times n$. This is obtained by using the singular value decomposition algorithm (Song 02b).

$$\begin{aligned} K_1 &= U_1 S_1 V_1' \\ K_2 &= U_2 S_2 V_2' \end{aligned} \quad (3.27)$$

where U and V are unitary matrices. The reduced matrix is determined according to:

$$\begin{aligned} \tilde{K}_1 &= U_1' K_1 \\ \tilde{K}_2 &= U_2' K_2 \\ \tilde{Y} &= U_1' Y U_2 \end{aligned} \quad (3.28)$$

Equation 3.26 can be rewritten:

$$\tilde{Y} = \tilde{K}_1 X \tilde{K}_2' + E. \quad (3.29)$$

This 2D form can be reduced to a 1D form after the tensor product of the matrices $\tilde{K}_1$ and $\tilde{K}_2$ and the data can be processed in the same way as for the 1D case mentioned above (Eq. 3.23, 3.24).

3.4 Results

Numerical simulations, which are presented in (Landeghem 10), were made for both 1D and 2D experiments. From the simulations, the following important conclusions can be made:

In case of 1D two site exchange, the influence of the exchange rate on peak positions and peak integrals, revealed the following:

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- when the exchange time becomes short compared to the relaxation times, one of the peaks grows at the expense of the other, until only a single peak is observed but its integral corresponds to the number of spins from both sites.
- when the exchange time is longer than the longest relaxation time the integrals provide the right population values. (In the case presented in (Landeghem 10) the exchange time was 10 times longer than the longest relaxation time).
- in the 1D relaxation distributions (after the ILT) the relaxation times extracted from the maxima of each distributions are just apparent relaxation times given by the eigenvalues of the $\mathbf{R} + \mathbf{K}$ matrix from eq. 3.4. Depending on whether or not it is fast exchange or slow exchange the distributions will be shifted on the relaxation time axis. It was shown that just for really slow exchange an unbiased estimation of the relaxation times and distributions was obtained.

In case of 2D two-site exchange, the following conclusions can be made:

- when the exchange is very fast, the exchange map exhibits a single diagonal peak in accordance with the 1D study. When the exchange time is increased and becomes comparable to the relaxation times, cross peaks appear as well as two distinct diagonal peaks and when the exchange time becomes longer than the relaxation times, the exchange is too slow to be seen and the cross peaks disappear.
- the peak intensities in the 2D maps vary with the mixing time t_m . The diagonal peaks decay and the cross-peaks grow exponentially with the mixing time t_m until the the longitudinal relaxation time influences this mixing period and the intensities start to decay towards 0.
- regarding the sensitivity of the ILT to noise (which sometimes give rise to artifacts in the 2D distribution maps), the simulations showed that the noise affects strongly the integral values. For only a 2% noise level in the NMR signal the relative deviations of the integral can go up to 10% and

also one of the diagonal peaks is shifting away from the diagonal for a noise level higher than 2%.

3.4.1 Two-site 2D relaxation exchange experiments

The results obtained from the simulations were compared to experimental data. For this purpose silica particles were synthesized according to the Stober reaction (Green 03, Stober 68). The reason for choosing these particles is the known and tested mono-disperse particle size distribution. The particles size was determined with a SEM to be 500 nm with a standard variation of 40 nm. These mono-disperse particles are good candidates for two site exchange. They just need to be water saturated and the two sites would be the bulk water between the particles and the water at the surface of the particles (Landeghem 10).

The experiments were performed in a Halbach magnet presented before in section 2.2. The field strength was $B_0 = 0.51$ T, defining a proton resonance frequency of 21.85 MHz. Figure 3.1 show the spin-lattice and spin-spin relaxation decays measured with a saturation recovery and a CPMG sequence. These pulse sequences were presented before in Fig. 2.4 and Fig. 2.3 in Section 2.4.2 and 2.4.1. This system of saturated silica particles exhibit two apparent T_2' s which

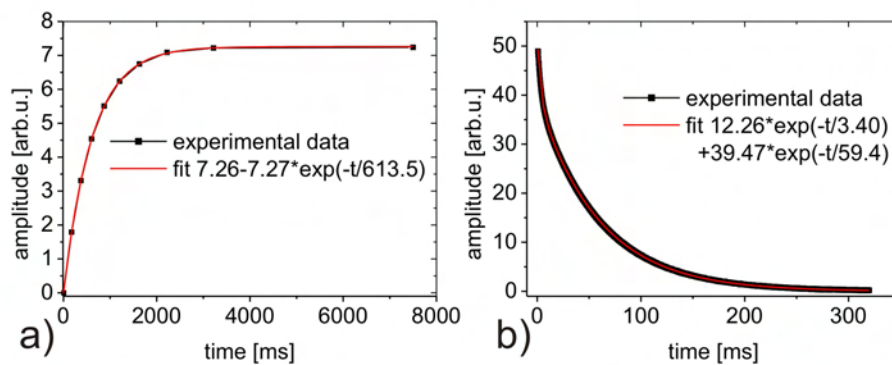


Figure 3.1: Water saturated silica particles. a) T_1 saturation curve and b) T_2 CPMG decay curve. Two apparent relaxation times can be distinguished.

correspond to two different sites, but has a single apparent T_1 . Taking into account

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the results from the 1D simulations we know that we are in the situation where the exchange time is small compared to the spin-lattice relaxation times.

T_1 - T_2 correlation maps were simulated and measured as a support material for the T_2 - T_2 exchange maps. The results are shown in Fig. 3.2. In the experimental

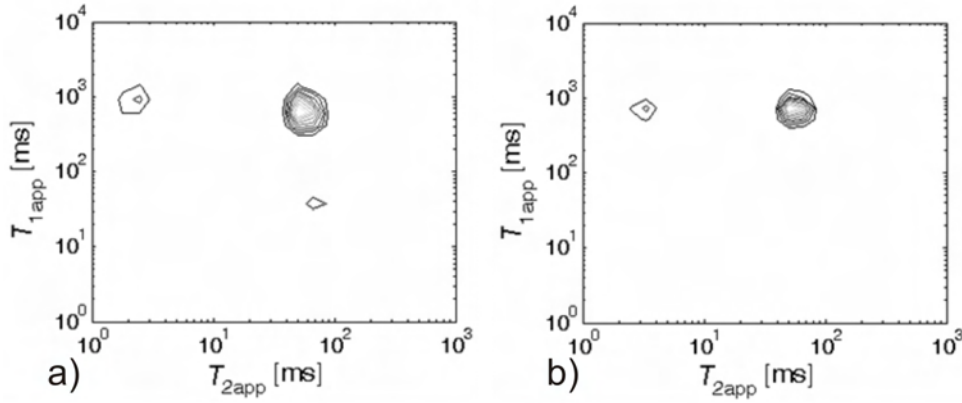


Figure 3.2: Experimental and simulated T_1 - T_2 correlation maps of water saturated silica particles. a) Experimental map and b) Simulated map. The experimental map shows a small peak which is attributed to noise in the experimental data.

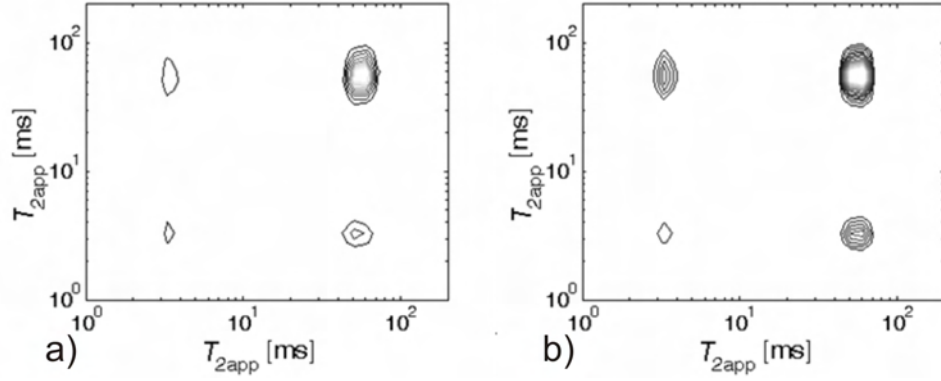
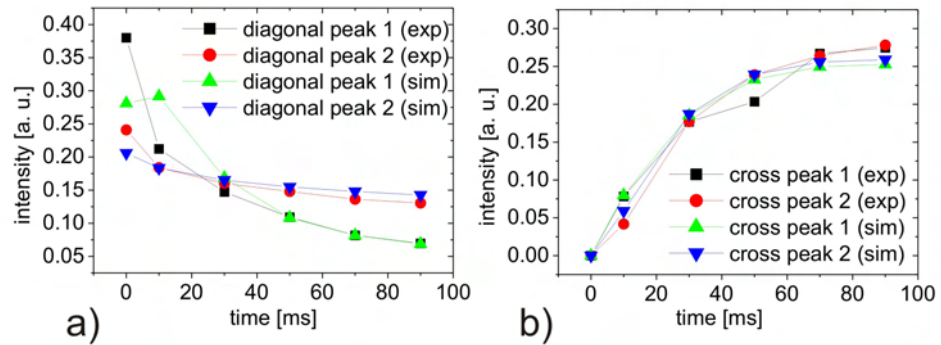
map a very small peak can be noticed which is not existent in simulated map. It is likely to be noise which gave raise to this artifact also because in the 1D fitted data just one component could be identified.

T_2 - T_2 relaxation exchange experiments were simulated and performed using the pulse sequence presented in Section 2.4.3. Six different mixing times were used (0, 10, 30, 50, 70 and 90 ms). The comparison between simulations and experiments is shown for one mixing time in Fig. 3.3. The parameters obtained by the simulations are summarized in Table 3.2.

Fig. 3.3a is obtained by inverting the experimental data and Fig. 3.3b by inverting the simulated data. The peaks of the simulated spectra are broader than the experimental ones. This is interpreted as an artifact due to regularization during the ILT. To verify the quality of the fitting, the peak integrals of both simulated and experimental exchange maps were compared. This evolution of the integrals over the mixing time is represented in Fig. 3.4. Except for a divergency for the diagonal peak 1 at low mixing time, the results present a pretty good

Table 3.2: Parameters obtained by the simulations.

Parameter	Value
M^A	5.5
M^B	18.4
T_2^A	3.9 ms
T_2^B	103.5 ms
T_1^A	610 ms
T_1^B	630 ms
k	$1/110 \text{ ms}^{-1}$


Figure 3.3: a) Experimental and b) Simulated T_2 - T_2 correlation maps for 70 ms mixing time. The agreement is evident.

Figure 3.4: Comparison between the peak integrals obtained experimentally and the simulated ones for different mixing times. a) Integrals obtained for the diagonal peaks and b) Integrals obtained for the cross peaks. Good agreement is observed.

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agreement between the mixing time dependence of the peaks intensities observed experimentally and the ones simulated.

3.4.1.1 Temperature dependence

The same system of water saturated silica particles was tested for its behavior at different temperatures. For this purpose the NMR system was put into a temperature chamber and the evolution of the T_2 - T_2 exchange maps at three different temperatures was monitored for different mixing times. The three temperatures were 5, 15 and 25°C.

First, after the whole system was equilibrated for the corresponding temperature, at each temperature the spin-lattice relaxation time was measured. T_1 behaviour at the three different temperatures is shown in Fig. 3.5. From the

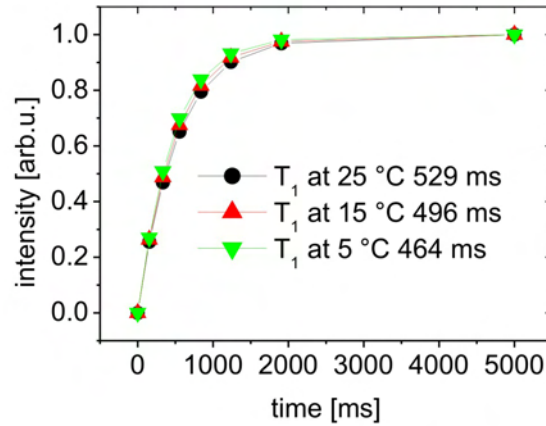


Figure 3.5: T_1 behavior at different temperatures. T_1 is increasing with temperature.

figure it is noticeable that the relaxation time is increasing with increasing temperature. This is a known fact (Levitt 08, Fukushima 81).

T_2 - T_2 relaxation exchange experiments were made at different temperatures for different mixing times, from 0 up to 160 ms. For 0 ms mixing time the ILT maps for all three temperatures were over plotted, and the 1D projection on both

relaxation time axis show the behavior of the spin-spin relaxation time with temperature. Figure 3.6 present the two main diagonal peaks, which are present

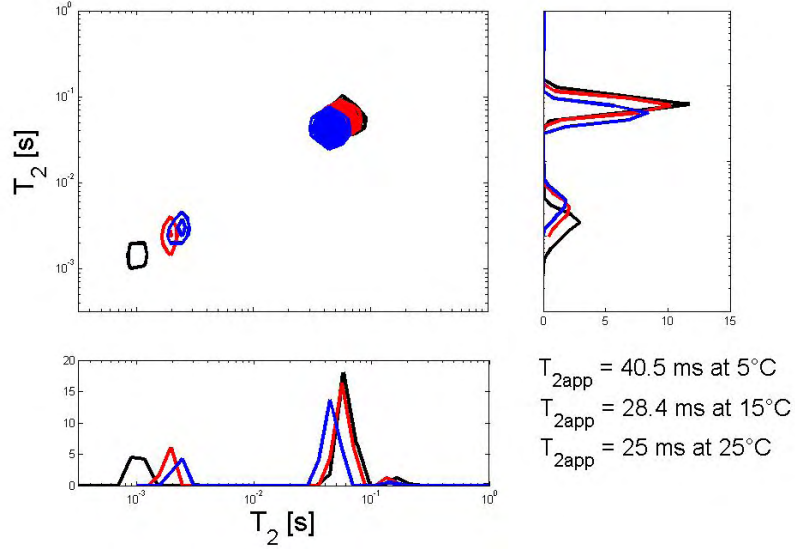


Figure 3.6: T_2 - T_2 correlation maps for different temperatures at 0 ms mixing time. The projection of the T_2 distributions on the relaxation axis show that T_{2app} is decreasing with increasing temperature.

at each temperature and their projections on each relaxation axis. It is observable that the T_{2app} is decreasing with increasing temperature. This may be due to higher molecular mobility. Higher molecular mobility gives a higher diffusion coefficient and this will induce a lower spin-spin relaxation time. The comparison between simulations and experiments is shown for 70 ms mixing time and at three different temperatures in Fig. 3.7. The peaks of the simulated spectra are broader than the experimental ones. This is interpreted as an artifact due to regularization during the ILT. The experimental spectra present an asymmetry for one of the cross peaks. This is due to noise artifacts, fact demonstrated (Landeghem 10). Nevertheless, good agreement between simulation and experimental data is observed. The parameters obtained by the simulations for the three different temperatures are summarized in Table 3.3.

To verify the quality of the ILT 2D correlation maps, the peak integrals of the exchange maps were compared. An overall evolution of the integrals over the

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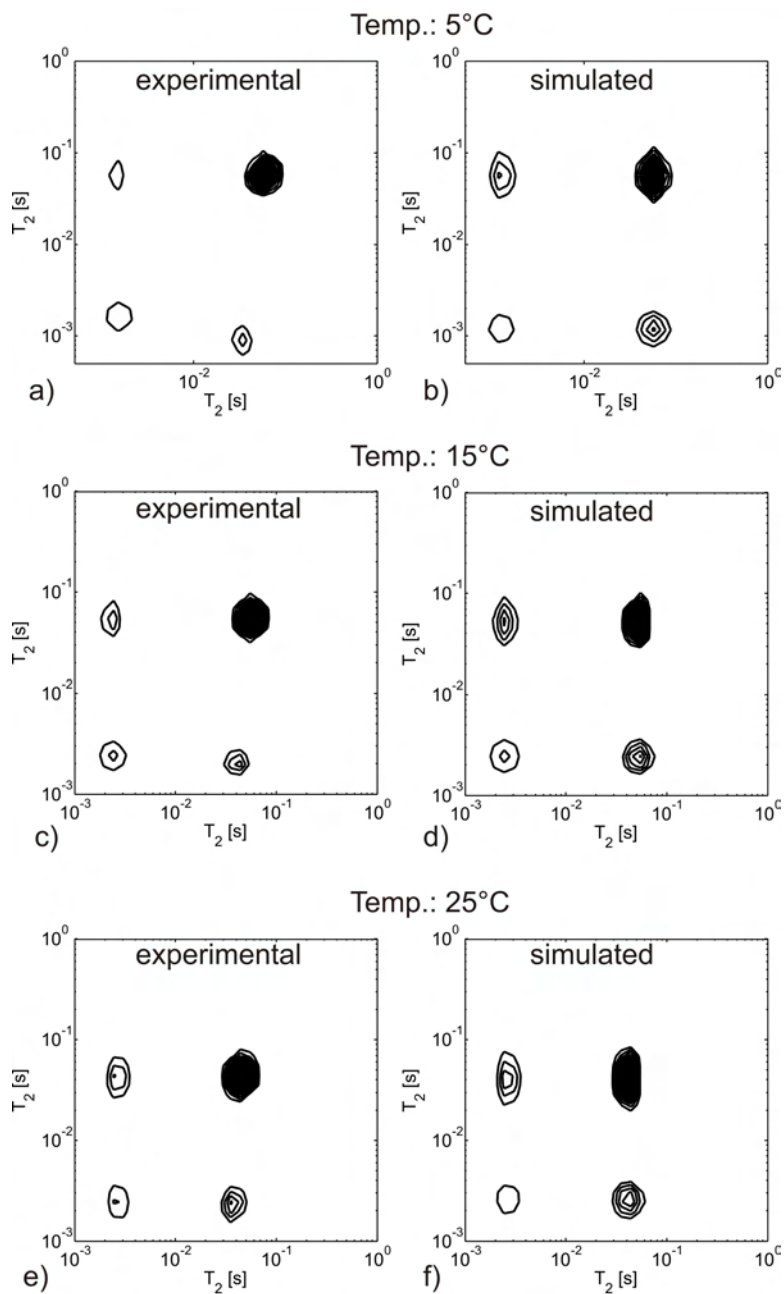


Figure 3.7: T_2 - T_2 correlation maps for different temperatures at 70 ms mixing time. a) Experimental 2D correlation map at 5°C. b) Simulated 2D correlation map at 5°C. c) Experimental 2D correlation map at 15°C. d) Simulated 2D correlation map at 15°C. e) Experimental 2D correlation map at 25°C. f) Simulated 2D correlation map at 25°C.

Table 3.3: Parameters obtained by the simulations for the three different temperatures.

Parameter	Value at		
	5°C	15°C	25°C
M^A	16	12	7.9
M^B	37	25	15.4
T_2^A	2 ms	2.7 ms	3 ms
T_2^B	130 ms	90 ms	70 ms
T_1^A	460 ms	500 ms	510 ms
T_1^B	470 ms	510 ms	540 ms
k	1/135 ms ⁻¹	1/121 ms ⁻¹	1/110 ms ⁻¹

mixing time at different temperatures is represented in Fig. 3.8. A detailed evo-

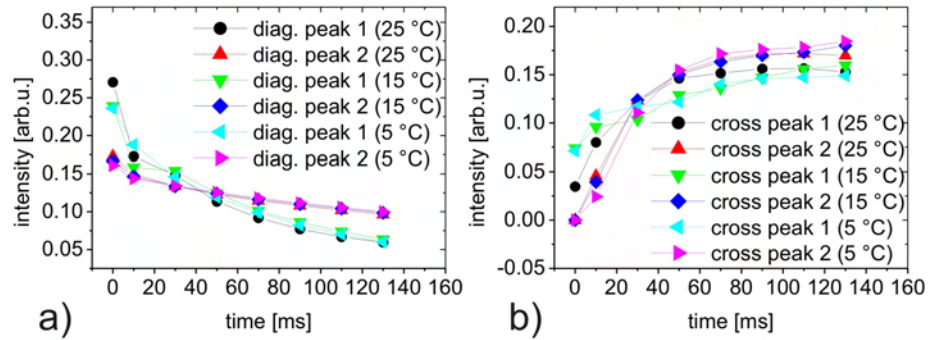


Figure 3.8: Comparison between the diagonal and cross peaks integrals for different mixing times at different temperatures. a) Diagonal peaks integrals at different temperatures. b) Cross peaks integrals behavior at different temperatures.

lution of the simulated and experimentally obtained peak integrals for different mixing times and at different temperatures is presented in Fig. 3.9. For all temperatures a slowly deviation of on cross peak integral evolution is observed. This is due to the noise artifact that influence the integral evolution (Landeghem 10). Good agreement between the simulations and experimentally obtained peak integrals is observed. From the dependence of the reaction rate, k , on temperature,

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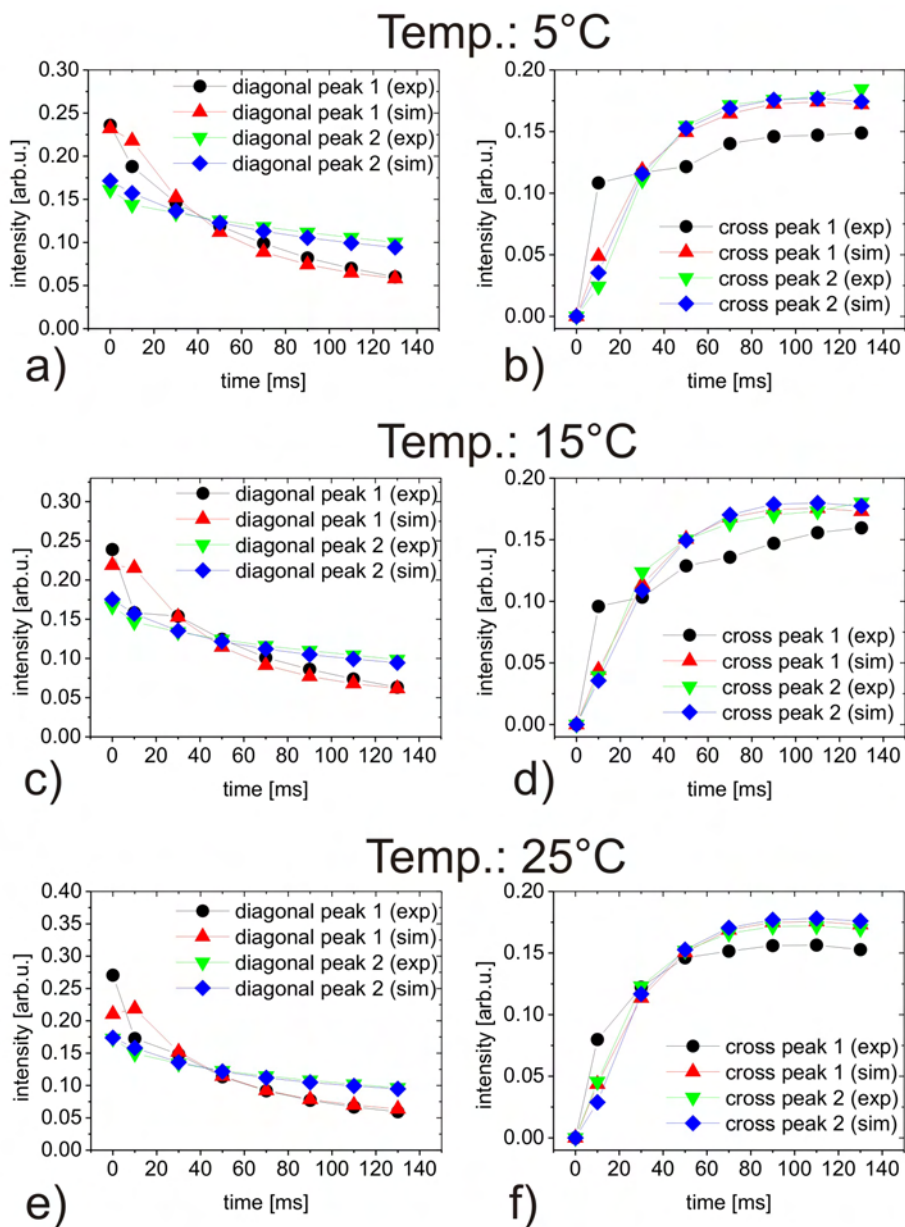


Figure 3.9: Comparison between the peak integrals obtained experimentally and simulated for different mixing times at different temperatures. a) Integrals obtained for the diagonal peaks at 5°C. b) Integrals obtained for the cross peaks at 5°C. c) Integrals obtained for the diagonal peaks at 15°C. d) Integrals obtained for the cross peaks at 15°C. e) Integrals obtained for the diagonal peaks at 25°C. f) Integrals obtained for the cross peaks at 25°C.

T , the activation energy can be estimated by the Arrhenius law (Laidler 65):

$$k = A \exp\left(-\frac{\Delta E}{RT}\right) \quad (3.30)$$

In this equation, A is the pre-exponential factor, R is the gas constant and ΔE is the activation energy (Atkins 02). The pre-exponential factor, is a statistical parameter which gives us information about the collision frequency. The dependence is plotted in Fig. 3.10. From the fit an activation energy of 7.059 kJ/mol and a pre-exponential factor of 1.036 ms^{-1} could be extracted. The value obtained for the activation energy is higher than the energy for dipole-dipole interaction and the Van-der-Waals interaction (2 kJ/mol) but lower than the energy for ion-dipole interaction and the hydrogen bond (15 respectively 20 kJ/mol) (Holleman 95, Moore 86). This activation energy is the energy that the hydrogen atom needs for moving from one environment to the other, in other words the diffusion activation energy (Stallons 01).

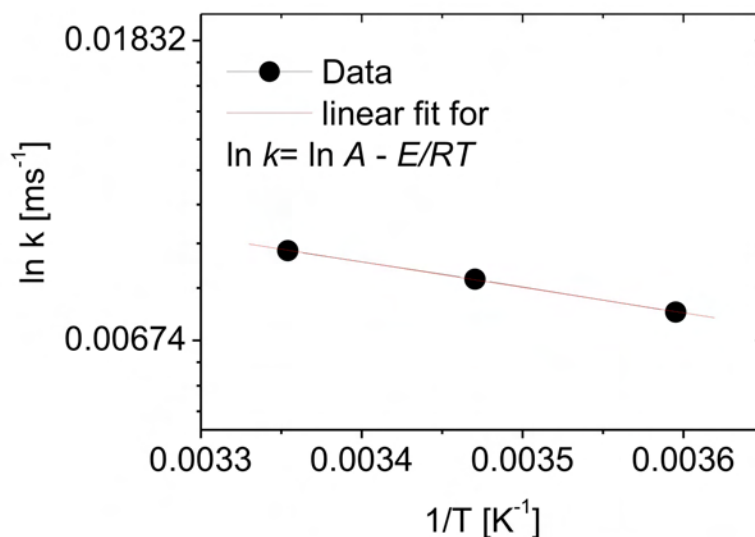


Figure 3.10: Dependence of the reaction rate on temperature. The activation energy can be extracted by the fit.

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3.4.2 Multi-site 2D relaxation exchange

There is a difference between two- and n -site exchange. For two-sites, the exchange maps are symmetric as mass balance requires the forward and backward exchange to be identical. When exchange takes place between more than two sites, detailed mass balance applies to the overall exchange but not to subsets consisting of only two of n sites, so that asymmetric exchange maps can arise. To obtain a symmetric map, all exchange times have to have the same value, whereas they differ strongly in the non-symmetric case.

To test this, another system was used, water saturated aluminium oxide. The substance was provided by Merck KGaA. Aluminium oxide is a hygroscopic, white powder used usual as adsorbent in chromatography in specifically defined pH ranges, with a mean pore size of 9 nm and particle size range of 60- 200 μm . T_2 - T_2 experiments were performed for different mixing times, from 0 to 150 ms.

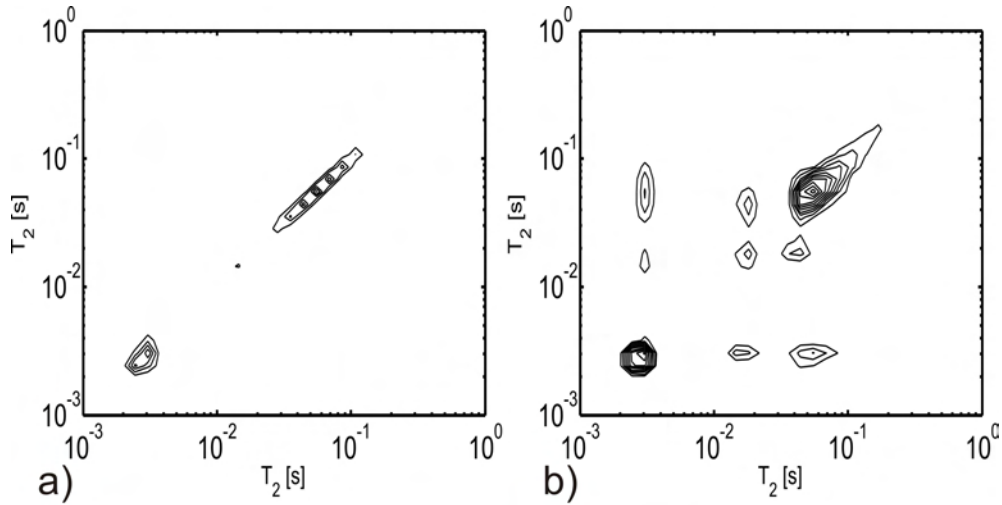


Figure 3.11: T_2 - T_2 exchange maps for different mixing times. a) Mixing time 1 ms, showing the multiple main peaks. b) Mixing time 40 ms showing a symmetric exchange map.

Figure 3.11 shows the 2D T_2 - T_2 exchange distributions for two different mixing times. For the left plot where mixing time is 1 ms, three, or possible more, main peaks without exchange between the different sites can be observed. The three components have $T_{2\text{app}}$ values of 3.4, 11.5 and 73.6 ms. We assume that the

shortest T_2 value arise from the water molecules inside the pores. The middle value arises from the water layers surrounding the Aluminium Oxide particle and the longest and broadest T_2 distribution arise from the bulk water. In the second plot, the mixing time used was 40 ms, and the exchange between the environments is quite noticeable. Water migration between different environments is observed.

To verify the quality of the ILT 2D correlation maps, the peak integrals of the exchange maps were compared. This evolution of the integrals over the mixing time is represented in Fig. 3.12. It shows how the intensity of the integrals of the

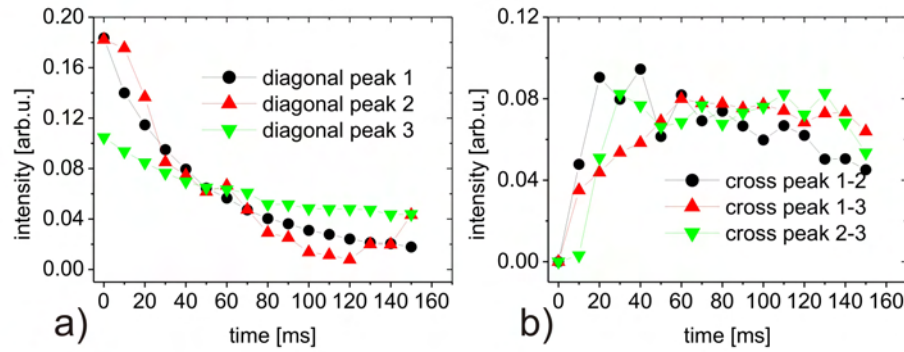


Figure 3.12: Evolution of the diagonal and cross peaks integrals for different mixing times. a) Evolution of diagonal peaks integrals for different mixing times. b) Evolution of cross peaks integrals for different mixing times.

diagonal peaks and off-diagonal peaks evolve with the mixing time. The intensity of the diagonal peaks decrease exponentially, and the intensity of the off-diagonal peaks are increasing with time, but after a time they start to decrease, this is because of the influence of the longitudinal relaxation time during the mixing time, fact that was demonstrated in the simulations.

To prove the accuracy of our dataset several other experiments were made. One of them was a drying profile for a 3 mm thick water saturated sample. The drying process was monitored during several hours with a Profile NMR-MOUSE with a proton resonance frequency of 17.1 MHz. The profiles shown in Fig. 3.13 exhibit a strange behavior at the beginning with water evaporating at the surface (surface being at 0 μm in this figure) and the capillary forces are moving the

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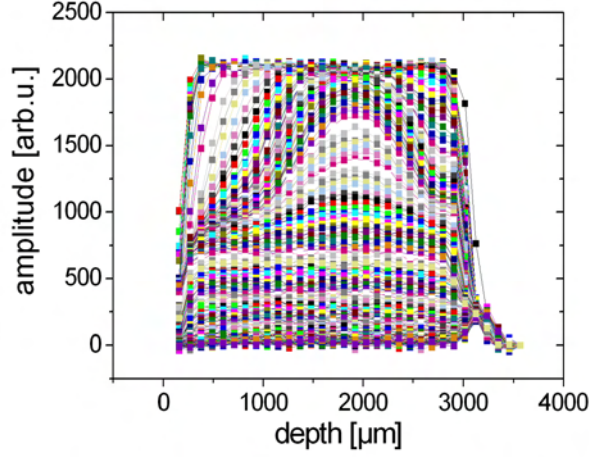


Figure 3.13: Drying evolution of the sample observed over several hours for a depth of 3 mm. At the beginning it shows a different behavior than after it is half dried.

water from the bottom to the top but keeping the intensity almost constant in the centre. After enough time the water in between particles evaporates and due to the capillary forces inside the particles, these remain filled. It leads to a constant profile with amplitude of about 700 arb. u.. At this time the sample is removed and introduced in the NMR tube to make an experiment similar to the one carried out for a saturated sample. A CPMG was measured on the partially dried sample and compared to the saturated sample.

Fig. 3.14 shows the comparison between the saturated and partially saturated sample. An Inverse Laplace Transformation was applied on these two exponential decays to see the T_2 relaxation distributions corresponding to the different environments. For the partially saturated sample we see only one distribution in comparison with the saturated sample. This shows that in the drying process the bulk and interstitial water vanished but the water inside the pore remained.

Another experiment that we made, to prove the water dynamics in the sample, was a 2D D - T_2 correlation experiment. This experiment was performed with the Profile NMR- MOUSE, with a proton resonance frequency of 18.1 MHz and a constant field gradient of 22 T/m. The correlation map is illustrated in Fig. 3.15. The correlation distributions are situated under the dotted line, which is the

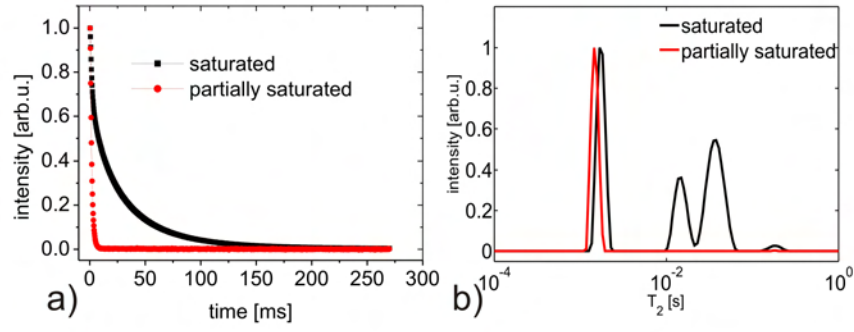


Figure 3.14: CPMG decays and the ILT distributions for the saturated and partially saturated aluminium oxide sample. a) CPMG decays showing that the spins relax much faster in the partially dried sample. b) ILT relaxation distributions for both samples. Only the water inside the pores remains.

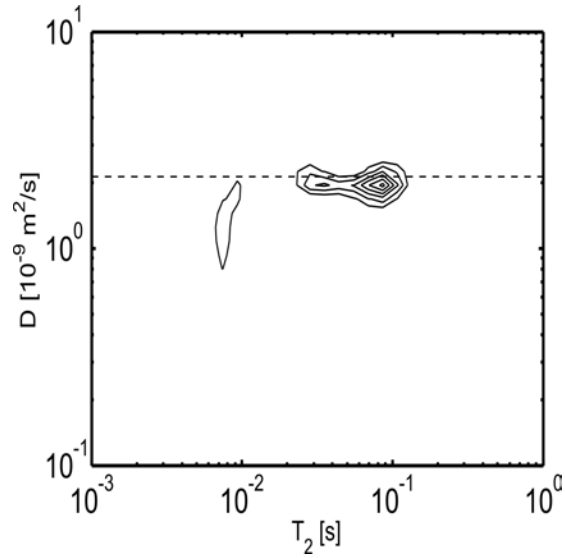


Figure 3.15: D - T_2 correlation map for the water saturated aluminium oxide system. Restricted diffusion is observable.

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line corresponding to the self-diffusion coefficient of water, $2.13 * 10^{-9} m^2/s$. This means that the water in the system is restricted, more restricted in the pores than in-between the particles and even the so considered bulk water. So the water that is the most restricted is situated inside the pores.

3.5 Conclusion

A general analytical solution for multi-site exchange NMR was presented, with a focus on a simple two-site exchange problem. Even though the theory allows studies on n -site systems, the more sites exist the more complicated gets the problem since we have more and more independent parameters. In order to understand the simple systems, experimental results were compared to simulated ones for a simple two-site system made of water saturated silica particles. T_1 - T_2 and T_2 - T_2 exchange correlation maps were compared. Moreover, the integrals of each relaxation distributions and their cross peaks were compared for different mixing times. The overlapping of the two evolutions obtained by simulations and experiments is observable. Complementary experiments were made at different temperatures for the same system. A temperature dependence of the main diagonal peaks integrals and the cross peaks integral could be noticed and compared with the simulation results. From the dependence of the reaction rate on temperature the activation energy could be extracted.

These 2D exchange experiments were tested further on a more complex system, a water saturated aluminium oxide system, which showed a more complex relaxation sites distribution. As the simulations showed that for more than two site systems, asymmetric or symmetric exchange maps can occur, for this system a symmetric 2D correlation map was obtained. In the same way as for the silica system, the peaks integral evolution over the mixing times was extracted. With these two types of information a multi-site system is expected, as the simulations showed that the maps can be symmetric just if the relaxation rates are equal. This is not the case here. In comparison with the simple system, the mixing time period is affected by the longitudinal relaxation time. This has the consequence that the integral values obtained for the cross peaks start to decrease towards

3.5 Conclusion

zero for too long mixing times. The restricted diffusion of the water molecules inside this sample is evidenced by the D - T_2 correlation map.

3. SIMPLE POROUS SYSTEMS

Chapter 4

Natural porous systems

Soils possess a complex composition with high variability in their properties and appearance. To the most important soil function belong filtering of water on its way down to aquifers and controlling the supply of plant nutrients. They relate to the structure of the pore system, which is characterized by the pore size distribution, pore connectivity, water content, and water dynamics. The understanding of the water transport dynamics is important in water resource management in agriculture, industry, household and environment.

4.1 Introduction

Nuclear magnetic resonance has proven to be suitable to investigate water in natural porous media, since NMR methods are non-invasive and directly probe the substance of interest, water. In the past, such natural pore systems have mainly been characterized by one-dimensional NMR relaxometry (Dunn 02, Todoruk 03, Kleinberg 05, Schaumann 05, Pohlmeier 09, Stingaciu 09), leading to pore size distribution functions by using the well-established Brownstein–Tarr relationship in combination with a uniform surface relaxivity as scaling parameter (Hinedi 97, Barrie 00). One drawback of soil is its heterogeneity in pore size, pore geometry, mineral content, and organic material content. This complicates the interpretation of relaxation time distribution functions, which may contain modes resulting from different pore sizes confounded by modes resulting from surface relaxation effects. Since a few years new techniques are used which

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can characterize much better the porous system, the two- and three-dimensional inverse Laplace correlation NMR methods (Lee 93, Godefroy 03, McDonald 05, Washburn 06, Song 08, Arns 07). In the 2D exchange NMR experiments, the magnetization of a spin system is monitored successively in two different time periods, called the encoding and detection period separated by a so-called mixing, exchange, or storage time, t_m as shown in Fig. 2.5 in chapter 2.4.3. These methods yield information far more detailed than the respective one-dimensional experiments. At present, longitudinal relaxation time T_1 - T_2 , diffusion D - T_2 , T_2 - T_2 and D - D correlation methods are gaining popularity for studying artificial porous media and rocks (Song 02b, Washburn 06, Hürlimann 02).

In T_2 - T_2 maps, the transverse relaxation time distributions from the first and the second encoding periods are plotted vs. each other. We have determined that this simplistic view can be used for a qualitative analysis, although it is flawed by the fact that the encoding and mixing times are on the same order of magnitude so that a quantitative analysis needs to be based on computer simulations of the experimental data (Landeghem 10). In this simple picture, water molecules that remain in or return to the same pore-size class during the mixing interval contribute to peaks on the diagonal, while cross-peaks indicate water molecules that have exchanged between pores of different sizes. Performing such experiments in a low magnetic field of 2 MHz are advantageous because heterogeneous structures are probed, which gives rise to local distortions of the magnetic field with so-called internal magnetic-field gradients, which may be small or even negligible at low magnetic field.

For the first time, T_2 - T_2 exchange experiments were applied to water-saturated soil systems to probe the water dynamics in the pore space. This project is part of a transregional collaborative research project (SFB TR32) between four different universities from Nord-Rhine Westphalia. The overall aim to find patterns in soil-vegetation-atmosphere systems, and to combine the information from the microscopic scale up to the macroscopic scale, in order to be able to predict water, CO₂ and energy transfer between these different scales. Our research group was involved in characterization of flow, transport and structure of saturated and unsaturated soils at the pore to meter scale. For this purpose 1D and 2D

NMR relaxation experiments were performed. In the past, fast transverse relaxation times have been assigned to immobile and clay-bound water in porous rocks (Straley 97). The question is whether two-dimensional relaxometry can give information on whether such water is really immobile or exchanges after all.

4.2 Saturated soils

Three repacked soil materials were chosen that represent pure fine sand and two silt loams with different sand and silt contents. The fine sand was purchased from Quarzwerke GmbH, Frechen, Germany, and the soil materials were from Merzenhausen and Selhausen, Germany. The names refer to the locations where the materials are taken from. The textural composition of each sample is presented in Table 4.1. To have an idea how different the textural compositions can be, CT images are shown in Fig. 4.1. The material was air dried, sieved in order to remove stone particles and to obtain an aggregate size smaller than 2 mm, packed into 9-by 100-mm cuvettes, centrifuged for 30 min at $1000 \times g$, and finally saturated with tap water for 48 h.

Table 4.1: Soil texture according to the USDA and maximum volumetric water content (θ) of the investigated soil samples.

Soil type	Sand %	Silt %	Clay %	($\theta_{\max}$) cm^3/cm^3
FH31 fine sand	100	0	0	0.36
Selhausen silt loam	15	68	17	0.4
Merzenhausen silt loam	4	81	15	0.43

1D experiments

Initially, one-dimensional CPMG experiments were performed to obtain transverse relaxation decays. The parameters used for the experiments for each soil type are listed in Table 4.2. The raw data were fitted with multi-exponential decays and subsequently inverted by one-dimensional inverse Laplace transformation (Godefroy 03) to obtain a transverse relaxation time distribution (Fig.

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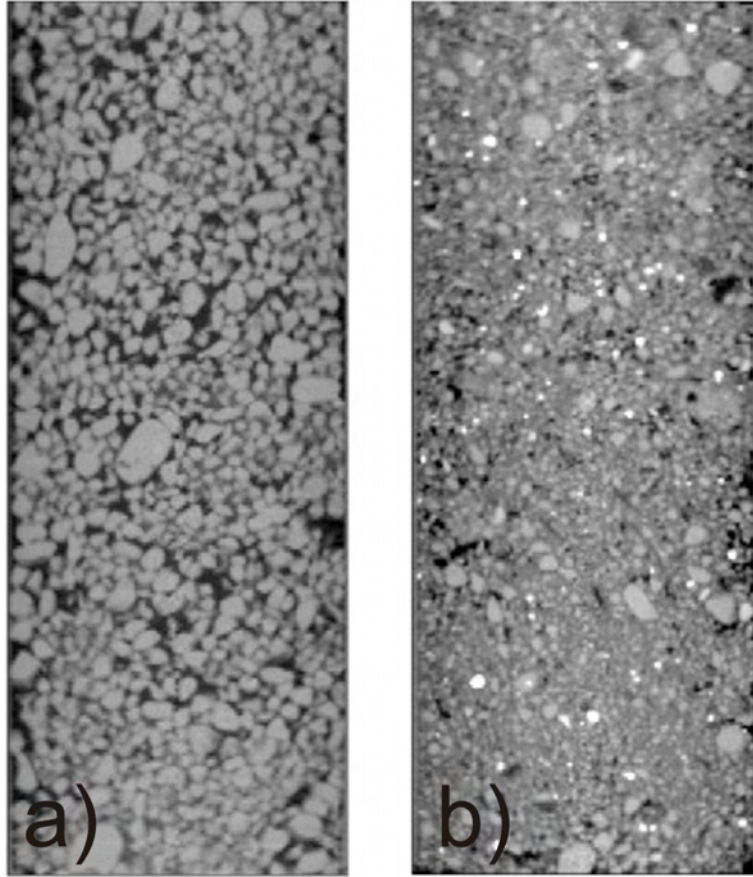


Figure 4.1: CT images of soil samples showing the different textural compositions. a) CT image for sand FH31. b) CT image for Merzenhausen silt loam. The resolution is 0.033 mm per pixel, the field of view is 6 by 15 mm. Data from Dr. Sabina Haber-Pohlmeier.

Table 4.2: NMR parameters used for the experiments and the obtained longitudinal as well as transverse relaxation times.

Soil type	NMR parameter			Relaxation times	
	t_E [ms]	nr. echoes	nr. scans	T_2 [ms]	T_1 [ms]
Sand FH31	0.5	7000	4	220/700	840/1650
Selhausen	0.2	600	8	1.3/7/40	21/114
Merzenhausen	0.1	500	8	0.3/2/8.6	4/16

4.2). Moreover, such experiments reveal information about pore connectivity. The relaxation times listed in Table 4.2 were obtained by fitting the raw data. The

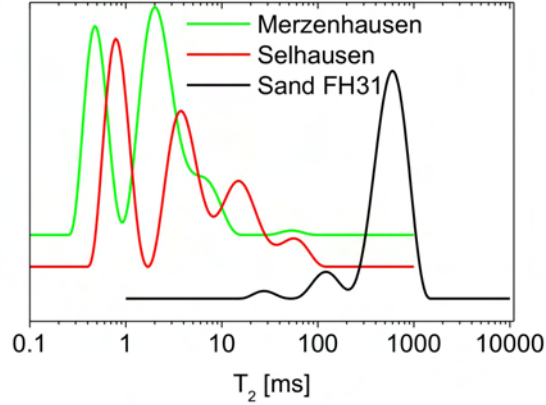


Figure 4.2: One-dimensional T_2 distribution functions obtained by inverse Laplace transformation for different water-saturated soil types : the black line is the FH31 fine sand, red line represents Selhausen silt loam, and the green line Merzenhausen silt loam.

longest T_2 relaxation distribution obtained for FH31 fine sand was also the simplest distribution, with maxima at 230 and 700 ms. The soil materials exhibited relaxation times much faster than those of the fine sand, covering ranges between 1 and 50 ms for the Selhausen soil and 0.5 to 10 ms for the Merzenhausen soil. Furthermore, the simple behavior found for the sand changed for the soils to multi-modal distribution functions. Comparing the relaxation times with the textural composition of each sample, it can be seen that with increasing silt and clay content, the T_2 relaxation times decreased significantly. The relaxation times decreased with smaller pore sizes under the assumption of a constant surface relaxivity (Barrie 00).

Another set of experiments was to check if the transverse relaxation is or not affected by the diffusion term, as shown in equation 4.1:

$$\frac{1}{T_{2,\text{app}}} = \frac{1}{T_{2\text{bulk}}} + \rho_2 \frac{S}{V} + \frac{D}{12} (\gamma G t_E)^2 \quad (4.1)$$

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where, ρ_2 represents the surface relaxivity, S/V represents the surface to volume ratio, D the diffusion coefficient, G the gradient strength, γ the gyromagnetic constant and t_E the echo time. In principle, T_2 experiments will be done at different echo times, and plot $1/T_2$ against the echo time square, t_E^2 . A linear dependence is expected, and from the slope of the fit the gradient strength could be extracted, under the assumption that D is constant and can be estimated reliably. Working at low fields will help to reduce the internal gradient strength of the samples, as known that the higher the magnetic field strength the higher the internal gradient effect. This is due to the magnetic susceptibility contrast between the grain material and the pore fluid (Kleinberg 90). Based on a model by Glasel (Glasel 74), Kleinberg (Kleinberg 90) mentions the average gradient for a region close to a grain wall as

$$G = \mu_0 B_0 \Delta\chi / 4R_g, \quad (4.2)$$

where μ_0 is the magnetic permeability of vacuum, B_0 the applied magnetic field strength, $\Delta\chi$ the difference between the volume magnetic susceptibilities of the grain and the external fluid and R_g the radius of the grain. Hürlimann mentioned in his paper (Hürlimann 98) the upper limit for the local effective gradient as :

$$G_{\text{eff,max}} \approx (\frac{\gamma}{D_0})^{1/2} (\Delta\chi B_0)^{3/2} \quad (4.3)$$

The largest possible effective gradient is scaling with a power of 3/2. The reason is that the largest effective gradient comes from the smallest pore.

First the dependence was checked for bulk water. In this way the inhomogeneity of the magnet was tested. The result is shown in Fig. 4.3. A linear dependence is observed and the field gradient was estimated using eq. 4.1 to be 180 mT/m. The T_2 dependence for the sand FH31 is shown in Fig. 4.4. In Fig. 4.4a the relaxation distributions for different t_E are over plotted and in Fig. 4.4b the dependence of $1/T_2$ with t_E^2 is represented. In case of the logarithmic mean T_2 (marked as $T_{2\text{LM}}$ in the figure), the whole relaxation distribution has been taken into account, whereas for the other dependencies (marked as T_{2l}) the value at maximum height has been taken. In case of the relaxation distributions a shift of the longer T_2 component towards shorter values is observed with increasing echo time. The dependence of $1/T_2$ with t_E^2 presents a linear dependence for the

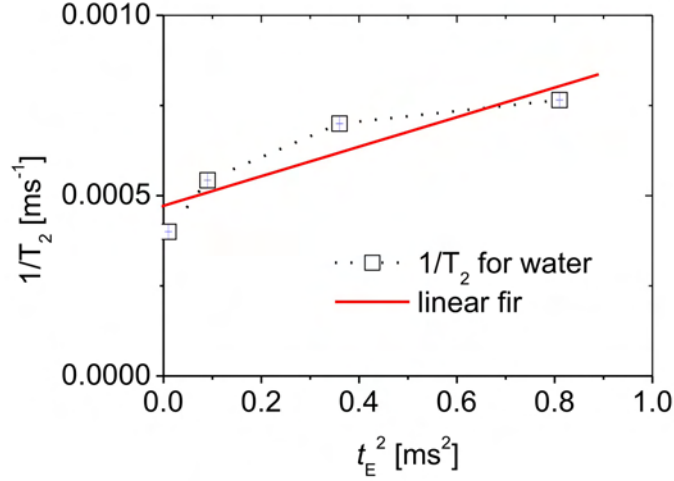


Figure 4.3: Influence of the echo time on the T_2 relaxation distribution for bulk water. The line between the points serves as a guide to the eye.

logarithmic mean T_{2LM} value and for the long T_{2l} value. The field gradient was estimated using eq. 4.1 to be 0.269 T/m for T_{2LM} and 0.227 T/m for T_{2l} . These values are not so high if we extract from them the value that already exists representing the field inhomogeneity.

In case of the Selhausen silt loam, Fig. 4.5, the same behavior is observed. But this time, the majority of the spins are trapped in the smaller pores, which is influencing the relaxation distribution function in a different way. Fig. 4.5a presents the behavior of relaxation distribution functions for different t_E and Fig. 4.5b the dependence of $1/T_2$ with t_E^2 . For this type of soil, it looks like all the different T_2 components are shifting to higher values with increasing echo time, but still the longest T_2 value, T_{2l} obtained for the shortest echo time is not exceeded in either of the distributions. In case of the $1/T_2$ dependence with t_E^2 , a linear dependence is noticed only for the longest component, T_{2l} , whereas the other components show an exponential decay behavior, marked with T_{2sh} for the shortest component, T_{2m1} for first middle component, T_{2m2} the second middle component and T_{2LM} the relaxation distribution logarithmic mean value. The fastest decay rate is noticed again for the shortest T_2 component. The field

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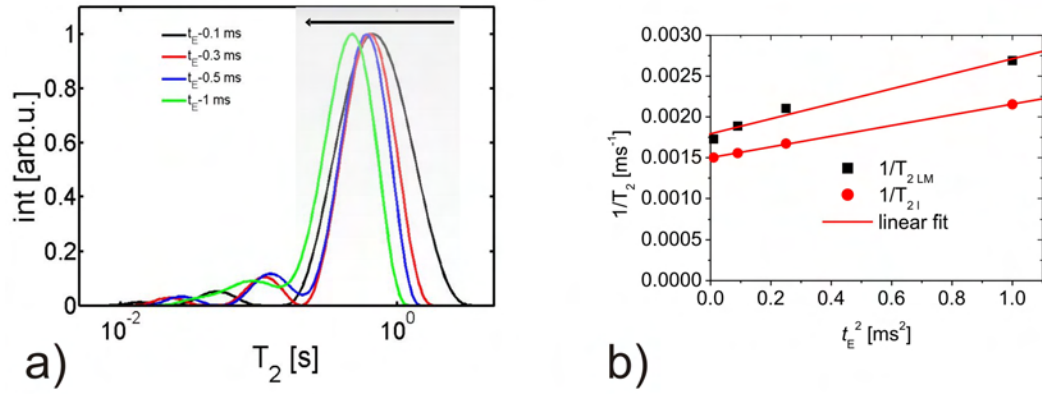


Figure 4.4: Sand FH31. Influence of the echo time on the T_2 relaxation distribution. a) T_2 relaxation distributions for different echo times. b) $1/T_2$ dependence with t_E^2 .

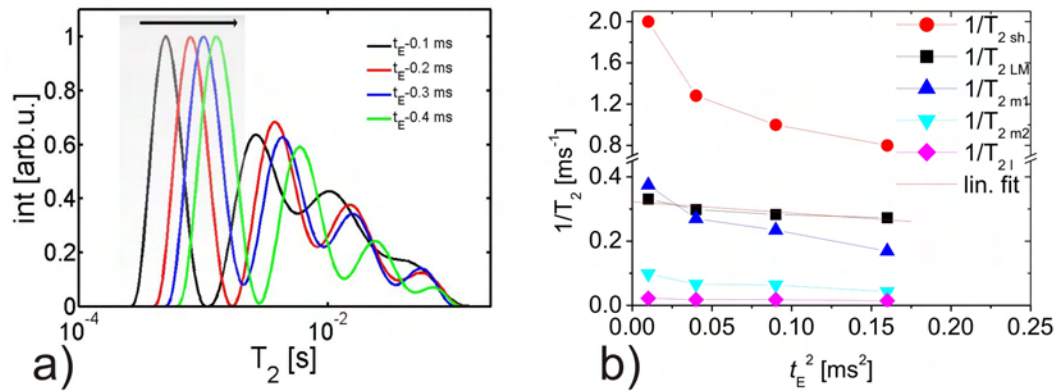


Figure 4.5: Selhausen silt loam. Influence of the echo time on the T_2 relaxation distribution. a) T_2 relaxation distributions for different echo times. b) $1/T_2$ dependence with t_E^2 . The lines connecting the points are just as a guide to the eye.

gradient was estimated using eq. 4.1 to be 5.25 T/m for T_{2LM} .

The T_2 dependence on t_E^2 for Merzenhausen silt loam is presented in Fig. 4.6. Fig. 4.6a presents the behavior of relaxation distribution functions for different t_E and Fig. 4.6b the dependence of $1/T_2$ with t_E^2 . It behaves in the same way as for the Selhausen silt loam. In this case, all the components shift to higher T_2 values. The $1/T_2$ dependence with t_E^2 , show linear dependence just for the longest

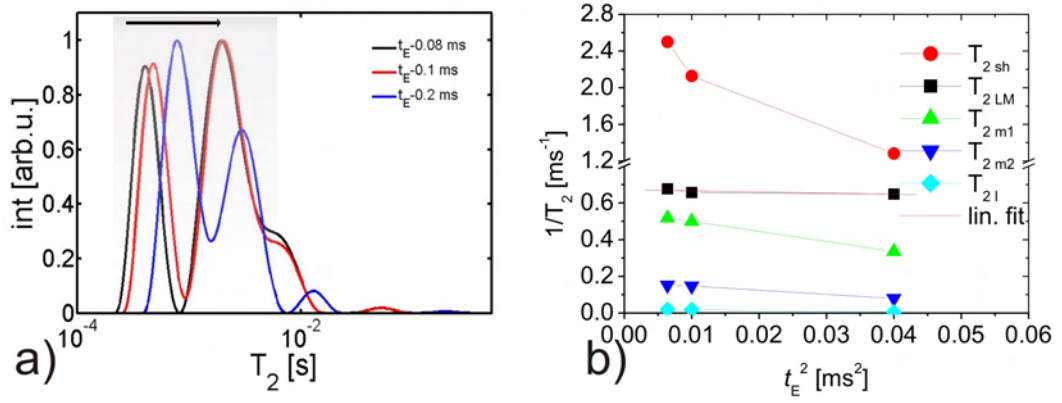


Figure 4.6: Merzenhausen silt loam. Influence of the echo time on the T_2 relaxation distribution. a) T_2 relaxation distributions for different echo times. b) $1/T_2$ dependence with t_E^2 . The lines connecting the points are just as a guide to the eye.

component, T_{2l} , and all the other components show an exponential behavior. The fastest decay rate is noticed again for the shortest T_2 component. The field gradient was estimated using eq. 4.1 to be 6.98 T/m for T_{2LM} .

From the results shown above, it can be seen that sand behaves differently than the other two soil types. Actually the sand behaves as we would expect it. The relaxation rate, $1/T_2$ present a linear dependence with t_E^2 as predicted from equation 4.1. Moreover the gradient strength extracted is not even double the value obtained for pure water measurements. Taking a closer look to the T_2 distribution function dependence with t_E , for the two silt loams, a difference is noticed. The distribution function components are shifting to higher values with increasing the echo time. This will affect of course also the $1/T_2$ dependence with t_E^2 . A linear dependence is noticed just for the long relaxation components. But

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this linear dependence has a negative slope, and not a positive as for sand. The shorter the relaxation components get, the more curved this dependence gets. A possible reason for this might be the non-linear behavior of the field. The CPMG signal is not able to refocus the effects coming from this quadratic field, this can be achieved with other pulse sequences (Blümich 05a). Another explanation would be the exchange of magnetization components for longer t_E^2 . This would explain the decrease in relaxation rates. This results make us understand that the texture and connectivity of sand is simple and different from that of the two silt loams. In the case of the sand we can assume that we have grains but in the case of the silt loams we find a more heterogeneous distribution, a sponge-like matrix with grains but mostly thin films.

2D experiments

To test if the two-dimensional exchange experiment and the data processing by the two-dimensional inverse Laplace transformation program worked reliably and to ascertain that the observed cross peaks were not artifacts from poor signal/noise ratios, a reference experiment with two stacked, individual, 5 mm NMR tubes was performed. Each tube was filled with different aqueous CuSO_4 solutions, doped water, which gave different T_2 values. Figure 4.7 shows the results from the relaxation–relaxation exchange experiments that were performed for mixing times of 1 and 30 ms. As expected, only diagonal peaks were observed and no cross peaks were detected. The loss of intensity of the fast diagonal peak at $T_2 \sim 10$ ms at a mixing time of 30 ms is remarkable. This was due to the fact that longitudinal relaxation is taking place during the mixing time, during which the magnetization was stored along the z axis.

Once the measurement procedure was tested, relaxation–relaxation exchange experiments were conducted for different mixing times for all samples (Fig. 4.8). The numbers of echoes used in the first encoding period was increased logarithmically in 64 steps so that the maximum first encoding time covered the same range as the second encoding time at the end. The mixing times used between the two T_2 encoding times were varied from 0 ms up to 1 s for the FH31 sand, up to 55 ms for the Selhausen soil, and up to 10 ms for the Merzenhausen soil.

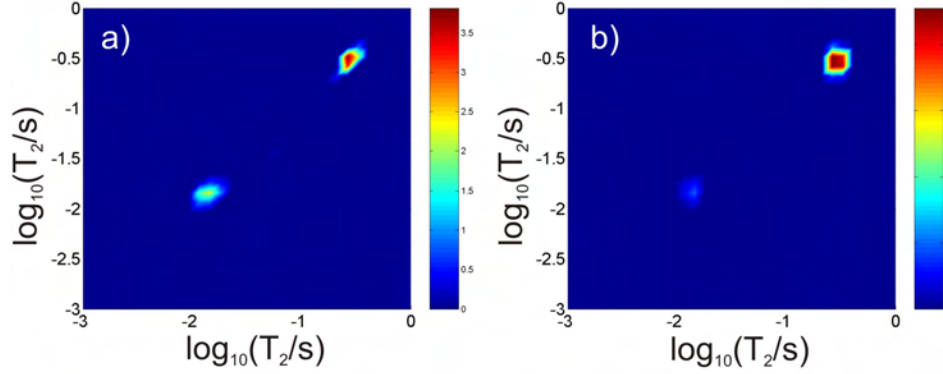


Figure 4.7: Test of the relaxation exchange method : transverse relaxation–relaxation (T_2 - T_2) correlation experiments of two aqueous CuSO_4 solutions measured with a) 1 ms and b) 30 ms mixing times. The colors represent the intensity of the signal (arbitrary units).

Each line of the two-dimensional data matrix was scanned eight times, and the repetition time between scans was set to five times the longitudinal relaxation time T_1 . The experiments lasted between 2 and 30 min depending on the duration of the mixing time and the repetition time. In the corresponding exchange distributions, molecules residing in their original environment gave rise to signals along the diagonal, while those changing their environment during the mixing time produced off diagonal peaks. Summarizing the results of the T_2 - T_2 correlation maps for the systems with zero mixing time (Fig. 4.8a–c), all signals appear as diagonal elements and the projections (not shown here) to the coordinate axes are similar to the one-dimensional distributions shown in Fig. 4.2. As expected, no exchange is observed for zero mixing time.

For the FH31 sand, the exchange of water molecules is barely observed at mixing times of both 200 and 700 ms by the presence of very weak cross peaks (Fig. 4.8d and g). The relaxation distribution function of sand was quite homogeneous, with only a small fraction of narrow pores, as evident from Fig. 4.2a and confirmed by T_1 relaxometry data (Pohlmeier 09). This is why exchange processes in the sand were caused mainly by molecular movement within pores of the same

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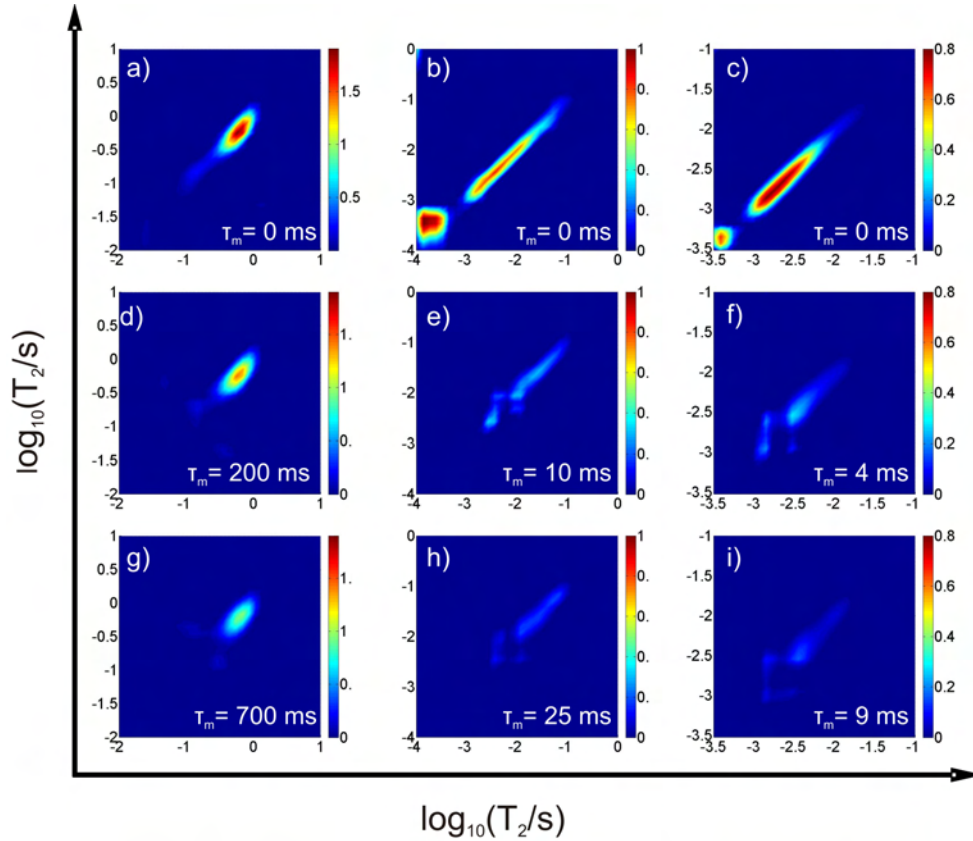


Figure 4.8: Two-dimensional transverse relaxation–relaxation (T_2 – T_2') correlation spectra obtained for the different soil types : a) FH31 fine sand, b) Selhausen silt loam, and c) Merzenhausen silt loam at mixing time $t_m = 0$ ms; d) FH31 fine sand at $t_m = 200$ ms; e) Selhausen silt loam at $t_m = 10$ ms; f) Merzenhausen silt loam at $t_m = 4$ ms; g) FH31 fine sand at $t_m = 700$ ms; h) Selhausen silt loam at $t_m = 25$ ms; and i) Merzenhausen silt loam at $t_m = 9$ ms. The colors represent the intensity of the signal (arbitrary units).

size class.

Figures 4.8e and h present the distributions for the Selhausen soil at mixing times of 10 and 25 ms. The observed values of T_2 are widely distributed and range between 1 and 50 ms, showing up to four components. Unlike the results for the FH31 sand, cross peaks appear at mixing times of 10 and 35 ms, indicating strong exchange between the shortest components and weak exchange to sites with longer relaxation times. These cross peaks identify migration of water molecules between relaxation environments. It is also noted that the distribution matrix is rather asymmetric, which is a strong indication of multi-site exchange (Landeghem 10). The fastest component on the diagonal completely vanishes at long mixing times (Fig. 4.8h) due to T_1 relaxation effects.

Like the Selhausen soil, the Merzenhausen soil shows a multi-modal distribution with short T_2 components ranging from 0.5 to 10 ms (Fig. 4.8c, f, and i). As in the Selhausen soil, cross peaks between the shortest components arise already at a mixing time of 2 ms (data not shown here) and became stronger at a mixing time of 4 ms (Fig. 4.8f). At a mixing time of 9 ms, the shortest diagonal peaks vanish due to a short T_1 relaxation of about 2 ms (Pohlmeier 09), and exchange with the longer components becomes observable. In addition, with increasing mixing times, the two-dimensional spectra become slightly asymmetric, which indicates the onset of multi-site exchange.

For all samples, the evolution of each T_2 component integral over the mixing time was tried to be plotted, but problems occur when trying to determine each component limits in the 2D correlation map. In the case of these soils we cannot talk any more about relaxation sites, because we have a distribution of relaxation sites, and the theory presented in Chapter 3 needs to be extended. This work is now in progress at another research group within this transregional research project TR32 (Mohnke 10).

4.3 Unsaturated soils

1D experiments

In case of the unsaturated soils, differences can be observed, as for all of these three samples, the long T_2 components vanish with decreasing the saturation level. This fact is evidenced very well in Fig. 4.9, where for each soil type the 1D transverse relaxation distribution is shown at different saturation levels. In case

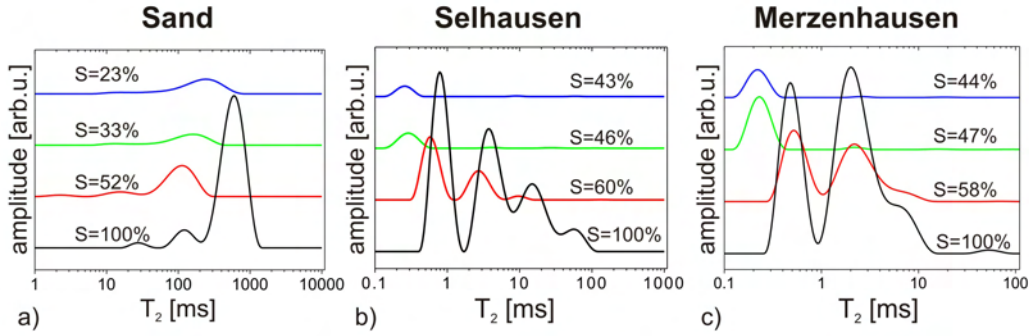


Figure 4.9: 1D relaxation distributions for different saturation levels for each soil type. a) Sand FH31, the T_2 values are shifting to shorter values for lower saturation levels. b) Selhausen silt loam, the relaxation distribution components present at fully saturated soil are vanishing with decreasing saturation level. Only the fastest component remain. c) Merzenhausen silt loam, the same behavior as for Selhausen soil is noticed.

of the sand FH31, the T_2 values are shifting to shorter values for lower saturation levels. The amplitude is decreasing dramatically as the hydrogen spins are getting less and less. This is in agreement with the data of Stangaciu (Stingaciu 09). For the Selhausen silt loam, the relaxation distribution components present at fully saturation level are vanishing with decreasing saturation level. Only the fastest component remain at a very low amplitude, which is shifting slightly to even faster values. For the Merzenhausen silt loam, the same behavior as for Selhausen soil is evidenced.

The same dependence of $1/T_2$ on t_E^2 was tested for the unsaturated samples too. For sand FH31, these dependence is shown in Fig. 4.10. Fig. 4.10a presents

the behavior of relaxation distribution functions for different t_E and Fig. 4.10b the dependence of $1/T_2$ with t_E^2 . In Fig. 4.10a a shift of the longer T_2 component towards shorter values is noticed. Plotting the $1/T_2$ against t_E^2 , a linear dependence for the logarithmic mean T_2 value and for the long T_2 value is noticeable. The field gradient was estimated using eq. 4.1 to be 0.414 T/m for T_{2LM} and 0.385 for T_{2l} . This values is higher than the value obtained for the saturated sample.

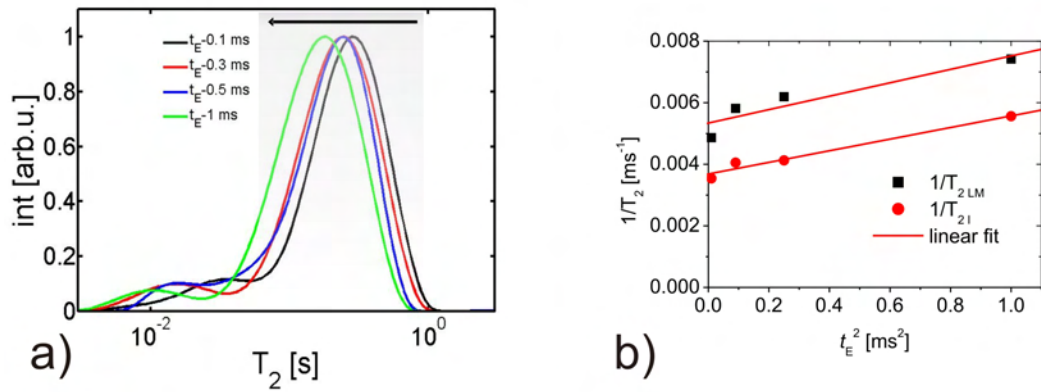


Figure 4.10: Partially saturated Sand FH31. Influence of the echo time on the T_2 relaxation distribution. a) T_2 relaxation distributions for different echo times. b) $1/T_2$ dependence with t_E^2 .

In case of the Selhausen silt loam, the same behavior is observed as for the saturated case. Fig. 4.11a presents the behavior of relaxation distribution functions for different t_E and Fig. 4.11b the dependence of $1/T_2$ with t_E^2 . The majority of the spins are in the smaller pores trapped, which is influencing in a different way the relaxation distribution. In Fig. 4.11a all the different T_2 components are shifting to higher values with increasing the echo time. In case of the $1/T_2$ dependence with t_E^2 , a linear dependence is noticed only for the longest T_2 value, whereas the other components show an exponential behavior. The fastest decay rate is noticed again for the shortest T_2 component. The field gradient was estimated using eq. 4.1 to be 2.39 T/m for T_{2LM} . This might be incorrect as theoretically it should have a higher value than for the saturated case.

The Merzenhausen silt loam, Fig. 4.12, behaves in the same way as the Selhausen silt loam. Fig. 4.12a presents the behavior of relaxation distribution functions

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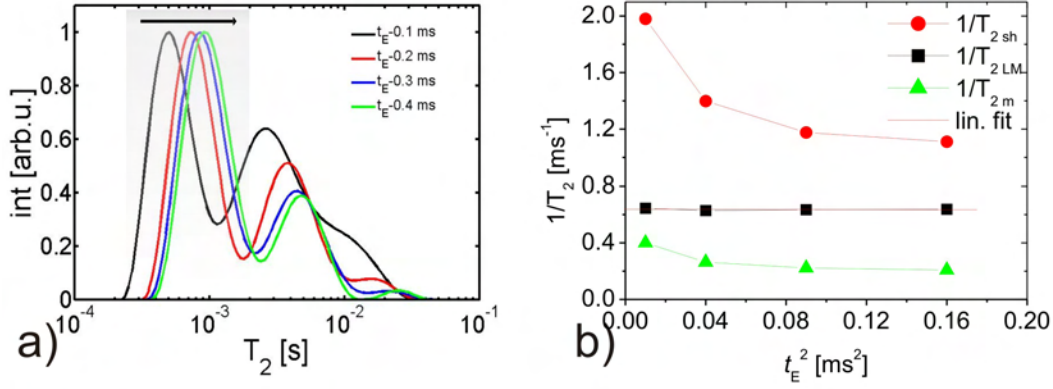


Figure 4.11: Partially saturated Selhausen silt loam. Influence of the echo time on the T_2 relaxation distribution. a) T_2 relaxation distributions for different echo times. b) $1/T_2$ dependence on t_E^2 . The connecting lines serve as a guide to the eye.

for different t_E and Fig. 4.12b the dependence of $1/T_2$ with t_E^2 . In Fig. 4.12a it is noticeable how all the relaxation components shift to higher T_2 values. The $1/T_2$ dependence on t_E^2 , show linear dependence only for the longest T_2 value, and all the other components show an exponential behavior. The fastest decay rate is noticed again for the shortest T_2 component. The field gradient was estimated using eq. 4.1 to be 8.76 T/m for T_{2LM} .

The results obtained for the partially saturated samples are in good agreement with the results obtained for the saturated samples. The behavior of the silt loams is different from the one for sand, and the gradients extracted are even higher. This is a confirmation for the presence of thin films in the silt loams, that is getting more accentuated when these are not fully saturated.

2D experiments

The parameters used in obtaining the 2D relaxation exchange maps are listed in Table 4.3. These parameters as well as the 2D maps will show the comparison between the fully saturated and partially saturated soil sample.

For the FH31 sand, 2D relaxation exchange maps are plotted for two saturation levels and in each case for three different mixing times in Fig. 4.13. With

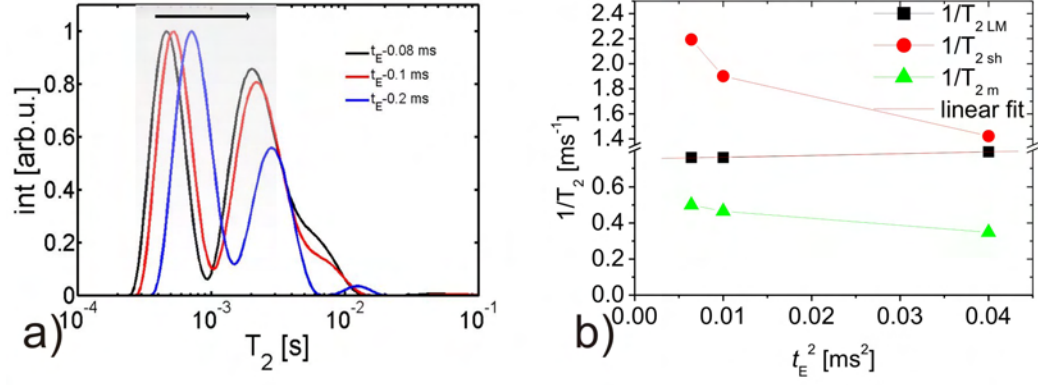


Figure 4.12: Partially saturated Merzenhausen silt loam. Influence of the echo time on the T_2 relaxation distribution. a) T_2 relaxation distributions for different echo times. b) $1/T_2$ dependence with t_E^2 . The connecting lines serve as a guide to the eye.

Table 4.3: NMR parameters used for the experiments and the obtained longitudinal as well as transverse relaxation time values.

Soil type and Saturation		NMR parameter			Relaxation time	
		t_E [ms]	nr. echoes	nr. scans	T_2 [ms]	T_1 [ms]
Sand	$S = 100\%$	0.5	7000	4	220/700	840/1650
FH31	$S = 23\%$	0.5	6500	12	88/330	160/750
Selhau	$S = 100\%$	0.2	600	8	1.3/7/40	21/114
sen	$S = 60\%$	0.2	300	24	0.85/4/14	1.4/11
Merzen	$S = 100\%$	0.1	500	8	0.3/2/8.6	4/16
hausen	$S = 58\%$	0.1	300	16	0.4/2/7	7

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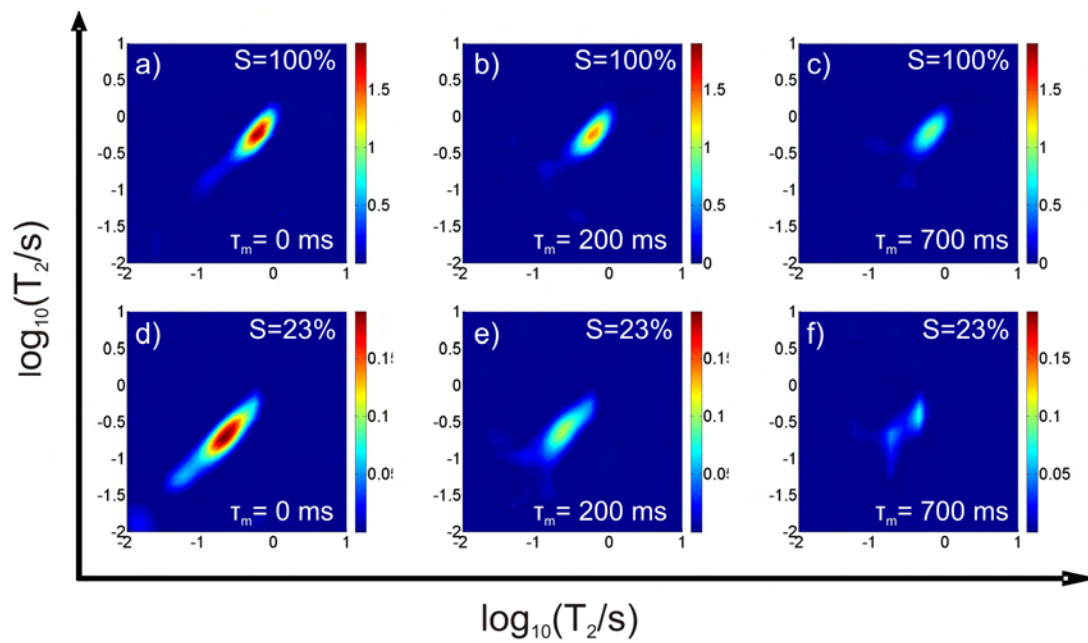


Figure 4.13: Two-dimensional T_2 - T_2 correlation spectra obtained for sand FH31 :a) Saturation (S) $S = 100\%$ and $t_m = 0$ ms; b) $S = 100\%$ and $t_m = 200$ ms; c) $S = 100\%$ and $t_m = 700$ ms; d) $S = 23\%$ and $t_m = 0$ ms; e) $S = 23\%$ and $t_m = 200$ ms; f) $S = 23\%$ and $t_m = 700$ ms. The color bar was kept fixed in order to be able to compare the amplitudes.

decreasing water content the relaxation times are shifted to faster values. Exchange processes are observed but weakly, which indicates the fact that water molecules move within pores of the same size class.

In the case of the silt loam Selhausen, the 2D relaxation exchange maps are plotted as well for two saturation levels and in each case for three different mixing times in Fig. 4.14. Different pore size classes are observable and moreover,

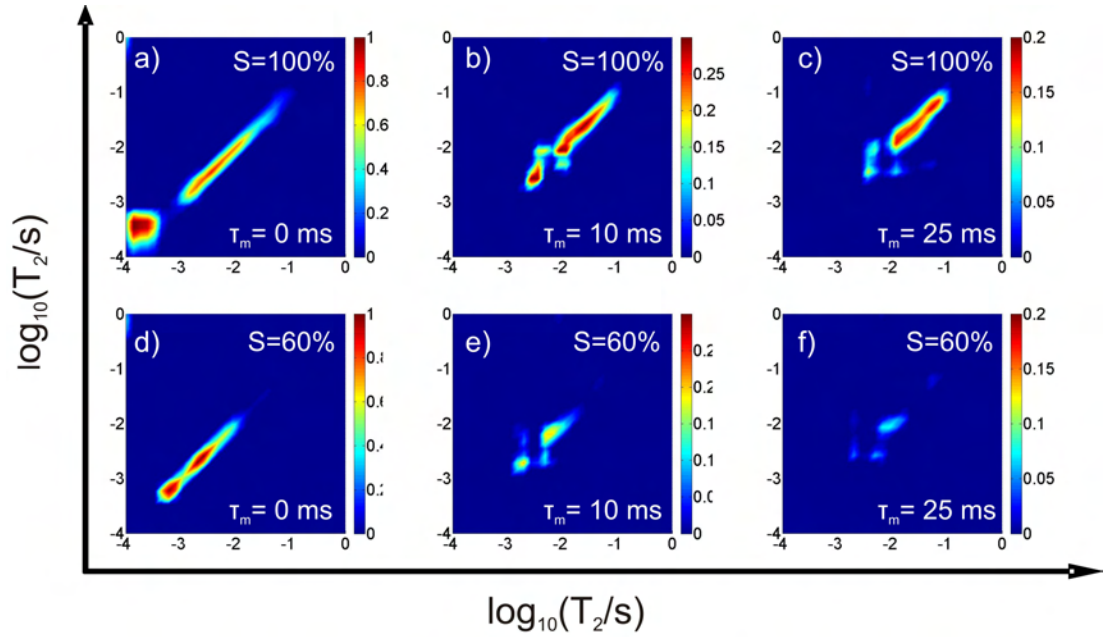


Figure 4.14: Two-dimensional T_2 - T_2 correlation maps obtained for Selhausen silt loam : a) Saturation (S) $S = 100\%$ and $t_m = 0$ ms; b) $S = 100\%$ and $t_m = 10$ ms; c) $S = 100\%$ and $t_m = 25$ ms; d) $S = 60\%$ and $t_m = 0$ ms; e) $S = 60\%$ and $t_m = 10$ ms; f) $S = 23\%$ and $t_m = 25$ ms. The color bar was kept fixed in order to be able to compare the amplitudes.

distribution of relaxation times are noticed. Water exchange occurs between the smallest pore classes. With decreasing water content first the more mobile water, with longer T_2 values vanishes, and at the same time the relaxation times shift to faster values.

For the Merzenhausen silt loam, the 2D relaxation exchange maps are plotted for two saturation levels and in each case for three different mixing times in Fig. 4.15. Water exchange occurs, as well as for the Selhausen silt loam, between the

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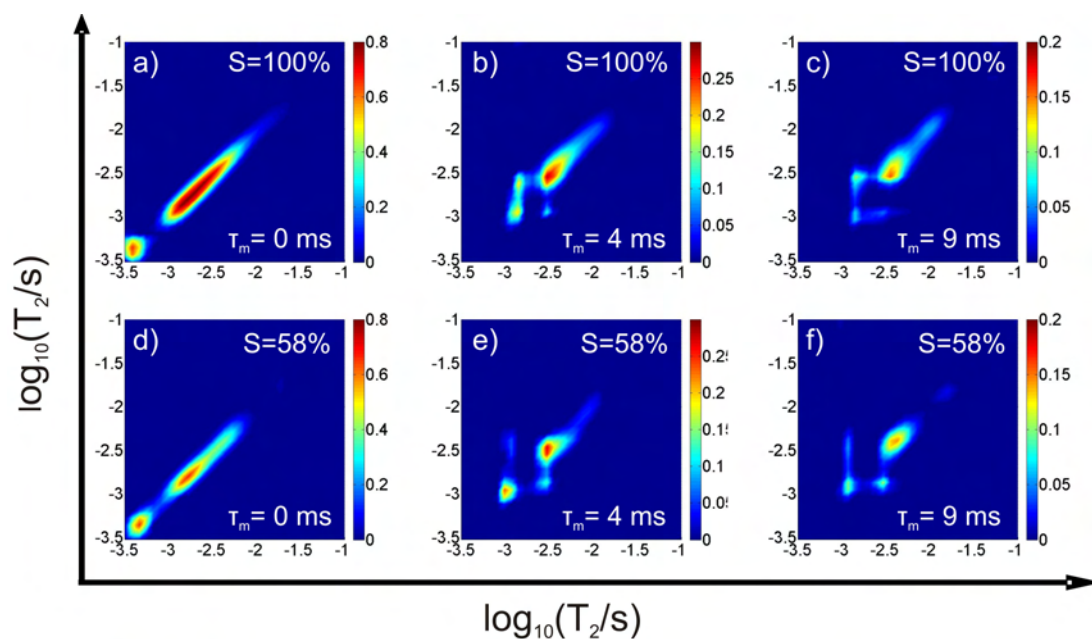


Figure 4.15: Two-dimensional T_2 - T_2 correlation spectra obtained for sand Merzenhausen silt loam :a) Saturation (S) $S = 100\%$ and $t_m = 0$ ms; b) $S = 100\%$ and $t_m = 5$ ms; c) $S = 100\%$ and $t_m = 9$ ms; d) $S = 58\%$ and $t_m = 0$ ms; e) $S = 58\%$ and $t_m = 5$ ms; f) $S = 58\%$ and $t_m = 9$ ms. The color bar was kept fixed in order to be able to compare the amplitudes.

smallest pores. Unsaturated samples present a shift in the relaxation times but only at saturations under 55 %. Water is predominantly lost from the big pores where the water is more mobile.

4.4 Conclusion

The results obtained from these experiments prove that two-dimensional exchange relaxometry methods are suitable to observe moisture migration in loose, natural, porous media like soils. For sand, only weak and slow exchange was found, while in more complex, textured soils, exchange of water molecules was observed for short mixing times between the fastest components and between slower components at longer mixing times. It is not surprising that, in such heterogeneous systems, the water molecules exchange between more than two sites as the mixing time is prolonged. By such two-dimensional experiments, the relaxation modes identified in one-dimensional experiments can be correlated and interpreted in terms of distances travelled by water molecules during the exchange experiment once the exchange rates are quantified by numerical simulation of the experimental exchange maps. This is in line with the assumption incorporated in the Brownstein Tarr model, where the observed relaxation modes can be assigned to pores of different sizes. In particular, it was shown that the existence of cross peaks identifies the motion of the water molecules between different pore size classes in a qualitative analysis. If exchange between two pore size classes is observed, water in these classes should be regarded as mobile, not immobile.

Depending on the soil texture the pore size distribution varies from a simple distribution, obtained for sand, to multi-modal distributions, obtained for the two different silt loams. A non linear behavior, in case of the silt loams, for the dependence of the relaxation rates with t_E^2 was observed which give us an idea about the texture and pore connectivity inside the sample. Water exchange is observable mostly within the small pores. Exchange within the big pores may occur, but due to the T_1 relaxation the exchange between large pores is hard to observe and the small components vanish at longer mixing times. Once the saturation level becomes lower the water from the bigger pores vanish and the T_2 values are shifting towards shorter values.

4. NATURAL POROUS SYSTEMS

Chapter 5

Complex porous systems

Concrete is a multi-length scale composite material first invented by the Romans. The name concrete comes from the Latin “concretus”, which means to grow together. Today, concrete has become ubiquitous — about five billion tones of concrete are used worldwide each year. The cement used mostly in today’s concrete is called Portland cement. The process to produce Portland cement was invented by Joseph Aspdin in the early 1800s in England. The name Portland may have been originally a marketing ploy, as Portland building stone was very popular in England at that time and Aspdin may have wanted people to favorably compare concrete made with his cement to the popular building stone (Garboczi 96).

It is important to remember that cement is the powder that reacts with water to form cement paste, a hard, solid material that forms the matrix for the concrete composite. The addition of sand (fine aggregates) that are up to a few millimeters in diameter forms mortar, and the addition of rocks (coarse aggregates) of up to a few centimeters in diameter forms concrete. It has been known that concrete is a porous material whose properties depend on its pore space. There are many different kinds of pores in concrete, ranging from the air voids that are entrapped in the mixing process, which can be quite large (up to a few millimeters in diameter) to the capillary pores, which are essentially the space occupied by the leftover water from mixing, down to the nanometer-scale pores that exist in some of the hydration products produced by the cement-water chemical reaction (Garboczi 96).

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5.1 Introduction

In order to have a realistic view about how much an outdoor concrete element is exposed to the environment and its influence on the quality and durability of the material, an accurate knowledge of the time evolution of the local temperature and moisture distribution in the material is required. For this purpose, the German Federal Institute for Materials Research (BAM or Bundesanstalt für Materialforschung) leads several research projects throughout Germany in order to determine the moisture content both inside and at the surface of concrete. For moisture measurements they have used the MRE (multi-ring electrode) method developed at IBAC RWTH Aachen University (Brameshuber 08). It allows indirect moisture measurements through the electrolyte resistance of the concrete. In order to be able to convert correctly the resistance into moisture, numerous calibration tests need to be done. The most important error in the measurements arrives from the salt concentration. For a correct evaluation of the quality of the MRE measurements, NMR measurements were performed, where only the signal from the hydrogen atoms is recorded, which can be converted to moisture content.

The NMR measurements were executed with the Profile NMR-MOUSE[®] for on-site measurements as well as for laboratory measurements. On-site measurements were performed at the Gäubahntunnel, near Stuttgart and at the Hohenwarthe lock, near Magdeburg. The aim was to compare concrete with different composition, but also concrete that is exposed to different environmental conditions. For the laboratory studies, two concrete blocks with the same composition as the on-site concrete were studied.

5.2 Experimental

The Profile NMR-MOUSE[®] sensor used for the measurements was mounted on a computer-driven positioning device. The NMR sensor and the positioning device used were manufactured by ACT GmbH, Aachen, Germany. The spectrometer was a Kea spectrometer manufactured by Magritek Ltd, New Zealand, connected to a Dressler 1 kW rf amplifier. In order to acquire depth profiles, the distance between sensor and object was changed in appropriate steps and the sensitive

volume was shifted through the object along the depth direction. This method can be applied to all objects which contain hydrogen nuclei and are neither electrically conductive nor magnetic. Depending on the penetration depth, different sensors are available. The greater the penetration depth, the larger is the sensor. Standard depths are 3, 5, 10, and 25 mm. Since the magnetic field decreases with increasing distance from the sensor, the field strength of sensors with larger depth access is smaller than that of sensors for smaller depth access. At the same time, the intensity of the received signals decreases with increasing depth. This loss is balanced for sensors with high depth access by an increase of the sensitive volume in width and thickness. The NMR measurements at the two different sites as well as in the laboratory were performed with a Profile NMR-MOUSE® with 21 mm depth access at a proton resonance frequency of 13.86 MHz. The echo time was kept as short as possible for this penetration depth, 140 μ s. 16 echoes were acquired and each CPMG echo train was repeated 512 times. Each profile was recorded in about 1 to 2 hours, depending on the on-site conditions. For the laboratory measurements the conditions were kept similar, except the number of scans was doubled as we had more time available.

5.2.1 On-site measurements

The first of the on-site measurements was performed in August 2009 at the Gäubahntunnel. This is a 450 m long tunnel where the temperature and moisture has been monitored for 8 years with the MRE method. In October 2009, the lock measurements at Hohenwarthe took place. This lock is a double-chamber water-saving lock that achieves a difference in water elevation of 18.55 m. The monitoring of temperature and moisture has been ongoing since 2003 (Weise 10). The two experimental setups are shown in Fig. 5.1. The composition of the two concretes is similar, except for the fact that the aggregates that consist of sand, pebble and chippings are from the respective regions. With the help of mercury intrusion porosimetry the pore radius was determined. In the case of Gäubahntunnel concrete, the distribution is more monomodal with a peak maximum at 25 nm, a mean radius of 32 nm and a limiting radius of 50 nm. In the case of the Hohenwarthe concrete, a bimodal distribution is evidenced with the two

5. COMPLEX POROUS SYSTEMS

peak maxima at 3 nm and 7 to 16 nm, a mean pore radius of 7 to 9 nm and a limiting radius of 16 to 25 nm (Weise 10). These values indicate a much more compact concrete in the Hohenwarthe case. This is necessary as the concrete wall is occasionally under water.

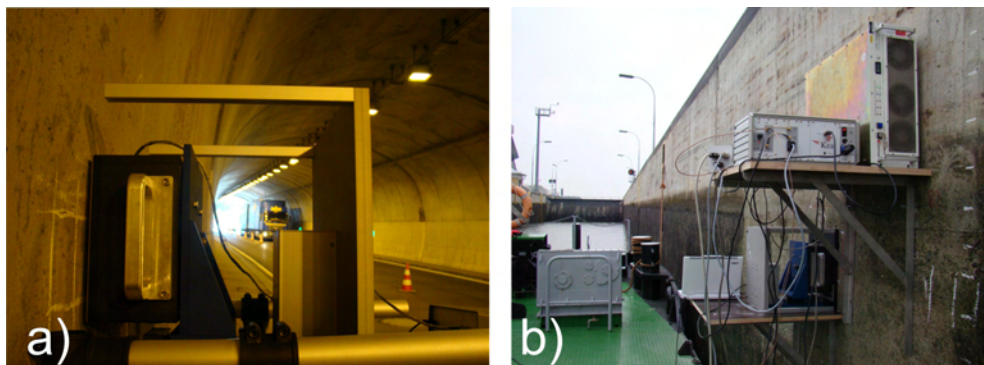


Figure 5.1: On-site measurements. a) The Gäubahntunnel setup, showing just the NMR-MOUSE positioned parallel to the measured point. b) The Hohenwarthe setup that needed to be fixed to the locks wall to avoid setup movements.

In case of the Gäubahntunnel, two NMR profiles were made: one at the entrance in the tunnel (marked as “Outside” in the Fig. 5.2 a and c) and another one at 120 m inside the tunnel (marked as “Inside” in the Fig. 5.2 b and d). To match the usually used units, the amplitude of the NMR signal was converted into grams of water per gram of dried cement. The conversion was done in the following way:

amplitude [g of w./g of dried c.] = (NMR signal / NMR signal of water) / density of concrete,

where the value for the density of concrete was evaluated before by the BAM to 2.1 g/cm³.

The results on the depth profiles and the extracted T_2 values for each depth are presented in Fig. 5.2. At each depth, a Carr Purcell Meiboom Gill (CPMG) measurement was performed. The real channel was recorded and fitted in order to extract the amplitude at zero echo time. Different fits were tried: single exponential, double exponential, Gaussian, etc. Ultimately, the best one was obtained

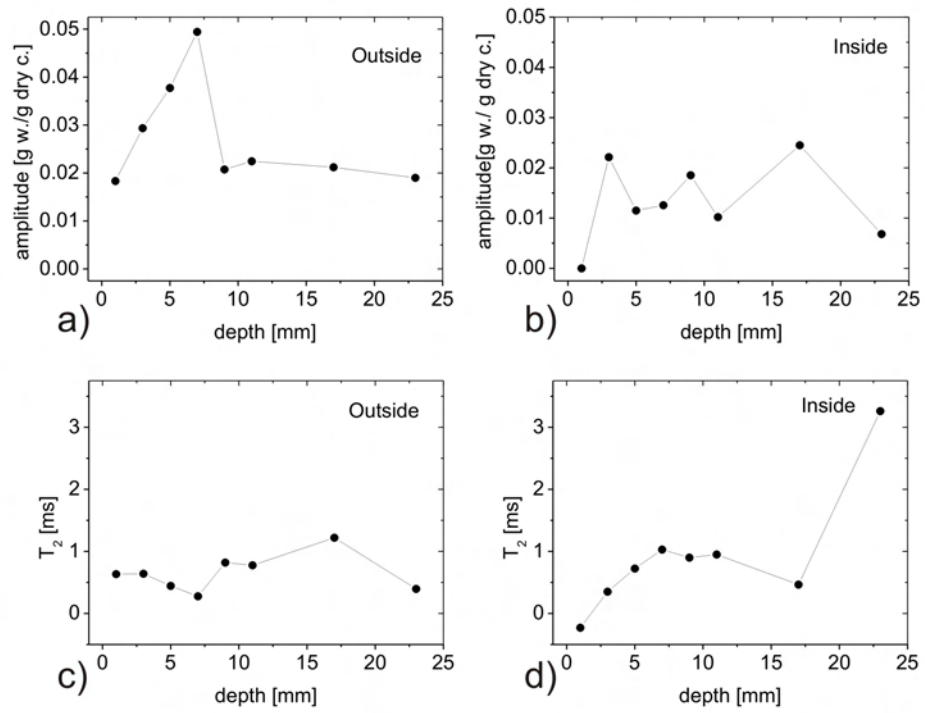


Figure 5.2: NMR depth profiles and extracted T_2 profiles for the two measured positions in the Gäubahntunnel. a) Depth profile for the outer measurement point at the entrance in the tunnel. b) Depth profile for the measurement point situated 120 m inside the tunnel. c) T_2 profile for the outer measurement point. d) T_2 profile for the inside measured point at 120 m inside the tunnel.

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for a so-called stretched exponential:

$$S(t) = \exp(-(1/b) * (t/T_2)^b) \quad (5.1)$$

The best value for b was found to be 1.2.

Comparing the two depth profiles obtained for the two different measurement points, differences can be observed. The amplitude at the entrance in the tunnel (marked as “Outside” in Fig 5.2a) is almost double for the first 7 millimeters as the amplitude for the profile measured 120 m inside the tunnel (marked as “Inside” in Fig 5.2b). After this point both amplitudes show a water content of up to 2%. One possible explanation for this could be that the wall at the entrance is more exposed to the outer environment and to the salt laid during the winter; these may induce cracks in the wall and water accumulation. The accumulation of salt, especially chloride, is confirmed by measurements performed in 2003 by BAM (Weise 10). The results show that the chloride percentage is five times bigger “Outside” than “Inside” but is not exceeding 1% out of the total concrete mass. This is the reason why the water content that was found through the NMR measurements is less than the value found by the MRE measurements. The NMR water content was with 1-1.5% less than the MRE values. In case of the extracted T_2 profiles, the values in both cases increase slightly with depth (Fig. 5.2c and d) which means that the pores are larger deeper in the sample. This can be understood as an effect of the CO_2 in air which leads to CaCO_3 formation, by a chemical reaction with the calcium ions of the C-S-H hydrates (Calcium Silicate Hydrate, components of every cement paste). This leads to a smaller porosity as some of the free space is now occupied by CaCO_3 (Ngala 97).

In the case of the Hohenwarthe lock, two profiles were planned: one in the lock wall part which is just exposed to ambient air, and the other one at a part which is occasionally under water (when the lock is full). One difficulty that was encountered during this on-site measurements was the reinforcement located 60 mm inside the wall. Because of this, the NMR-MOUSE was magnetically attracted and we expected alterations to the position of the sensitive slice and to the final results. After looking for a position where the magnetic attraction was the lowest, the measurements were performed. To match with the usual units and with the other on-site measurements, the amplitude of the NMR signal was

converted into grams of water per gram of dried cement, as shown above. Figure 5.3 exhibits the profile amplitudes and T_2 values of the measurements performed under water level (marked with blue in the figure) and above (marked with red in the figure). For each measurement, a Carr Purcell Meiboom Gill (CPMG) measurement was performed. Both channels were recorded and a fit using the two channels but a single relaxation rate was performed in order to extract the amplitude at zero echo time. The fitting was made again with the stretched exponential function, because it was the best fit obtained and for the consistency of the results. The function was mentioned before in equation 5.1. The best value for b was evaluated to 1.2.

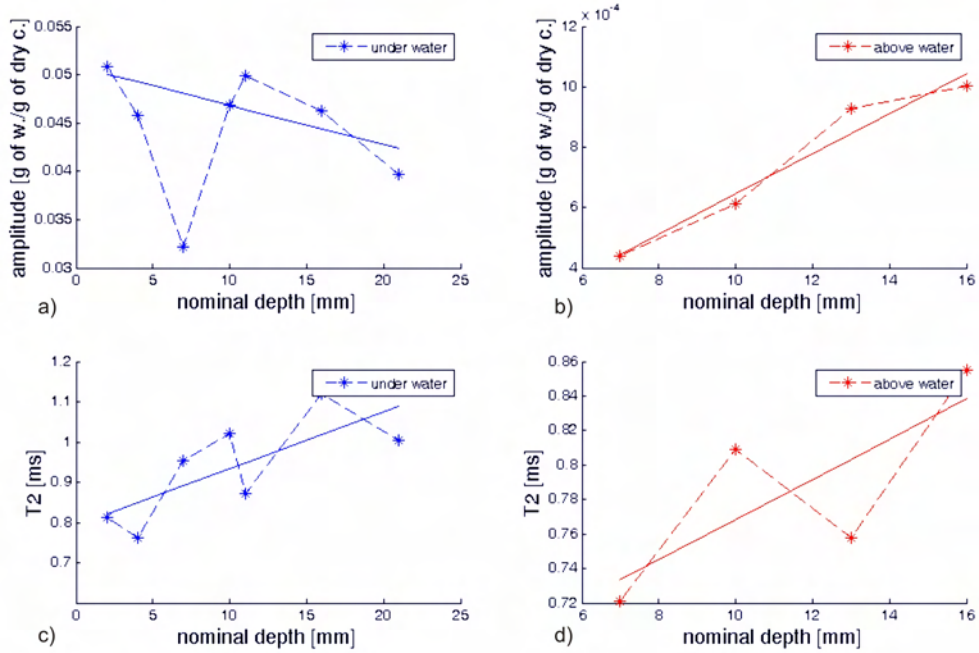


Figure 5.3: NMR depth profiles and extracted T_2 profiles measured at the Hohenwarthe lock. a) Depth profile measured for the part that is exposed to water. b) Depth profile for the part that is exposed just to ambient air and no water. c) T_2 profile for the part exposed occasionally to water. d) T_2 profile for the part that is exposed just to the ambient air.

For the measurements in the region of the lock wall that is occasionally under water, we can notice that the T_2 values increase steadily (Fig. 5.3c), whereas

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the amplitude slightly decreases (Fig. 5.3a). Only the third point in amplitude exhibits much lower amplitude. This is interpreted as heterogeneity of the material. Indeed, this concrete contains stones from 0 to 16 mm, and the maximum thickness of the sensitive layer which can be used at this depth is 150 μm which is probably not enough to average out all heterogeneities. The profile amplitudes measured in the region that is exposed to ambient air show a very low value (Fig. 5.3b). This low signal intensity may have occurred due to bad reference data that was used for normalizing the results. The measurements for obtaining the reference data were performed one hour before the lock wall measurements took place. Between these two experiments, the whole setup needed to be taken off and then put back as a ship needed to go through the lock. So the reference data amplitude had probably changed. However, we can state that the trend is good and the wall becomes more moist when we go deeper. In both cases, T_2 increases with the depth (Fig. 5.3d), which means that the pores are larger deeper in the sample. This trend was noticed also at the Gäubahntunnel measurements.

5.2.2 Laboratory measurements

For the laboratory measurements two concrete blocks were examined, which had the same composition as the concrete from the on-site measurements. These blocks were used also by the IBAC group and had inside electrodes at different depths (7 mm, 12 mm and 22 mm from the bottom surface) for the MRE measurements. In this way two types of measurements could be performed at the same time. The MRE experiments were monitored on a different PC as the NMR measurements. The cement blocks were named after the origin of the aggregates included in the composition, that is Gäubahn and Hohenwarthe. Figure 5.4a shows the container where the blocks were put in and kept in water. They were took out it just for the time of the experiment. The measurement setup is shown in figure 5.4b. All the block faces, except the bottom and the top, were covered with a watertight substance, so the water can ingress into the block only from the bottom through the capillary forces. The NMR experiments lasted between 45 and ninety minutes, depending on the water saturation level in the blocks.

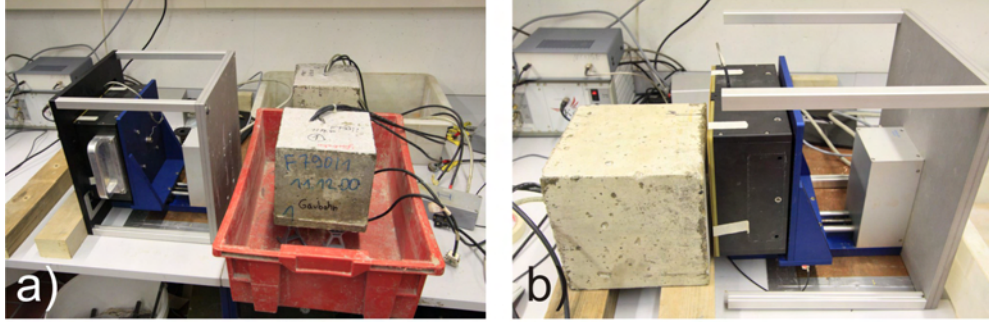


Figure 5.4: NMR experimental setup for the laboratory experiments. a) Preparation of the concrete blocks for immersing in water. b) NMR experimental setup.

The first step of the study was to perform the measurements on the dry blocks. These measurements were done for a depth of 17 mm, because one spacer was still between the magnets and the coil. Later on, for having a better comparison between the NMR and the MRE results, we took the spacer out and measured the water ingress up to 21 mm deep. Even though the differences between the pore diameters are not that big, the two blocks behaved differently, during the water ingress processes. The Hohenwarthe block was saturated in four days; after that, the drying process was observed during our study. At the end of the study (after 63 days), the Gäubahn block was still not saturated at a depth of 21 mm. To have a good comparison between the on-site measurements and the laboratory studies, first the Gäubahn block study will be presented and then the one for the Hohenwarthe block.

The Gäubahn block was very dry at the beginning, as seen from the black curve in Fig. 5.5a. After two days in water, the saturation level reached above 4% in the first two millimeters. A constant gradient of water content with depth can be observed, the amplitude at a depth of 21 mm being always lower than in the rest of the sample. The constant gradient is even more visible in the case of the T_2 depth profiles. The values in the first six millimeters are three times bigger than in the last five measured millimeters. We split the amplitude and T_2 depth profiles into six regions, where the amplitude as well as the T_2 values stay constant. These six regions are 0 to 3 mm, 4 to 7 mm, 8 to 11 mm, 12 to 15 mm, 16 to 19 mm and 20 to 21 mm. In this way we can plot the mean values

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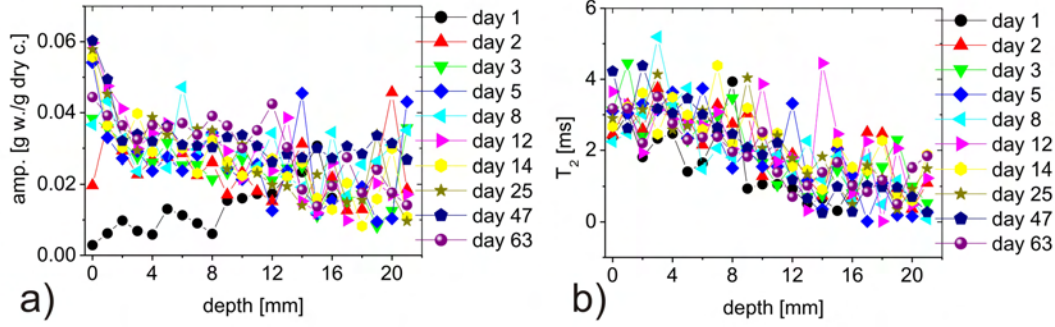


Figure 5.5: Water saturation levels and T_2 depth profiles for the Gäubahn block.
a) Water content for the different measurement days during the study. After 64 days, the amplitude still increased. b) T_2 depth profiles for the Gäubahn block.

over these six regions for each measurement day. The obtained plots are shown in Fig. 5.6. In Fig. 5.6a the evolution of the amplitude over the study days are plotted. Considering just the black curve, representing the mean amplitude value over the first three millimeters, it is clear that it reached saturation after 15 days. This means that the pores inside the block were saturated and had a 4% water content. Observing the other curves, it is clear that, deeper regions in the sample are less saturated. Additionally, lower proton signal leads to a poorer signal to noise ratio, so the points scatter more. The differences between the six regions are clearer in the mean T_2 evolution curves, where the T_2 value for the saturated region is three times higher than for the deepest measured region. The MRE measurements reveal the same water content.

Both plots in Fig. 5.6 are a function of time. In this way we can plot the mean amplitudes for each region against their mean T_2 values, to show the dependence of the saturation level with the T_2 values as mentioned by Halperin (Halperin 94). The results are plotted in Fig. 5.7a. All points for all depths are on the same line, except the points for the deepest 2 mm measured. A reason might be the bad signal to noise for this region or the fact that values are an average of two values, as the other points are obtained by averaging 4 points. The results of Halperin (Halperin 94) were obtained on a white cement setup and the linear dependence between the T_2 and the saturation level is going through the origin. In

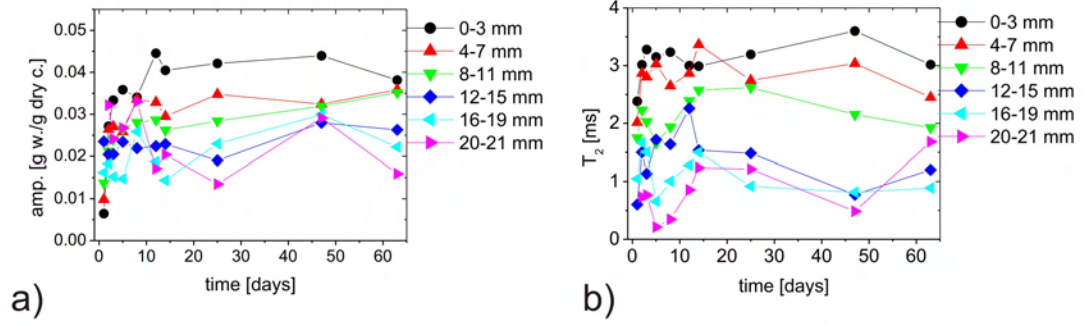


Figure 5.6: Evolution of the water saturation and T_2 values over the study time. a) For the first seven millimeters the saturation is reached already after 15 days, but for the last millimeters the pores are still not saturated with water. b) T_2 evolution for different regions of the measured depth during the study days. The values for the unsaturated region are clearly lower as for the saturated region.

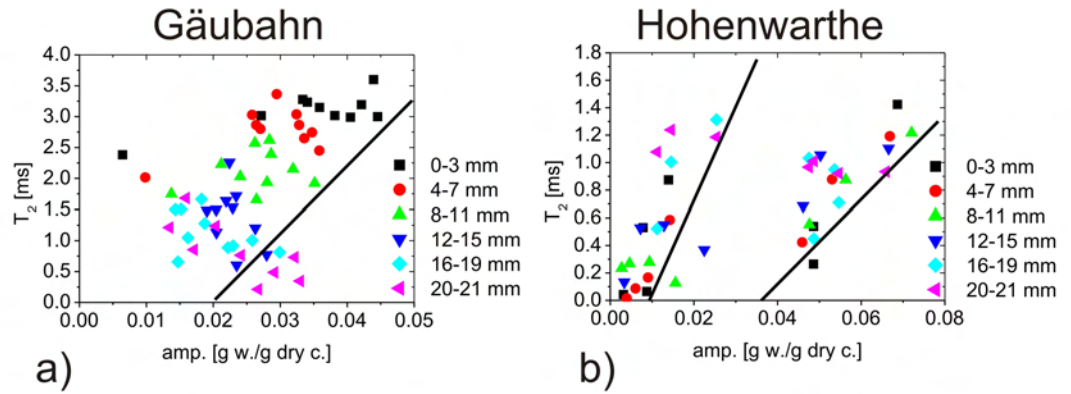


Figure 5.7: Linear dependence between the saturation level and the T_2 values. a) Dependence showed for the Gäubahn block. b) Dependence for the Hohenwarthe block. The linear trend is for both of the blocks noticeable but they have a different offset.

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our case an offset is observed. A reason for this discrepancy might be the material composition, as our sample is much more heterogeneous. In order to confirm the accuracy of our work, performed both on-site as well as in the laboratory, we took the T_2 values obtained at the on-site measurements and from those values we extracted the amplitudes from the correlation plot shown before in Fig. 5.7a. The comparison between the measured amplitude and the calculated one is represented in Fig. 5.8. The comparison was made for both on-site measured

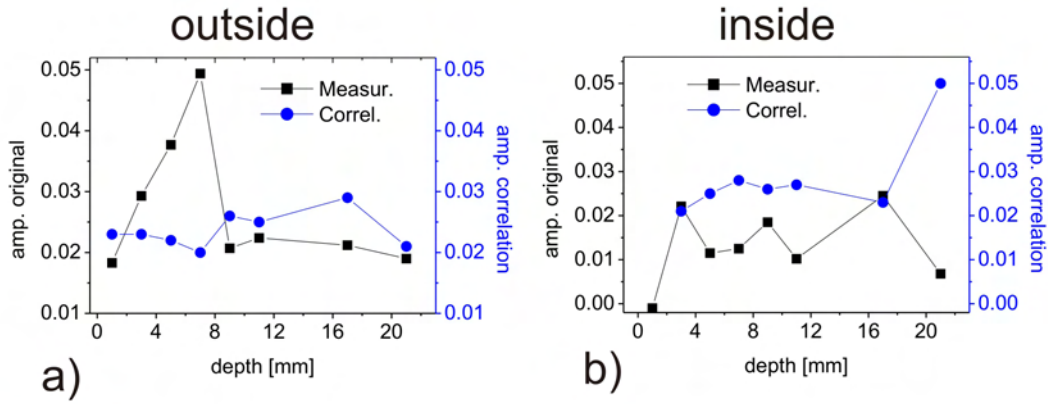


Figure 5.8: Measured and calculated values for the saturation levels. a) Comparison between the amplitudes measured and calculated at the entrance in the tunnel. b) Comparison between the amplitudes measured and calculated at 120 m inside the tunnel.

points, at the entrance into the tunnel (Fig. 5.8a) and 120 m inside the tunnel (Fig. 5.8b). The values obtained from the correlation are slightly higher than the on-site values. This is expected as we do not take into account the environmental effects and the poorer signal to noise ratio from the on-site measurements. And this values are closer to the values obtained by the MRE measurements. Exception are the first 7 mm at the entrance point, but this way to high values can be explained by the chloride problem that exists in this region (Weise 10).

The Hohenwarthe block looked at the first sight more homogeneous in consistency than the Gäubahn block. Then, during the measurements, the differences between the two blocks were even more clear when, after only four days, the saturation level was reached even in the deepest region of the block. Because of

this fast rate of water ingress and too few measurement points to make conclusive claims, we decided to monitor the water desaturation of the concrete block. Figure 5.9 shows the saturation as well as the desaturation of the Hohenwarthe block. In both cases the gradient that was seen at Gäubahn is not that high, and

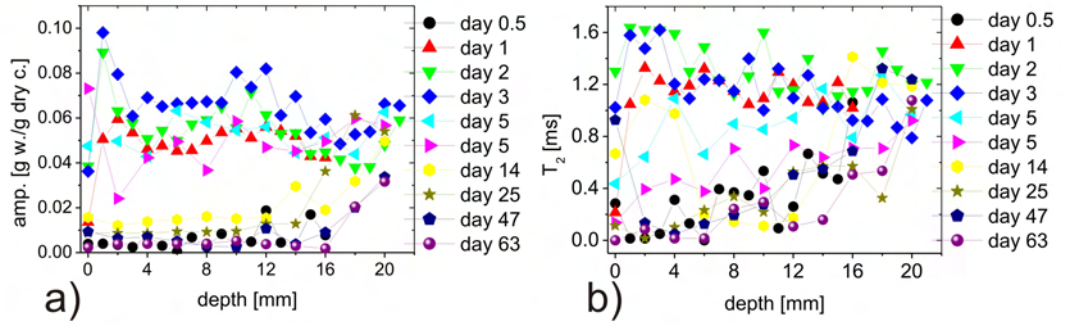


Figure 5.9: Water saturation levels and T_2 depth profiles for the Hohenwarthe block. a) Water content for the different measurement days during the study. After the fourth day the block was saturated, so the drying of the block was monitored. b) T_2 depth profiles for the Hohenwarthe block.

is more visible in the T_2 desaturation part. The water saturation is at around 5% which is slightly higher than for the Gäubahn block. However, the T_2 values are lower by a factor of two. This could be explained by the fact that the pore size of the Hohenwarthe block is much smaller than for the Gäubahn block. The T_2 depth profiles show also that the block was not completely desaturated during this time, the values at higher depth being still larger than those near the surface. If we divide now the depth profiles in the same six regions, as before for the other block, the mean values of the amplitudes and for T_2 over the study period can be obtained. These results are reported in Fig. 5.10. In Fig. 5.10a the evolution of the amplitude over the study days are plotted. For the first 15 mm of depth, the amplitude grows until the maximum water saturation is reached (in four days) and then it decreases to a minimum value, suggesting this region's desiccation and drying. Only the last two measured region did not dry out completely during this time. In case of the T_2 evolution curves (Fig. 5.10b), only the values for the

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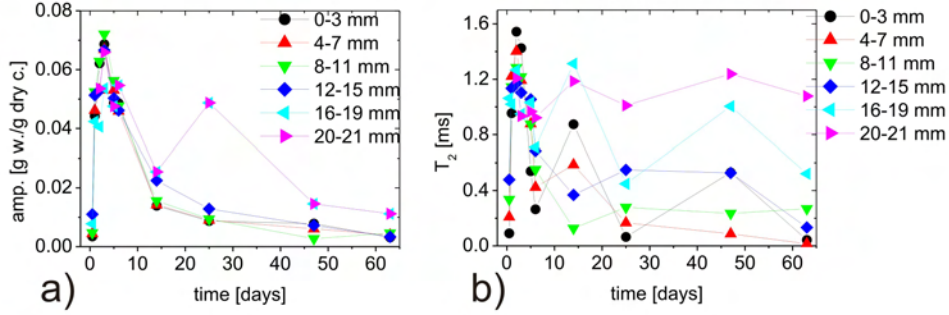


Figure 5.10: Evolution of the water saturation/desaturation and T_2 values over the study time. a) The fast saturation and the slow desaturation is visible. b) T_2 evolution for different regions of the measured depth during the study days.

deepest measured region is still increased. These observations are in accordance with the MRE results, where the maximum water content was found to be 5.8 %.

Regarding the plot of the dependence of the saturation level with the T_2 values, only the desaturation part was taken into account. This result is shown in Fig. 5.7b. The dependence show a different behavior than the one for Gäubahn. It looks like we would have two dependencies. This is maybe because we have a bimodal pore distribution as mentioned before. Gäubahn has just one pore size. As we have two pore sizes, we have also two different surface relaxivities and this influence in different ways the relaxation mechanism. Both linear dependencies present an offset. The reason for this might be again, as for Gäubahn too, the sample heterogeneity. The accuracy of our work was studied again, and we made the same correlation as before for the other concrete block. For this comparison we took into consideration the linear dependence line with the highest offset from Fig. 5.7b. The comparison between the measured amplitude and the calculated one is represented in Fig. 5.11. The correlation was performed again for both on-site measured points, the region which is flooded with water named as “under” in Fig. 5.11a and the region that is exposed only to the environment named as “above” in Fig. 5.11b. In both cases the correlated amplitudes are much higher. This might be because we did not consider the right linear dependence from Fig 5.7b. The MRE water content for the “under” region is close to 6%. A significant

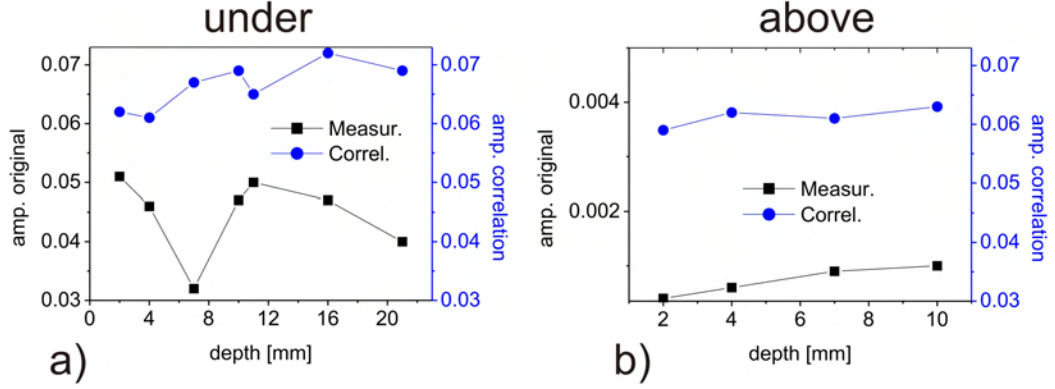


Figure 5.11: Measured and calculated values for the saturation levels. a) Comparison between the amplitudes measured and calculated for the region that is occasionally under water. b) Comparison between the amplitudes measured and calculated for the region that is exposed only to the environment.

difference can be noticed for the “above” case, where the difference is a factor of sixty. This can be explained by the reposition of the equipment between two measurements. As mentioned before, between the experiments the whole setup needed to be set up and taken down in order not to disturb the naval traffic. Nevertheless, this factor is consistent for all points and the trend is similar for both measured and calculated amplitudes.

5.3 Conclusion

Two different concrete types with different composition were examined on-site as well as in the laboratory. Because of their different composition, especially the aggregate composition, the water ingress in the saturation process behaves completely differently. This study was relevant both for the industry partner, who could compare two different analytical techniques based on different methods, and for us in order to understand more about complex porous systems. The use of the NMR-MOUSE proved appropriate, especially for the on-site measurements, where the measurements could be performed quickly, non-invasively and without affecting traffic, both road and the naval.

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In the case of the on-site measurements, the tunnel and the lock measurements, differences were noticed both between the two different concrete types as well as between the points measured at each site. For the tunnel measurements, differences in profile amplitudes and T_2 values were observed, evidencing the influence of the environment on concrete quality and lifetime. At the lock site, differences were already expected, as one part of the concrete wall is exposed more often to water.

The laboratory measurements proved to be efficient and informative. Not only could the differences in the rate of water uptake be evidenced, but the results could be compared with the MRE method and showed a good agreement. During 63 days, the total duration of the study, the Gäubahn concrete block did not reach complete saturation while the Hohenwarthe block was saturated and subsequently almost dried out. It is impressive how a difference in pore size of less than one order of magnitude can influence so dramatically the water dynamics in the concrete blocks.

The complexity of the composition of these concrete blocks, poured and in active use prior to our study, makes it difficult to attribute cause to artifacts in our results. It would be interesting to make a new case study with experiments performed just on different types of cement pastes, and then modify these samples to include the fine sand particles and the aggregates, and in different percentages, to see which components influence the most.

Chapter 6

Cultural Heritage

Cultural heritage is a vast and fascinating field. It tells the story of our society, from the beginning until now, giving us insight into various ways of life. Starting with objects like monumental remains, the concept now includes new categories like ethnographic heritage (Unesco 11). In this chapter, we focus on the tangible heritage: physically extant cultural property like wall paintings from Herculaneum, master oil paintings of Rembrandt, or an unknown Cologne painter. Knowing the fact that tangible cultural heritage is unique and cannot be replaced, it is a big responsibility for conservators to maintain art objects in their original state. Therefore, the use of non destructive testing techniques is a must. Furthermore, in many cases the transport of the artwork is not possible and the use of portable equipment is required.

Single-sided NMR (the Profile NMR-MOUSE) has been employed for non-destructive testing of different types of objects of cultural heritage (Blümich 10a), including investigations of the structure of paintings (Proietti 05, Proietti 07), paper (Blümich 03, Casieri 04a, Proietti 04, Federico 10), bones and mummies (Rühli 07), porous materials (Proietti 06, Sharma 03, Blümich 05b, Blümich 08), and the moisture distributions of wood (Casieri 04b) and wall paintings (Proietti 05, Proietti 07, Blümich 10b).

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6.1 Ancient Roman wall paintings

6.1.1 Introduction

Herculaneum is an ancient city located in the bay of Naples. The city gained its name from the Greeks and became a Roman municipium under Titus Didius in 89 BC. In 62 AD an earthquake destroyed parts of the city and in 79 AD it was completely buried under volcanic material by the eruption of the volcano Vesuvius. The city was at that time a summer residence for wealthy Romans. The best known residence is the Villa of the Papyri, so named due to a library of 1800 papyrus scrolls found in the Villa during the 18th century. Excavations of Herculaneum started in 1737 and they are still continuing today. They have led to a wealth of information about the Roman life in general, but have also brought deterioration and general decay to the site and its precious wall paintings and mosaics. In 2001, the Herculaneum Conservation Project was launched in order to address and halt the serious decay conditions found at the site (Project 07).

Two major types of wall paintings can be differentiated, fresco and secco. In the case of a fresco, the paint pigment is mixed with water and applied on a fresh (not dried) lime mortar. For this technique a binder is not required, as the pigment will sink into the mortar. On the other hand, in case of a secco, the paint pigment is applied on a dry mortar or plaster. In this case the pigments need a binder (usually egg, glue or oil) in order to attach to the wall (Procacci 69).

Moisture is a key concern in Herculaneum, especially in the conservation of wall paintings and mosaics. Moisture triggers the leaching of salt ions from the ground, from new consolidation building as well as from depositions of a polluted atmosphere or from biological activity which are then transported through the walls by capillary forces. Transported salts can accumulate in areas of water evaporation and, once saturation is reached, they precipitate. Salts generally become concentrated on the surface of paintings as efflorescences or below the surface as sub-efflorescences. The highest damage is caused when salts precipitate below the paint layer as it causes the painting to flake and disintegrate. Gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$) is one of the most damaging salts present in deteriorated wall paintings, and most wall painting restoration procedures have been developed to remo-

ve this salt followed by the reconsolidation of the wall (Matteini 87, Matteini 99). It is now known that the organic consolidants used in early conservation treatments cause more damage in the long term because they occlude the natural porosity of a wall painting and interfere with salt transport, causing salt precipitation where it otherwise would not occur. Therefore, the use of inorganic substances, such as barium hydroxide or ammonium oxalate for the consolidation of mural paintings is favored over treatments with primarily organic substances (Matteini 99). Moreover, the identification of past restoration efforts is equally as important as the assessment of the state of conservation and the effectiveness of new conservation treatments.

The experimental results about the stratigraphy of decorated walls in ancient Herculaneum are presented here. Depth profiles were recorded from the Mosaic of Neptune and Amphitrite, and from the wall paintings located in the House with the Black Room and in the Villa of the Papyri. These measurements were made to identify the type of information available from unilateral NMR with the NMR-MOUSE and their significance to the conservation and restoration of mosaics and wall paintings.

6.1.2 Experimental

The NMR measurements at the excavation site were conducted with a Profile NMR-MOUSE[®] with 21 mm depth access at a proton resonance frequency of 13.86 MHz. The measurements of the Mosaic of Neptune and Amphitrite and of the wall paintings at the Villa of the Papyri were recorded with a 5 mm spacer between the sensor and the rf coil to reduce the depth access to 16 mm for the benefit of improved sensitivity. For the measurements in the Black Room, a 15 mm spacer was introduced to gain even higher sensitivity at the expense of the depth access being reduced to 6 mm. The sensor was mounted on a computer-driven positioning device. The NMR sensor and the positioning device were manufactured by ACT GmbH, Aachen, Germany. The spectrometer was a Kea spectrometer manufactured by Magritek Ltd, New Zealand, connected to a Dressler 1 kW rf amplifier. The measurement parameters of all experiments are summarized in Table 6.1. Complementary to the NMR measurements, reflectance Fourier transform

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Table 6.1: Acquisition parameters for depth profiles for the measured points

Parameters	Mosaic of Neptune and Amphitrite	House of the Black Room	Villa of the Papyri
Maximum depth [mm]	16	6	16
Pulse width [μ s]	21	6.5	19
Echo time [μ s]	140	145	140
Number of echoes	16	16	256/64
Number of scans	512	256	32/64
Time/profile [min]	80	60	90

mid-infrared spectroscopy (FTIR) and X-ray fluorescence (XRF) experiments were performed. Recently, reflectance FTIR and XRF analysis has been shown to be very valuable for the in situ, non-destructive analysis of the organic material present on wall paintings (Miliani 07b, Rosi 09, Miliani 07a). These techniques were applied in situ with portable instrumentation by the group of Prof. Eleonora Del Federico. Some results are presented in this chapter, more results can be found in the paper (Haber 11).

6.1.3 On site studies

6.1.3.1 Mosaic of Neptune and Amphitrite

The mosaic at the house of Neptune and Amphitrite (Fig. 6.1a) is one of the most famous ancient Roman mosaics, still preserved in its original setting on the south wall of a triclinium (dining room) of a middle-class household. The room is decorated with wall paintings with garden scenes, a nymphaeum, creating an illusion of the outdoors.

The mosaic was examined twice in 2009, once in March and once in October, in order to observe the climate influence over the moisture distribution in the mosaic. A mosaic is a decorative piece of art, where on a mortar bed cubiform, colored glass pieces (called tesseræ) are assembled, such that a picture or a pattern is created. Two regions can be identified in the profiles, one from 0 to 6 or 7 mm depth and one at greater depth. They are assigned to the tesseræ with some

mortar between them and to the mortar embedding the tesserae, respectively. Points 1 and 2 were measured in March, and all other points including point 1 were measured in October (Fig. 6.1a,b). Point 1 was measured in both March and October to investigate the reproducibility of the measurements and the influence of seasonal moisture variation. The locations of the measurement points were chosen depending on the composition and color of the tesserae and their height from the bottom of the mosaic, as both may have an impact on the moisture content.

The depth profiles shown in Fig. 6.1b compare the results from March and October. The two profiles obtained for point 1 overlap almost identically in the region of the tesserae from 0 to 7 mm, but they differ at higher depths: the amplitude of the profile measured in the fall is lower than that of the profile measured in the spring. The reason for this difference is that the wall dried during the hot summer. This interpretation is supported also by the fact that points 1 and 2 show identical profiles for the mortar bed in the March measurements, being also at the same height from the ground (Fig. 6.1a). The profiles from all the points, except point 1, overlap well in the region of the tesserae (Fig. 6.1c). Differences are detected only at higher depths and are characteristic of the mortar bed. Point 6 is the highest one measured in the mosaic and shows the lowest signal in the mortar region. Point 7, on the other hand, is the lowest and shows the highest mortar signal. Points 3 and 4 are positioned at the same height and overlap in the mortar region. In summary, the higher the measurement position, the lower the profile amplitude in the region of the mortar bed. This is a strong indication that the source of moisture is located on the ground.

Regarding the T_2 profiles (Fig. 6.1d), they look very different from the amplitude profiles. They show the highest T_2 value at the interface between the tesserae and the mortar bed with a broad maxima. While the amplitude in the regions of the tesserae at point 1 is very intense, the corresponding T_2 values are very low. Together with the fact that the first 7 mm of the profile did not change during more than 6 months is a strong indication that the ^1H signal does not derive from water but from an organic coating at the surface, possibly an undocumented early consolidation treatment. Also noteworthy is the extreme behavior of the amplitude and T_2 profiles at point 7. The signal amplitude increases strongly with depth

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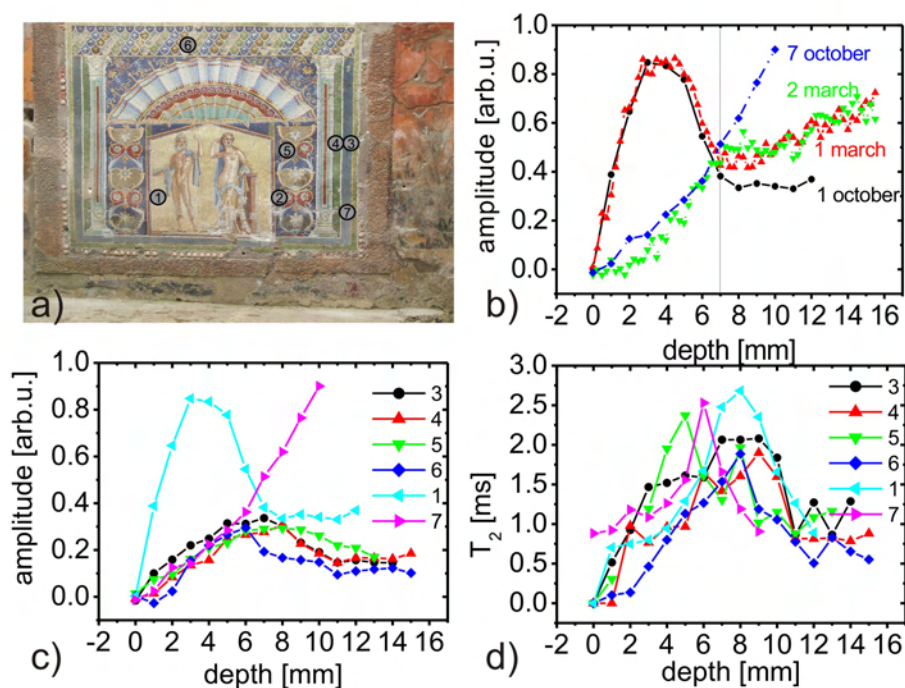


Figure 6.1: Mosaic of Neptune and Amphitrite. a) The mosaic showing the position where the NMR depth profiles were recorded. Point 2 was measured just in March, while point 1 was measured both in March and October. b) Depth profiles for point 1 (black and red), recorded more than 6 months apart. The first parts of the profiles overlap identically while inside the mortar bed they differ. The mortar signal from point 2 (green) overlaps with the mortar signal from point 1, both measured in March, evidencing the same moisture conditions for this section of the mortar bed. The depth profile for point 7 (blue) being at the lowest point measured reports an even higher moisture content of the mortar bed. The line indicate the boundary between tesserae and mortar. c) Depth profiles for all the points measured in October. The moisture content in the mortar decreases with the increasing height above the floor. d) T_2 profiles for all measured points.

(Fig. 6.1b), while the T_2 profile exhibits a narrow peak at the interface between tesserae and mortar bed. This indicates high moisture deep inside the wall and possibly a beginning detachment of the tesserae from the mortar bed.

6.1.3.2 The Black Room

Across the street from the House of Neptune and Amphitrite is the back entrance to the House of the Black Room, one of the most luxurious houses in Herculaneum. The House owes its name to a large main room that is decorated with pillars and chandeliers painted with a black background in the third Pompeian style. The room was excavated between 1933 and 1961. Although not documented, it is believed that all wall paintings excavated during that time period were treated with a mixture of beeswax and paraffin as a preventative conservation measure to avoid further damage. Over time, changes in relative humidity, the use of modern cement for restoration of the walls, and air pollution have caused damaging salts to leach from the walls and precipitate as efflorescence either on the surface or just below the paint layer. This efflorescence has caused significant damage, including delamination and pulverization of the paintings. The presence of earlier conservation treatments containing an organic coating have contributed to this decay. The north-western wall of the main room (marked as wall B in the figures), behind which a modern Portland cement wall was built, has been damaged the most by the salt, with its paint layer delaminated in many areas. Salt efflorescence on this wall appears especially around the area behind which a new wall was built.

Recent conservation efforts involved the removal of the organic consolidants from early restorations and application of test treatments in small areas of the cleaned walls. The three painted walls labeled A, B, and C (Fig. 6.2a,b) were analyzed in order to record depth profiles at different positions identified by Prof. Mauro Matteini, a professor involved for several years in applying different inorganic conservation treatments in Herculaneum. On the walls A and B (Fig. 6.2a), measurements were made at points where the organic fixative treatment was still present and at points where it had been removed. On wall C (Fig. 6.2b), ammonium oxalate and barium hydroxide inorganic treatments were tested. To maximize the sensitivity for the measurements at natural proton content, the depth profiles

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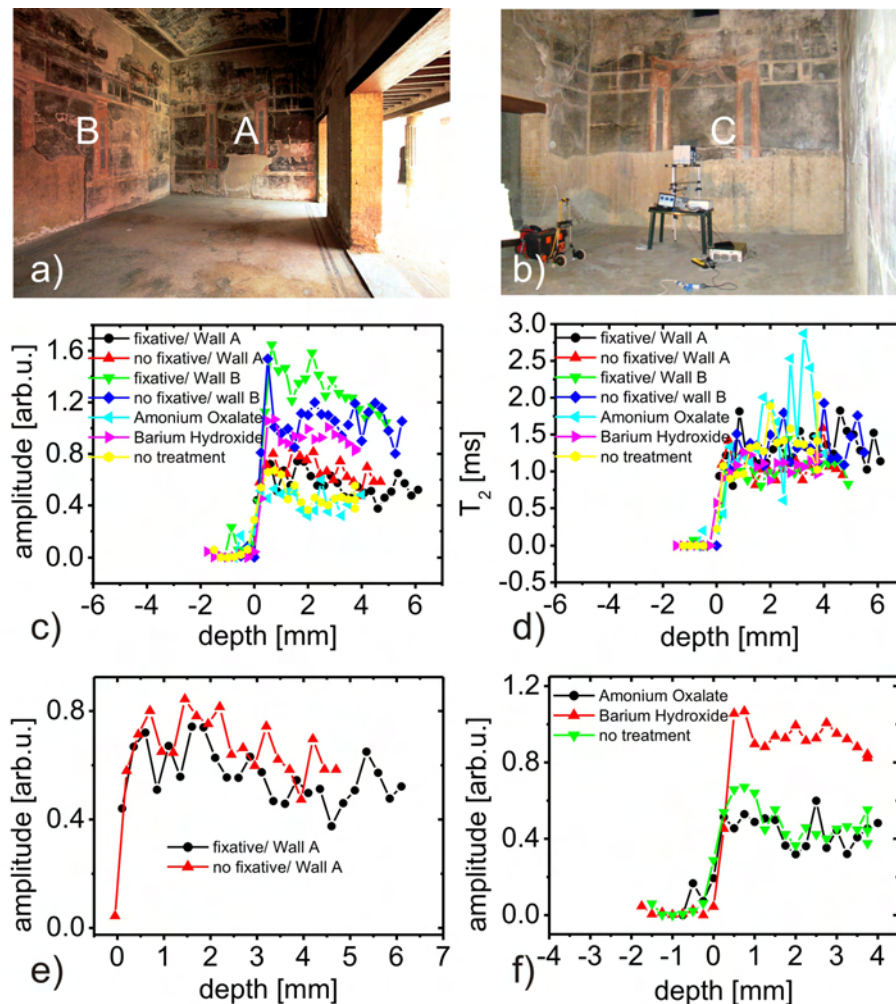


Figure 6.2: NMR measurements in the Black Room. a) The measurement points were on walls A and B. b) Wall C, where sections have been treated with barium hydroxide and ammonium oxalate. The experimental setup is seen in the foreground. c) Depth profiles for all measured points. Depending on the wall and the conservation technique, differences in moisture content are found. d) T_2 profiles for all measured points. All points show mostly the same T_2 value. e) Depth profiles for wall A showing the proton content with and without fixative on the wall painting. f) Depth profiles for wall C showing the difference in proton content for the treatments with barium hydroxide and ammonium oxalate and their comparison with the proton content for an untreated region from the same wall.

were taken with all spacers placed in the magnet. This resulted in a maximum accessible depth of 6 mm.

The proton density profiles (Fig. 6.2c) differ in the absolute signal amplitudes but are similar in shape: they show the highest signal at the surface, a peak at a depth of about 2 mm, with the signal amplitude decreasing with increasing depth. The T_2 profiles (Fig. 6.2d) show little variation between different positions which is mainly attributed to the low signal-to-noise ratio of the measurement. The profiles acquired on wall A (Fig. 6.2e), both for the area which is coated with organic consolidants and the area where this organic coating was removed, do not differ significantly in signal amplitude.

The NMR depth profiles from wall B show a proton density significantly higher than in depth profiles from walls A or C. This observation is consistent with the existence of visible salt layers at the surface, due to the hindered moisture exchange by the modern concrete wall (Fig. 6.2c). For this wall, with the exception of the first two profile points at the surface, the amplitudes of the profiles measured at locations with organic consolidant treatment and with the consolidant removed are quite different. There, the amplitude is high and the same for both profiles suggesting incomplete removal of the organic treatment. The profiles of wall C (Fig. 6.2f) present the proton density amplitude from a point where the older organic consolidant was removed (no treatment) and from the points with inorganic conservation treatments (ammonium oxalate, barium hydroxide) are all very different, although they are positioned really near one to each other. The point where the organic consolidant was removed shows a profile similar to that at the point treated with ammonium oxalate. However, in the first millimeter, the higher signal is given by the cleaned area, probably due to incomplete removal of the older organic treatment. At larger depths, the amplitude of the area treated with the barium hydroxide displays the highest intensity and, therefore, is the area with the highest moisture content. Although the higher moisture content in the barium hydroxide treated area may suggest that the treatment with barium hydroxide is inferior to the treatment with ammonium oxalate, it is not the moisture content but the moisture volume flux which should be assessed to rank the effectiveness of the treatments. These observations are consistent with laboratory studies of moisture in mock-up mortar beds presented later in this chapter and

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with the reflectance FTIR spectra obtained by the Pratt Institute research group, led by Prof. Eleonora Del Federico (Haber 11).

6.1.3.3 The Villa of the Papyri

The Villa of the Papyri was first explored from 1750 to 1765 through a network of tunnels burrowed into the volcanic rock by Karl Jacob Weber, a Swiss architect and engineer, under the patronage of Charles III of Spain. From his detailed drawings an accurate map of the Villa could be established. Today, a large part of the Villa remains underground. In 1826, a library of about 1800 papyrus scrolls was discovered, giving the Villa its name. It is believed that the house belonged to a wealthy and learned Roman noble man, probably Lucius Calpurnius Piso Caesoninus, father-in-law of Julius Caesar and consul in 58 BC. Recent excavations have revealed a small room with brightly colored wall paintings in a great state of conservation. Therefore, this room is of major interest to conservators, conservation scientists, archaeologists, and historians as its surfaces remain as they were left in 79 AD. It is in this room and on the above terrace where our NMR measurements were taken.

As a benefit to our NMR measurements, this room is kept moist at 96 % relative humidity to direct the moisture migration into the walls and keep salts from crystallizing. Two of its walls were examined (Fig. 6.3a). In all, four positions were studied in detail, three inside the room and one on the outside wall (Fig. 6.3b) facing the terrace above that room. Interestingly, two types of stratigraphy can be identified, each being represented by two different profiles (Fig. 6.3c,d). One type was measured at the points A and B on a wall dating from the time before the earthquake of 62 AD (Fig. 6.3a), and the other on a wall rebuilt after the earthquake (without any wall paintings) and the outside terrace wall (Fig. 6.3b). The amplitude profiles look similar, though with an offset in the depth, which reveal the layer structure of the mortar bed. The fabrication of these layers has been described by Vitruvius (Pollio 99) and more recently in many scientific studies of wall fragments found in different locations (Donner 69, Bearat 96).

The profiles of the older wall (A and B) are thicker and more intricate than those of the newer wall (C) and the terrace wall (D). This may be because either

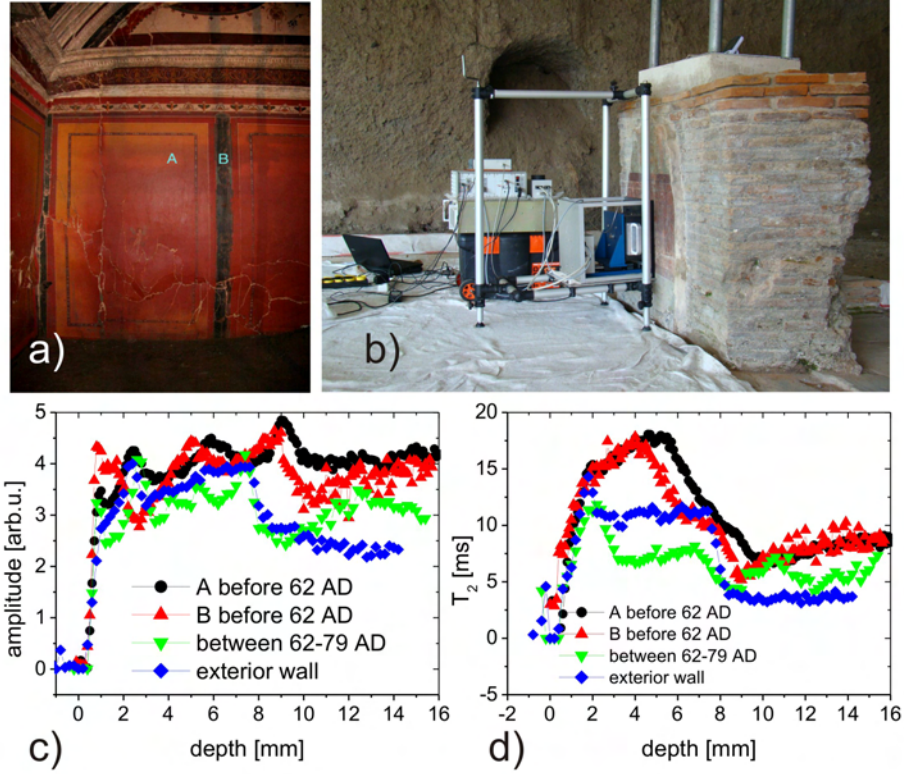


Figure 6.3: NMR measurements at the Villa of the Papyri. Location of the measured points dating from before 62 AD, when the earthquake took place. b) Experimental setup at the exterior wall on the terrace. c) Depth profiles at the four measured positions identifying different mortar layers and two different types of layer structures. The profiles for A and B differ from the other two. d) T_2 profiles for all measured points. Different T_2 values are assigned to different mortar materials

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the more recent method of preparing the mortar for the wall painting was less expensive, as much reconstruction work had to be completed after the earthquake, or the refined technique of crafting the more intricate mortar layers was lost. This interpretation suggests that either the outside wall of the terrace (D) may have been reconstructed after the earthquake, or that the technique used for outside walls had been applied for the reconstructed wall inside (C).

The NMR depth profile results clearly suggest that two different techniques were employed, which is in agreement with the XRF studies of the walls recorded by the research group of Prof. Del Federico (Haber 11). Comparing the XRF spectra of the red pigments used for the restored and the original paintings shows that the restorers used cheaper red pigment as the spectrum of the restored wall shows a high iron content suggesting the use of iron oxide pigment to replace the highly expensive, precious and intensely colored cinnabar (HgS), which is inferred by the presence of mercury and sulfur lines in the second spectrum. The overlay of the XRF spectra of both walls is reported in figure 6.4a.

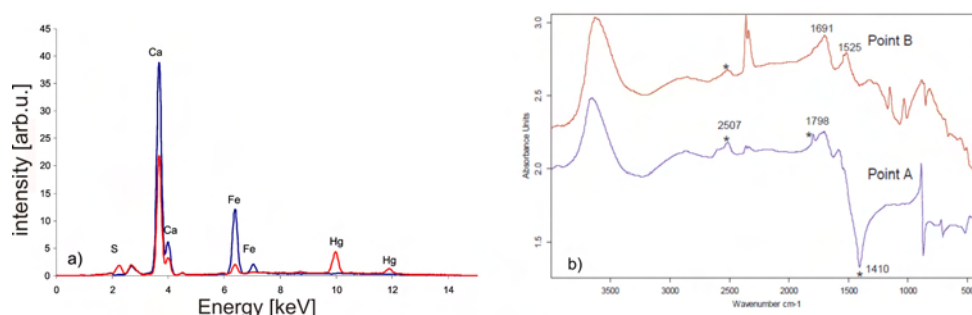


Figure 6.4: XRF and FTIR measurements at the Villa of the Papyri. a) Red: XRF spectrum of the red pigment on the wall built before 62 AD showing the presence of mercury (Hg) and sulfur (S) indicating that the red pigment used was cinnabar. Blue: XRF spectrum of red pigment on the wall built after 62 AD, showing the absence of mercury and sulfur and the high content of iron (Fe), suggesting an iron red pigment such as hematite or red ochre. b) FTIR spectra. Point B reveals the possible presence of organic binder and oxalates, known to be degradation products of organic substances. Point A reveals stronger calcium carbonate bands, suggesting that the technique in this case is fresco. Data by E. del Federico and collaborators from Pratt Institute.

6.1 Ancient Roman wall paintings

It should be mentioned that the two depth profiles recorded at the positions A and B differ in their moisture content in the first 3 mm. Profile A has been measured at a blank section of the painted wall, while profile B was taken on the painted vertical decoration. It is speculated that the decoration is a secco so that the moisture uptake from the humid air is different in the two sections. This is also consistent with FTIR results of both surfaces (Fig. 6.4b).

Following Vitruvius (Pollio 99), the mortar bed for wall paintings consists of three mortar layers prepared with sand followed by the application of three lime plaster layers prepared with marble dust. The painting is applied on the last layer while still wet. Vitruvius recommends that the marble plaster layers be prepared with decreasing particle size towards the surface and to press them with a trowel to improve the packing density (Pollio 99). The packing of the layers coupled with the use of different aggregates (sand vs. marble dust) is likely to give rise to differences in the pore sizes across the mortar-layer structure. We postulate that the last three layers give rise to the top three peaks observed up to a depth of 9 mm, beneath which the sand-mortar layers follow.

Similar to the amplitude profiles, the T_2 profiles can also be grouped into the same two pairs, first are the ones obtained for position A and B and the second the other two (Fig. 6.4d). The T_2 profiles from points A and B differ only in the thickness of the second mortar layer, shown by the shift of the end of the profile of point B to the left when compared to the profile of point A. It is possible that the mortar with the second types of profiles was prepared with a different aggregate or mixture of aggregates. Based on the T_2 profiles at least two types of mortar layers can be identified. One type was used for the layers between 0 and 8 mm and another one for the deeper layers. However, the layer between 6 and 8 or 9 mm (for the wall reconstructed between 62 - 79 AD) may indicate an inconsistency to Vitruvius' rule because the measured T_2 values are smaller, there could be smaller particles or lower moisture content in the inner layers, especially in the case of the interior walls in the newly excavated room which is kept at above 95 % RH. Furthermore, a period of rain during the days before the measurements cannot be dismissed and could also account for the higher moisture content in the surface layers of the outer wall. The T_2 profile from the interior wall reconstructed between 62 - 79 AD shows high T_2 values in a thin surface layer, a

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lower T_2 plateau between 3 and 8 mm depth and even lower values further down. This general trend is followed by the profile from the terrace wall; however, the initial peak is broader and the T_2 plateau is higher which is likely to be due to the different moisture content of the outside wall.

Given the good signal-to-noise ratios of the data measured at the Villa of the Papyri, the measured CPMG decays could be inverted by inverse Laplace transformation to produce distributions of relaxation times T_2 (Fig. 6.5). This

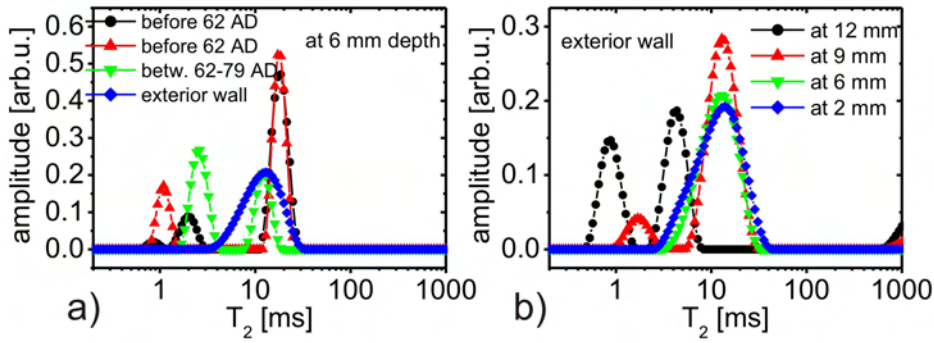


Figure 6.5: Distributions of relaxation times obtained by inverse Laplace transformation of T_2 decays. a) T_2 distributions for all measured points at 6 mm depth. b) T_2 distributions at different depths for the exterior wall.

is a more refined analysis than fitting the measured signal with a model decay function, such as a monoexponential decay, to extract a single relaxation time T_2 (cf. Fig. 6.3d). The T_2 distributions at a depth of 6 mm at the four positions reveal the similarities of the mortar structures identified in the depth profiles above. Two peaks are observed, one at short relaxation times and one at long relaxation times. The one at short times can be assigned to bound water confined to small pores, while the one at long relaxation times is assigned to more mobile water that partially fills larger pores. At a depth of 6 mm, this peak is basically identical for both positions A and B of the original wall and very similar for the restored wall and the outside wall (Fig. 6.5a). The peaks at short times seem to vary in an uncorrelated way, which may be due to the noise of the measurement data. The change of the T_2 distribution with depth reports the moisture content

and pore structure in the different layers (Fig. 6.5b). All peaks at large T_2 arise at the same position (up to a depth of 9 mm), indicating similarity of the mortar material across that range and a change towards larger depth.

The use of different types of mortar can also be seen by visual inspection of mortar pieces broken off the walls (Fig. 6.6a). A section from the Villa of

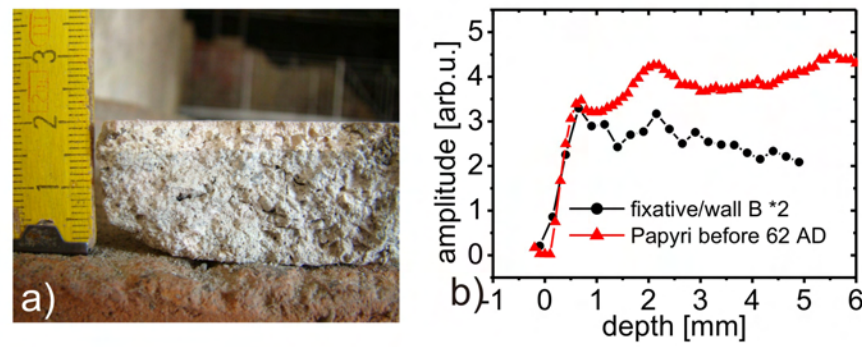


Figure 6.6: Mortar layer comparison. a) Photo of a mortar fragment from a fresco of the Villa of the Papyri showing the layer structure. b) Comparison of depths profiles recorded from walls in the Villa of the Papyri and the Black Room for the first 6 mm.

the Papyri clearly shows a whitish first layer about 4 mm thick, followed by a somewhat darker layer up to a depth of 9 mm and slightly lighter colored mortar beyond that; this matches the profiles A and B from the older wall inside the Villa of the Papyri. As profiles ranging up to 16 mm depth were not acquired from the much dryer walls of the Black Room, only the initial parts of the profiles from the Villa of the Papyri (point A) and the Black Room can be compared (Fig. 6.6b). The initial contours match, showing, that at least the outer mortar layers onto which the paintings were drawn are similar.

6.1.4 Laboratory studies

For a better understanding of the moisture flow rate inside the wall paintings, mockup samples were prepared by the group of Prof. Eleonora del Federico (Pratt Institute, New York) and shipped to Aachen to be characterized by low field

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NMR. The mockup samples were prepared with one or more mortar layers, and different conservation treatments, found also in the Black Room, were applied. One of these samples is shown in Figure 6.7. Already upon arrival in Aachen, the pigment layer of the sample was flaking off. A reason for this might be an incorrect concentration of components used for preparing the mortar layers and the pigment layer on top. Initial attempts to measure the samples dry yielded

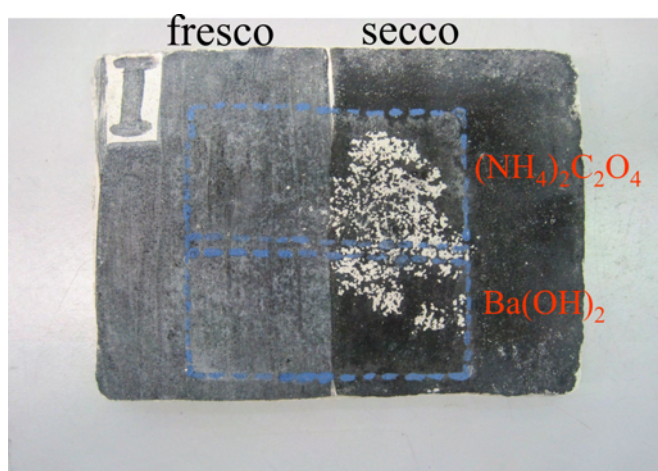


Figure 6.7: Mock-up sample. The mortar layers were applied with different techniques (fresco and secco) and different conservation treatments were applied on the pigment layer; $\text{Ba}(\text{OH})_2$ is barium hydroxide and $(\text{NH}_4)_2\text{C}_2\text{O}_4$ is ammonium oxalate. The pigment layer in the secco region was flaking off already before the measurements.

no signal. Consequently, they were put for 3 hours into a climate temperature chamber at 90 % relative humidity; still no signal could be recorded. Therefore, the samples were saturated with water and the drying process over 15 hours was observed through depth profile measurements. In order to have a better informative value of the moisture transport, the signal amplitude over a region of 1 mm was averaged for each sample and plotted against the drying time. The measurements were performed with the Profile NMR-MOUSE[®] with 5.4 mm depth access at a proton resonance frequency of 17.1 MHz. The sensor was mounted on a computer-driven positioning device. The NMR sensor and the positioning device were manufactured by ACT GmbH, Aachen, Germany. The spectrometer

hardware was a Minispec spectrometer manufactured by Bruker Biospin (Rheinstetten, Germany). The 90° pulse width was $5 \mu\text{s}$, the echo time was $45 \mu\text{s}$ and 512 scans were repeated for each depth position. The averaged signal amplitudes in a range of 1 mm for different samples with different conservation treatments are plotted in figure 6.8.

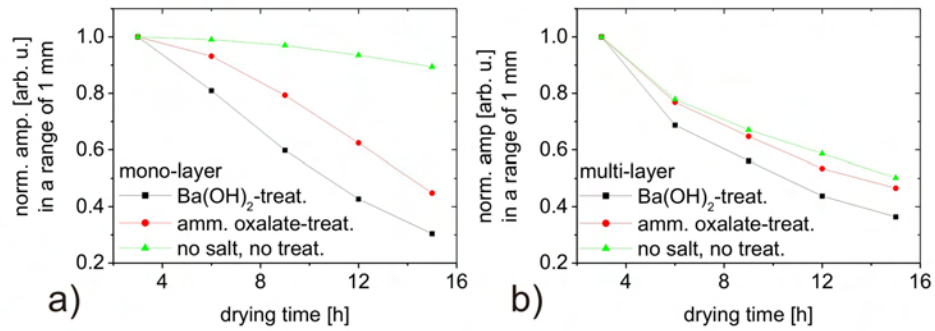


Figure 6.8: Fresco drying experiments. a) Drying process of the mono-layer mortar for different conservation treatments. b) Drying process of the multi-layer mortar for different conservation treatments.

Figure 6.8a shows the results for mono-layered fresco samples. It is noticeable that the ammonium oxalate treated region dries more slowly than the barium hydroxide region. This means that the moisture transport in the barium hydroxide treated frescoes is much higher than in the ammonium oxalate treated frescoes. The region that was not treated shows the slowest drying. But this is not relevant for wall paintings exposed to the environment as the wall paintings in Herculaneum, because there is salt in the air (being near to the sea) and the wall paintings will absorb the salt from the ambience. The same trend was observed also for the multi-layer fresco models. Here the salt-free, treatment-free region dries out much faster than in the mono-layer structure, but it is still slower than the ammonium oxalate treatment and the barium hydroxide treatment. In the first 5 hours of drying it behaves exactly the same as the ammonium oxalate treatment. But still the barium hydroxide treatment shows faster drying. As a comparison between the mono-layer structures and the multi-layer structures the ammonium

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oxalate conservation treatment shows the most stable behavior. The barium hydroxide treatment shows much faster drying in the mono-layer structure than in the multi-layer structure, as well as the salt-free and treatment-free region.

Along with the drying experiments, diffusion experiments were performed, in order to get a better understanding of the moisture transport. The diffusion experiments were executed with the same Profile NMR-MOUSE[®], having a constant magnetic field gradient of 20 T/m. The pulse sequence used is a stimulated echo experiment with a constant gradient. The pulse sequence was presented before in figure from the first chapter. The time between the first two 90° pulses is 180 μ s and the signal from 1024 scans was averaged. The diffusion coefficients obtained for the ammonium oxalate treatment are similar in the mono-layer and the multi-layer sample (7.8×10^{-10} m²/s and 7.7×10^{-10} m²/s). In case of barium hydroxide treatment differences in diffusion coefficient between mono-layer and multi-layer are observed (8.5×10^{-10} m²/s compared to 7.5×10^{-10} m²/s). By comparison of the two diffusion coefficients obtained for the mono-layer fresco model, a higher diffusion coefficient is obtained for the barium hydroxide treatment than for the ammonium oxalate treatment. This means that the molecular mobility in the ammonium oxalate treated region is lower than in the region treated with barium hydroxide, and thus was the moisture transport slower. These measurements are in good agreement with the results obtained for the drying experiments. In the case of multi-layer fresco models, the values are not that different, but they show that the diffusion coefficient for the ammonium oxalate treatment is higher than the diffusion coefficient for the barium hydroxide treatment. This is not in agreement with the drying experiments. A reason for this disagreement could be the inhomogeneity of the mortar layers or that the sensitive volume during the measurements was at the interface between two different mortar layers having different compositions.

In order to be able to decide which treatment is more effective, simple moisture studies or diffusion measurements are not sufficient because, the effectiveness is given by the overall flux of the liquid through the mortar. This flux is a product between diffusion coefficient and concentration variation. This means, a good estimation of treatment effectiveness is made when both, diffusion and moisture content, measurements are performed. Of course these complex measurements

will increase the measurements time as well on-site as in the laboratory, but the results will be more conclusive.

6.2 Old master paintings

6.2.1 Introduction

Rembrandt Harmensz van Rijn, a 17th century painter, is considered the greatest painter of the Netherlands and generally one of the greatest painters who ever lived (Gombrich 08). He is best known for his enormous number of self-portraits and for biblical illustrations. With the help of his self-portraits it is easy to make an autobiography. His sincere way of painting himself left us the story of his life, from his early years when he was a successful and wealthy master until his old age when he was a respected but lonely artist. Rembrandt made over his lifetime over 40 self-portraits, which is an astonishing number in art society generally. Rembrandt was also known for his etching technique. It has the same principle as printing letters, but here he used copper plates, which he then covered with wax and drew upon with a needle. The copper plate was placed in acid and, where the needle removed the wax, the acid attacked the copper. In this way only the printing of the drawing from the copper plate was needed (Gombrich 08).

6.2.2 Studied paintings

Within a collaboration with the Wallraf-Richartz-Museum & Fondation Corboud in Cologne, we had the opportunity to analyze the Rembrandt self-portrait as Zeuxis that belongs to this museum. The measurements were made in the museum, under the observation of the conservators. It is one of the last self-portraits of the great artist, showing him as an old man. The aim of the study was to analyze the stratigraphy of the painting and the NMR properties of the measured points, which is an oil painting on canvas. The conservators from the museum made a detailed evaluation of the painting and noticed that the painting suffered during his formation modifications (Schaefer 12). Another challenge in this project was to determine the age of painting and to confirm the results of the conservators

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with the help of the NMR measurements. Complementary NMR measurements were performed in order to enhance our age dating technique on another master painting dating from the 14th and 15th century made by an anonymous painter originally from Cologne.

The NMR measurements were executed with a Profile NMR-MOUSE[®] with 5.4 mm depth access at a proton resonance frequency of 17.1 MHz. The measurements of both paintings were recorded with a 2 mm spacer between the sensor and the rf coil, this increased the sensitivity at the expense of the depth access being reduced to 3.4 mm. The sensor was mounted on a computer-driven positioning device. The NMR sensor and the positioning device were manufactured by ACT GmbH, Aachen, Germany. The spectrometer hardware was a Minispec spectrometer manufactured by Bruker Biospin (Rheinstetten, Germany). The measurement parameters of all experiments are summarized in Table 6.2. Complementary to

Table 6.2: Acquisition parameters used in the NMR depth profile experiments for the measured points

Parameters	Rembrandt	Anonymous painter
Pulse width [μ s]	5	5
Echo time [μ s]	36	46
Number of echoes	16/32	50
Number of scans	256	256
Time/profile [min]	20	50

the NMR measurements, infrared reflectography (IRR), X-ray, UV-fluorescence and other invasive measurement results are documented.

6.2.2.1 Rembrandt

As mentioned before, this painting is one of the last Rembrandt's self-portraits. It has been part of the Wallraf-Richartz-Museum & Fondation Corboud Museum since 1936. A stir is caused by the fact that the painting presents numerous deviations between the incident light photography and the X-ray photography (Fig. 6.9). In the X-ray photography, Rembrandt looks more sad, the eyebrows and forehead are painted differently, he has longer hair, and the person to the left

is missing. The question is, where the changes made by him, or after his death by his students. It is also possible that he assigned the changes during his lifetime to a student, fact that explains the high number of self-portraits existing worldwide, but not attributed to him. Invasive techniques, like cross-sections from the right



Figure 6.9: Rembrandt self-portrait. a) Incident light photography, where the six points measured by NMR are pointed out. b) X-ray photography, where the differences are clearly evidenced (Pilz 07).

area of the background, evidence a very complex structure of up to seven layers of paint over the quartz-containing primer. In the X-ray photograph the mouth is closed. The current visible opening of the mouth is based on red and brown color jobs that diffusely describe a form of a smile (Pilz 07).

NMR depth profiles were recorded in the 6 different positions marked in figure 6.9a. The results of the depth profiles, in which the mean of the second and third echoes were taken into account are shown in Fig. 6.10a. In Fig. 6.10b the T_2 values over the depth are represented for all the measured spots. The depth profiles evidence differences in the stratigraphy as well as in paint thickness. The depth profile from the forehead (point 3) is the thinnest confirming that this paint layer is the thinnest. In contrast, the depth profile of the scarf (point 6) is the widest one, meaning that this paint layer is the thickest. The upper lip (point 2) and the forehead (point 3) profiles are the lowest ones in amplitude intensity in

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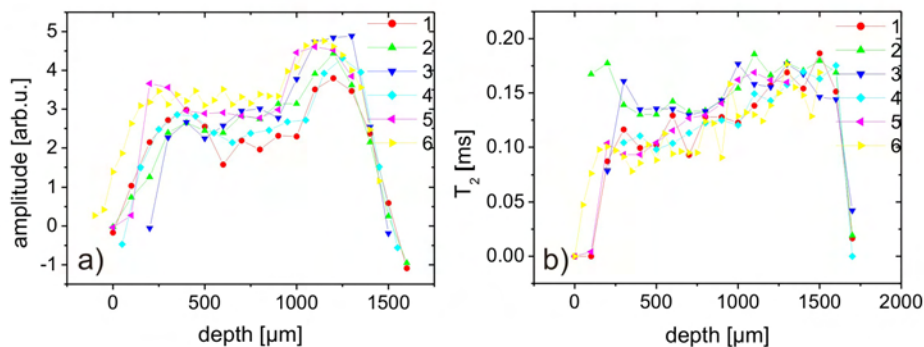


Figure 6.10: NMR depth profiles and T_2 values over the depth for all the measured points a) Depth profiles evidencing the paintings stratigraphy. Three main layers can be observed, the new canvas over a range of about 500 μm , the old canvas being around 600 μm thick and the paint layer at the surface 200 to 400 μm thick. b) T_2 depth profiles for each measured point. The T_2 values corresponding to the new canvas are slightly higher than that of the old canvas. Differences in the paint layer T_2 values are noticeable.

the region of the paint layer. The paint layer corresponding to the mouth center (measurement point 1) is thicker and more intense than the one corresponding to the upper lip (point 2).

The T_2 depth profiles underline the stratigraphy observed in the depth profiles: the T_2 values corresponding to the new canvas are higher than the values for the old canvas, and even differences in the T_2 values corresponding to the paint layers in different measured points are evidenced. The T_2 values in the paint layers corresponding to the upper lip and forehead are much higher than the other paint layers. For a better understanding and visibility, each of the depth profiles was plotted separately together with the corresponding T_2 depth profile in Fig 6.11. The amplitude corresponding to the paint layer is highest for the scarf measured point (Fig. 6.11f) and the background on the left side (Fig. 6.11e). This suggests that these paint layers are newer than the other ones, observation underlined by the X-ray photography, in which no other person except Rembrandt is visible on the painting, and the conservators' report, in which they remark that the paint layer in the region of the scarf is applied differently. Moreover, the depth profile

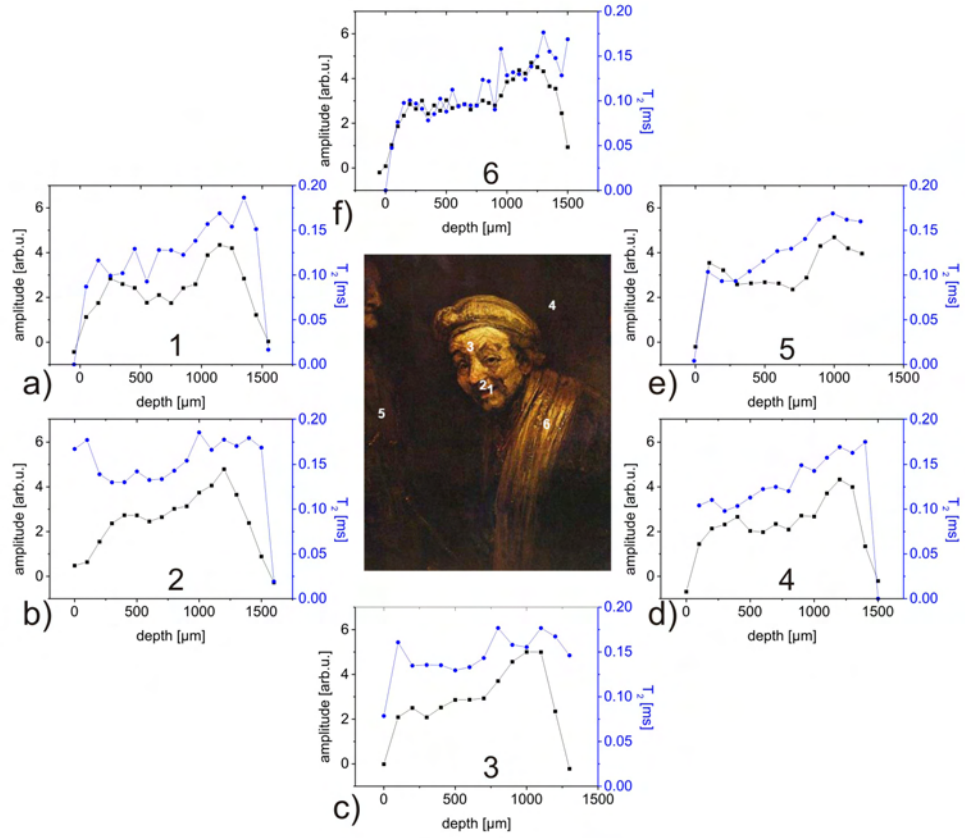


Figure 6.11: NMR depth profiles (black) and T_2 depth profiles (blue) for each of the measured points. Comparison between the depth profiles and T_2 profiles for the a) mouth, b) upper lip, c) forehead, d) dark background upper right, e) dark background middle left, and f) scarf.

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for the scarf is the broadest, revealing a thick paint layer. The higher T_2 values for the upper lip and forehead paint layer is a hint for the younger age of the paint layer in that region: the drier the paint layer, the lower the T_2 value. Of course, the T_2 value is influenced also by the composition of the paint. It may contain some paramagnetic elements which will lower the T_2 value. For example, the signal amplitude of the scarf paint layer is high compared with the other ones, but the T_2 values are low.

Along with the depth profiles and T_2 experiments, T_1 saturation recovery experiments were performed at the paint layer depth for each point. These experiments, together with the CPMG experiments, were made to investigate the aging of the binder. Similar experiments were reported in (Presciutti 08), where naturally aged old master paintings were compared to artificially aged tempera paints (Fig. 6.12a). The authors reported much lower relaxation times for natural aged master paintings. The T_1 relaxation times were between 3 and 10 ms,

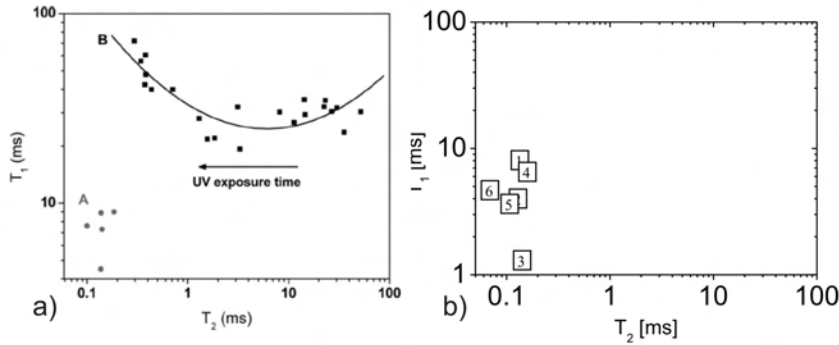


Figure 6.12: T_1 values compared with T_2 values for oil paintings a) Comparison for T_1 and T_2 values for naturally aged master paintings (point A) and artificially aged oil paintings (Presciutti 08). b) Comparison for T_1 and T_2 values for each measured point (1 to 6) for the Rembrandt self-portrait.

and the T_2 relaxation times were between 0.08 and 0.3 ms. In case of the Rembrandt self-portrait measurements the values for both relaxation times are in the same region. A challenge was to learn if it is possible to determine the age of a painting only by a T_1 NMR measurement. Several measurements on different old master paintings were already performed in our research group, and from these

measurements a T_1 contrast aging map was established (Fig.6.13). To obtain

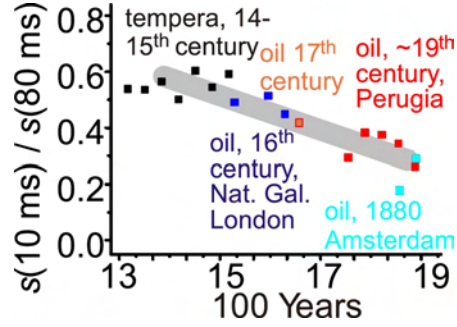


Figure 6.13: Natural aging by T_1 weighted data. Different oil and tempera paintings, measured over the years in our working group, age dated by the T_1 weighted data where the normalized signal at 10 ms is divided by the signal at 80 ms recorded in a T_1 saturation recovery experiment.

these T_1 weighted data, the signal from the saturation recovery measurement is first normalized, then the signal at 10 ms is divided by the signal at 80 ms. The results of the aging values for the Rembrandt are shown in Table 6.3. The va-

Table 6.3: T_1 weighted data for each measured position

Position	$S(10)/S(80)$	Position	$S(10)/S(80)$
1	0.469	4	0.38
2	0.6	5	0.68
3	0.67	6	0.54

lues for the T_1 weighted data presented in Table 6.3 raises some questions. If we compare the values for the T_1 weighted data in Fig. 6.13 and knowing that the Rembrandt self-portrait was made around 1668, the expected values should be between 0.37 and 0.5. But in the table only two of the measured points are in that range, point 1 and 4. One of the reasons why this is so could be the poor signal to noise ratio, due to lack of measurement time. Another reason could be the chemical composition of the paint: it may contain some compounds that are influencing the measurements. A third reason could be the fact that the several paint layers were not parallel and that our T_1 measured signal had a component from a different paint layer and this could influence the T_1 weighted data values.

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Another reason could be the fitting of the data. T_1 data was fitted biexponentially for the Rembrandt points, but all other data from the aging map was obtained by mono exponential fits of the T_1 data. The second component could come from the organic solvent vapor to which once the painting had been exposed. For the available data at least position 1 and 4 agree in this T_1 data with the aging curve.

6.2.2.2 Anonymous Cologne painting

The art of the 13th and 14th century was strong under the influenced of religion (Gombrich 08). In this period the big cathedrals were built all over Europe and the paintings illustrate scenes from the Bible. The painting we examined comes from Cologne region by an anonymous artist and is dated on 1330. It shows in the upper part Saint Elisabeth donating clothing to poor people and in the lower part nursing a sick man.

In order to test the aging curve, we performed depth profiles and T_1 measurements on this oil painting on wood. We chose this painting because it has fewer and flatter paint layers than the Rembrandt self-portrait, so the paint layers can be resolved much better with depth profiling. Another important reason is the difference of about 100 years between the first version and the second version of the painting (Fig. 6.14). In this way we can determine the age of the two different versions and fill the aging curve with another masterpiece.

The depth profiles recorded at the museum site are outlined in Figure 6.15, and the parameters used for the measurements are in Table 6.2. Because the measurements took place in the museum's laboratory (unlike the Rembrandt measurements, which were done in the exhibition room), the measurement time for one profile is almost three times longer. Depth profiles were recorded at the forehead of Jesus and the left part and right part of Saint Elisabeth's face. Both sides of St. Elisabeth's face were measured because a newer conservation treatment, mentioned in the archives, was applied on the right side of the face. Similarities between the stratigraphy of the measured points are noticeable and several layers are evidenced. The most intense signal amplitude is attributed to the wood on which the oil painting lies; this appears between 1500 and 2000 μm . The canvas produces a NMR signal between 1300 and 1500 μm . On the canvas, a



Figure 6.14: Anonymous Cologne painting a) Incident light photography showing Saint Elisabeth dressing poor people (top) and caring for sick people (bottom). b) Mapping of the first version and the second version of the painting, with a difference of 100 years between them.

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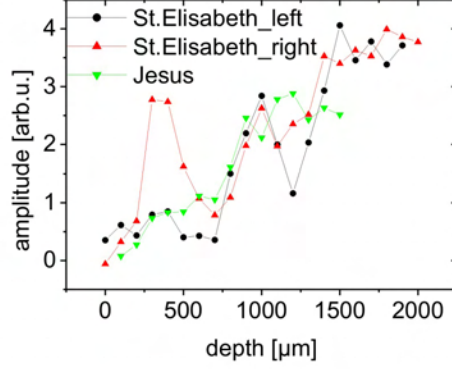


Figure 6.15: Measured depth profiles of Jesus' face and left and right of Saint Elisabeth face. The face of Jesus belongs to the first version of the painting and the face of Saint Elisabeth to the second version. Differences in stratigraphy and signal intensity are observed.

300 μm thick ground layer is applied. The region of interest is the 400 μm thick paint layer, then the varnish covers the whole surface of the painting. The signal intensity in the paint layer does not really differ from one profile to the next, which is understandable because the same paint color was applied. In order to resolve the difference in age, T_1 measurements were performed at the paint layer depth. Again, because we had much more time available to do our measurements, T_1 measurements were made at several depths. T_1 weighted data were obtained in the same way as for the Rembrandt measurements. The results, presented in Table 6.4, show clearly that different paint layers from the first version and the second version are from different ages. The T_1 weighted data for St. Elisabeth are around 0.58, corresponding to 14th century master paintings, and the T_1 weighted data for Jesus are around 0.7, corresponding to the 13th century. Of course, one of the T_1 weighted data values is much too high, but this is one reason why this project is still ongoing, to have a more accurate aging map.

Recent studies revealed that the positioning device present a hysteresis curve in its back and forth movement when it is not in the vertical position. This may induce an uncertainty in the results and in the aging map. It might be that the T_1 experiments were not performed exactly in the paint layer, that the sensitive slice

Table 6.4: T_1 weighted data for the paint layer depth for each measured profile.

Depth [μm]	St. Elisabeth left part	St. Elisabeth right part	Jesus
800			0.68578
900	0.58064		
1000		0.58779	0.74768
1100		0.55345	
1200	0.5881		

was in the paint layer part but also in the canvas layer and the signal recorded is from the spins in the paint layer as well as in the canvas. This could induce errors in the T_1 weighted data and make the paint layer appear to be older than it really is. However, the results obtained for the T_1 weighted data in the mouth region are in the correct age time. Moreover, with the results obtained on the other oil painting a better accuracy, lower than 100 years, for the T_1 weighted data and the aging map was obtained. These two facts lead us to the conclusion that the corrections made on the painting, near the mouth region, are less than 100 years away from the paintings original version.

6.3 Conclusions

A non-invasive investigation of ancient wall paintings and mosaics was performed using the NMR-MOUSE[®]. For the measurements at the Mosaic of Neptune and Amphitrite, differences between the tesserae and the mortar bed could be evidenced and strong indications were found for an impregnation of the tesserae with an organic substance. Different conservation treatments and their influence on the moisture content in wall paintings in the Black Room were evidenced and the efficacy of each of the conservation treatments was estimated. The study shows that the conservation treatment with ammonium oxalate leads to lower moisture content of the wall, while the treatment with barium hydroxide leads to higher moisture content.

For the first time, depth profiles of untreated, recently excavated wall paintings

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were recorded in the Villa of the Papyri, which identified two different building techniques. The results of all of these measurements are expected to help identify suitable conservation strategies and to detect undocumented restoration efforts of the past. These studies demonstrate that NMR is a suitable tool to analyze and characterize the layer structures of different wall paintings and mosaics. Measurements taken in the same point but at different time periods demonstrate the accuracy and reproducibility of the measurements. Moreover, in situ XRF studies coupled with reflectance FTIR spectra are in agreement with the NMR-MOUSE data and complemented the studies of the wall surfaces. It is shown that the NMR-MOUSE measurements coupled with in situ FTIR and XRF studies make a powerful, non-destructive methodology to access the conservation treatments and the techniques of wall paintings.

Further laboratory studies, to elucidate the mobility of the proton spins in terms of diffusion measurements, were in good agreement with the on-site measurement results. It was shown that treatment with ammonium oxalate leads to lower mobility of the proton spins inside the mortar, while treatment with barium hydroxide leads to higher mobility of the proton spins.

For the old master paintings, both of them being oil paintings, the stratigraphy could be elucidated. T_1 weighted data for the Rembrandt and the anonymous Cologne master paintings revealed the age of the painting and the aging map was filled with new data points. This shows that different stages of oil aging, or different aging conditions, can be discriminated by the NMR method. This indicates that NMR has the potential to differentiate between old and recently restored sections in a painting, thus raising the hope of identifying genuine and forged paintings.

Chapter 7

Conclusions

In this thesis, mobile, low-field NMR techniques were used to study porous system, from simple to complex structures, and objects of cultural heritage. It was shown that NMR relaxometry can be used to study the fluid dynamics inside a porous system. A simple theoretical model for multi-site relaxation exchange NMR permitted us to extract exchange kinetic parameters when applied on a model porous systems. It provided a first step towards the study of more complex systems, where continuous relaxation distributions are present, such as soil systems or building materials. The single-sided NMR sensor proved to be a useful tool for on-site measurements. This is very important also in the case of the cultural heritage objects, as most of the objects cannot be moved out of their environment.

At the beginning of this thesis, a general analytical solution for multi-site exchange NMR was established, with a focus on a simple two-site exchange problem. In order to understand the simple systems, experimental results were compared to results obtained from simulations for a simple two-site system made of water saturated silica particles. T_1 - T_2 and T_2 - T_2 exchange correlation maps were compared. Moreover, the integrals of each relaxation distributions and their cross peaks were compared for different mixing times. The overlapping of values obtained by simulation and experiments is remarkable. Complementary experiments were made at different temperatures for the same system and compared to simulations. The activation energy was extracted from the dependence of the exchange rate on temperature. These 2D exchange experiments were tested further on a

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more complex system, a water saturated aluminium oxide system, which showed a more complex relaxation-site distribution (as predicted in the simulations). The restricted diffusion of the water molecules inside this porous system was evidenced by a D - T_2 correlation map.

The efficiency of these 2D correlation experiments was proved furthermore on natural porous systems, on saturated and unsaturated soil systems. The results obtained for all three soil types proved that 1D and 2D NMR relaxometry methods are suitable to observe moisture migration in these types of porous media. Depending on the soil texture the pore size distribution varies from a simple relaxation distribution, obtained for sand, to multi-modal distributions, obtained for the two different silt loams. Water exchange is observable mostly within the small pores. Exchange within the big pores may occur, but the T_1 effect from the small pores is bigger and the small components vanish at longer mixing times. Once the saturation level becomes lower the water from the bigger pores vanishes and the T_2 values are shifting towards shorter values. All this results give us an insight about the pore connectivity and texture.

In case of the complex concrete samples, two different concrete types with different composition were examined on-site as well as in the laboratory. Because of their different composition, especially the aggregate composition, the water ingress in the saturation process did behave totally different. The use of the NMR-MOUSE showed to be appropriate, especially for the on-site measurements, where the measurements could be performed fast and non-invasive. In case of the tunnel measurements differences in profile amplitudes and T_2 values were observed, evidencing the influence of the environment onto concrete quality and lifetime. At the lock site, these differences were already expected, as one part of the concrete wall is exposed more often to water. The laboratory measurements proved to be efficient and informative. Not only the differences in water uptake rate could be evidenced, also the results could be compared with the MRE results and conclusions could be drawn. It is impressive how a difference in pore size, which was not even one order of magnitude, can influence so hardly the water dynamics in the concrete blocks.

Regarding the measurements on the cultural heritage objects, a non-invasive investigation of ancient wall paintings and mosaics was performed using the NMR-

MOUSE[®] in Herculaneum, Italy. For the measurements at the Mosaic of Neptune and Amphitrite, differences between the tesserae and the mortar bed could be evidenced. Different conservation treatments and their influence on the moisture content in wall paintings in the Black Room were evidenced and the efficacy of each of the conservation treatments was estimated. For the first time, depth profiles of untreated, recently excavated wall paintings were recorded in the Villa of the Papyri, which identified two different building techniques. Furthermore, laboratory studies were performed, to elucidate the mobility of the proton spins in terms of diffusion measurements in mock-up samples with different surface conservation treatments. In case of the old master paintings, the stratigraphy could be elucidated. T_1 weighted data for the Rembrandt and the anonymous Cologne master paintings could be used to make the painting dating method more accurate and the aging map was filled with new data points.

The results presented in this thesis, demonstrate that low-field NMR is a suitable tool to analyze different porous systems and characterize the layer structures of different cultural heritage objects. As a next step the numerical simulations could be expanded to study systems with continuous relaxation distributions and compared, for example, with the results that we obtained on the soils systems. A more detailed study of the soils systems, where clay and silt would be added gradually to sand, would help to understand how this components are influencing pore connectivity and texture. Complementary 2D correlation experiments, like D - T_2 correlation experiments, would lead to a better understanding of water transport between pores. This study could be applied also in the case of the concrete block. With the help of the on-site measurements the seasonal influence on the concrete could be monitored and compared with the MRE results. As for the cultural heritage objects, on-site diffusion and 2D correlation measurements would lead to a better understanding on moisture transport inside the wall painting.

7. CONCLUSIONS

References

- [Abragam 61] A. Abragam. The principles of nuclear magnetism. Clarendon Press, Oxford, UK, 1961. 5
- [Arns 07] C.H. Arns, K.E. Washburn & P.T. Callaghan. *Multidimensional NMR inverse Laplace spectroscopy in petrophysics*. Petrophysics, vol. 48, pages 380–392, 2007. 44
- [Atkins 02] P. Atkins & A. Hopfner. Physikalische chemie. Wiley-VCH, 2002. 35
- [Bain 03] A.D. Bain. *Chemical Exchange in NMR*. Progress in Nuclear Magnetic Resonance Spectroscopy, vol. 43, no. 16, pages 63–103, 2003. 18
- [Barrie 00] P.J. Barrie. *Characterization of porous media using NMR methods*. Annual Reports on NMR Spectroscopy, vol. 41, pages 265–318, 2000. 7, 17, 43, 47
- [Bearat 96] A. Bearat, M. Fuchs, M. Maggetti & D. Paunier. *Roman wall painting. materials, techniques, analysis and conservation*. In Proceedings of the International Workshop Fribourg, 7-9 March 1996, pages 35–62. Institute of Mineralogy and Petrography: Fribourg, 1996. 90
- [Blümich 03] B. Blümich, S. Anferova, S. Sharma, A.L. Segre & C. Federici. *Degradation of historical paper: nondestructive analysis by the NMR-MOUSE*. Journal of Magnetic Resonance, vol. 161, no. 2, pages 204 – 209, 2003. 81

REFERENCES

- [Blümich 05a] B. Blümich. *Essential nmr*. Springer, Berlin, 2005. 1, 5, 12, 52
- [Blümich 05b] B. Blümich, F. Casanova, J. Perlo, S. Anferova, V. Anferov, K. Kremer, N. Goga, K. Kupferschläger & M. Adams. *Advances of unilateral mobile NMR in nondestructive materials testing*. Magnetic Resonance Imaging, vol. 23, no. 2, pages 197 – 201, 2005. 81
- [Blümich 08] B. Blümich, J. Perlo & F. Casanova. *Mobile single-sided NMR*. Progress in Nuclear Magnetic Resonance Spectroscopy, vol. 52, no. 4, pages 197–269, MAY 2008. 8, 81
- [Blümich 09] B. Blümich, F. Casanova & S. Appelt. *NMR at low magnetic fields*. Chemical Physics Letters, vol. 477, no. 4-6, pages 231 – 240, 2009. 10
- [Blümich 10a] B. Blümich, F. Casanova, J. Perlo, F. Presciutti, C. Anselmi & B. Doherty. *Noninvasive testing of art and cultural heritage by mobile NMR*. Accounts of Chemical Research, vol. 43, no. 6, SI, pages 761–770, JUN 2010. 81
- [Blümich 10b] B. Blümich, A. Haber, F. Casanova, E. Del Federico, V. Boardman, G. Wahl, A. Stilliano & L. Isolani. *Noninvasive depth profiling of walls by portable nuclear magnetic resonance*. Analytical and Bioanalytical Chemistry, vol. 397, pages 3117–3125, 2010. 81
- [Blümich 11] B. Blümich, F. Casanova, M. Dabrowski, E. Danieli, L. Evertz, A. Haber, M. Van Landeghem, S. Haber-Pohlmeier, A. Olaru, J. Perlo & O. Sucre. *Small-scale instrumentation for nuclear magnetic resonance of porous media*. New Journal of Physics, vol. 13, pages 1–15, JAN 2011. 8

REFERENCES

- [Borgia 98] G.C. Borgia, R.J.S. Brown & P. Fantazzini. *Uniform-Penalty Inversion of Multiexponential Decay Data*. Journal of Magnetic Resonance, vol. 132, no. 1, pages 65 – 77, 1998. 24
- [Brameshuber 08] W. Brameshuber, F. Spörel & A. Rahimi. *Frost-exposure of concrete structures - determination of the exposure in dependence of the actual degree of water saturation*. Rapport technique Nr. F 965, Institute for Building Research, RWTH Aachen University, 2008. 66
- [Brownstein 77] K.R. Brownstein & C.E. Tarr. *Spin-lattice relaxation in a system governed by diffusion*. Journal of Magnetic Resonance, vol. 26, no. 1, pages 17–24, 1977. 7
- [Brownstein 79] K.R. Brownstein & C.E. Tarr. *Diffusional theory for proton spin-lattice relaxation in biological cells*. Bulletin of the American Physical Society, vol. 24, no. 3, page 346, 1979. 7
- [Burnett 80] L.J. Burnett & J.A. Jackson. *Remote (inside-out) NMR. 2. Sensitivity of NMR detection for external samples*. Journal of Magnetic Resonance, vol. 41, no. 3, pages 406–410, 1980. 5
- [Carr 54] H.Y. Carr & E.M. Purcell. *Effects of diffusion on free precession in nuclear magnetic resonance experiments*. Phys. Rev., vol. 94, pages 630–638, May 1954. 10
- [Casanova 11] F. Casanova, J. Perlo & B. Blümich. *Single-sided nmr*. Springer Berlin / Heidelberg, 2011. 1, 7, 17, 24
- [Casieri 04a] C. Casieri, S. Bubici, I. Viola & F. De Luca. *A low-resolution non-invasive NMR characterization of ancient paper*. Solid State Nuclear Magnetic Resonance, vol. 26, no. 2, pages 65 – 73, 2004. 81

REFERENCES

- [Casieri 04b] C. Casieri, L. Senni, M. Romagnoli, U. Santamaria & F. De Luca. *Determination of moisture fraction in wood by mobile NMR device*. Journal of Magnetic Resonance, vol. 171, no. 2, pages 364 – 372, 2004. 81
- [Cooper 80] R.K. Cooper & J.A. Jackson. *Remote (inside-out) NMR .1. Remote production of a region of homogeneous magnetic-field*. Journal of Magnetic Resonance, vol. 41, no. 3, pages 400–405, 1980. 5
- [Donner 69] O. Donner. Die erhaltenen antiken wandmalereien in technischer beziehung. Breitkopf und Härtel, Leipzig, 1869. 90
- [Dunn 02] K.J. Dunn, D.J. Bergmann & G.A. Latorraca. Nuclear magnetic resonance: Petrophysical and logging applications. Elsevier Science, Amsterdam, 2002. 43
- [Eidmann 96] G. Eidmann, R. Savelsberg, P. Blümmler & B. Blümich. *The NMR MOUSE, a Mobile Universal Surface Explorer*. Journal of Magnetic Resonance, Series A, vol. 122, no. 1, pages 104 – 109, 1996. 8
- [Ernst 87] R.R. Ernst, G. Bodenhausen & A. Wokaun. Principles of nuclear magnetic resonance in one and two dimensions. Clarendon Press, Oxford, 1987. 5
- [Federico 10] E. Del Federico, S. Centeno, C. Kehlet, P. Currier, D. Stockman & A. Jerschow. *Unilateral NMR applied to the conservation of works of art*. Analytical and Bioanalytical Chemistry, vol. 396, pages 213–220, 2010. 81
- [Fukushima 81] E. Fukushima & S.B.W. Roeder. Experimental pulse nmr. a nuts and bolts approach. Addison-Wesley Publishing Company, Inc, 1981. 30

REFERENCES

- [Garboczi 96] E.J. Garboczi & D.P. Bentz. *Multi-scale picture of concrete and its transport properties: Introduction for non-cement researchers*. Rapport technique, National Institute of Standards and Technology, 1996. 65
- [Glasel 74] J.A. Glasel & K.H. Lee. *Interpretation of water nuclear magnetic resonance relaxation times in heterogeneous systems*. Journal of the American Chemical Society, vol. 96, no. 4, pages 970–978, 1974. 48
- [Godefroy 03] S. Godefroy & P.T. Callaghan. *2D relaxation/diffusion correlations in porous media*. Magnetic Resonance Imaging, vol. 21, no. 3-4, pages 381 – 383, 2003. 24, 44, 45
- [Gombrich 08] E. H. Gombrich. The story of art. Phaidon Press Limited, 2008. 99, 106
- [Green 03] D.L. Green, S. Jayasundara, Y.-F. Lam & M.T. Harris. *Chemical reaction kinetics leading to the first Stober silica nanoparticles -NMR and SAXS investigation*. Journal of Non-Crystalline Solids, vol. 315, pages 166–179, 2003. 27
- [Haber 11] A. Haber, B. Blümich, D. Souvorova & E. Del Federico. *Ancient Roman wall paintings mapped nondestructively by portable NMR*. Analytical and Bioanalytical Chemistry, vol. 401, no. 4, pages 1441–1452, SEP 2011. 84, 90, 92
- [Halperin 94] W.P. Halperin, J.-Y. Jehng & Y.-Q. Song. *Application of spin-spin relaxation to measurement of surface area and pore size distributions in a hydrating cement paste*. Magnetic Resonance Imaging, vol. 12, no. 2, pages 169 – 173, 1994. 74
- [Hinedi 97] Z.R. Hinedi, A.C. Chang, M.A. Anderson & D.B. Borchardt. *Quantification of microporosity by nuclear magnetic resonance relaxation of water imbibed in porous media*. Water Resour. Res., vol. 33, pages 2697–2704, 1997. 43

REFERENCES

- [Holleman 95] A.F. Holleman & E. Wiberg. Lehrbuch der anorganischen chemie. W. de Gruyter Berlin, 1995. 35
- [Hürlimann 98] M.D. Hürlimann. *Effective gradients in porous media due to susceptibility differences*. Journal of Magnetic Resonance, vol. 131, no. 2, pages 232 – 240, 1998. 48
- [Hürlimann 02] M.D. Hürlimann, L. Venkataramanan & C. Flaum. *The diffusion-spin relaxation time distribution function as an experimental probe to characterize fluid mixtures in porous media*. Journal of Chemical Physics, vol. 117, no. 22, pages 10223–10232, 2002. 18, 44
- [Jackson 80] J.A. Jackson, L.J. Burnett & J.F. Harmon. *Remote (inside-out) NMR. 3. Detection of nuclear magnetic-resonance in a remotely produced region of homogeneous magnetic-field*. Journal of Magnetic Resonance, vol. 41, no. 3, pages 411–421, 1980. 1, 5
- [Kleinberg 90] R.L Kleinberg & M.A Horsfield. *Transverse relaxation processes in porous sedimentary rock*. Journal of Magnetic Resonance, vol. 88, no. 1, pages 9 – 19, 1990. 48
- [Kleinberg 05] R.L. Kleinberg & D.D. Griffin. *NMR measurements of permafrost: unfrozen water assay, pore-scale distribution of ice, and hydraulic permeability of sediments*. Cold Regions Science and Technology, vol. 42, no. 1, pages 63 – 77, 2005. 43
- [Kroeker 86] R.M. Kroeker & R.M. Henkelman. *Analysis of biological NMR relaxation data with continuous distributions of relaxation times*. Journal of Magnetic Resonance, vol. 69, no. 2, pages 218 – 235, 1986. 18
- [Laidler 65] K.J. Laidler. Chemical kinetics. McGraw-Hill, New-York, 1965. 35

REFERENCES

- [Landeghem 10] M. Van Landeghem, A. Haber, J.-B. d’Espinose de Lacaillerie & B. Blümich. *Analysis of multisite 2D relaxation exchange NMR*. Concepts in Magnetic Resonance Part A, vol. 36A, no. 3, pages 153–169, MAY 2010. 19, 25, 26, 27, 31, 33, 44, 55
- [Lawson 74] C.L. Lawson & R.J. Hanson. Solving least squares problems. Prentice-Hall, 1974. 24
- [Lee 93] J.H. Lee, C. Labadie, C.S. Springer & G.S. Harbison. *Two-dimensional inverse Laplace transform NMR: Altered relaxation times allow detection of exchange correlation*. J. Am. Chem. Soc., vol. 115, pages 7761–7764, 1993. 18, 44
- [Levitt 08] M.H. Levitt. Spin dynamics. basics of nuclear magnetic resonance. John Wiley & Sons, Ltd, 2008. 6, 30
- [Matteini 87] M. Matteini. The conservation of wall paintings. Getty Conservation Institute, California, 1987. 83
- [Matteini 99] M. Matteini. *The mineral approach to the conservation of mural paintings: Barium hydroxide and artificial oxalates*. In Wall Painting Conference: Conserving the Painted Past: developing approaches to wall painting conservation, pages 110–115. English Heritage, London, 1999. 83
- [McDonald 05] P.J. McDonald, J.-P. Korb, J. Mitchell L. & Monteilhet. *Surface relaxation and chemical exchange in hydrating cement pastes: A two-dimensional NMR relaxation study*. Phys. Rev. E, vol. 72, page 011409, Jul 2005. 18, 23, 44
- [Meiboom 58] S. Meiboom & D. Gill. *Modified spin-echo method for measuring nuclear relaxation times*. Review of Scientific Instruments, vol. 29, no. 8, pages 688–691, aug 1958. 10

REFERENCES

- [Miliani 07a] C. Miliani, F. Rosi, I. Borgia, P. Benedetti, B. G. Brunetti & A. Sgamellotti. *Fiber-optic Fourier transform mid-Infrared reflectance spectroscopy: a suitable technique for in situ studies of mural paintings*. Appl. Spectrosc., vol. 61, no. 3, pages 293–299, Mar 2007. 84
- [Miliani 07b] C. Miliani, F. Rosi, A. Burnstock, B. G. Brunetti & A. Sgamellotti. *Non-invasive in-situ investigations versus micro-sampling: a comparative study on a Renoirs painting*. Applied Physics A: Materials Science & Processing, vol. 89, pages 849–856, 2007. 84
- [Mitchell 07] J. Mitchell, J.D. Griffith, J.H.P. Collins, A.J. Sederman, L.F. Gladden & M.L. Johns. *Validation of NMR relaxation exchange time measurements in porous media*. Journal of Chemical Physics, vol. 127, no. 23, page 234701, 2007. 18
- [Mohnke 10] O. Mohnke & N. Klitzsch. *Microscale simulations of NMR relaxation in porous media considering internal field gradients*. Vandose Zone Journal, vol. 9, pages 846–857, 2010. 55
- [Monteilhet 06] L. Monteilhet, J.-P. Korb, J. Mitchell & P.J. McDonald. *Observation of exchange of micropore water in cement pastes by two-dimensional T2-T2 nuclear magnetic resonance relaxometry*. Phys. Rev. E, vol. 74, page 061404, Dec 2006. 23
- [Moore 86] W.J. Moore, D.O. Hummel, G. Trafara & K. Holland-Moritz. Physikalische chemie. W. de Gruyter Berlin, 1986. 35
- [Ngala 97] V.T. Ngala & C.L. Page. *Effects of carbonation on pore structure and diffusional properties of hydrated cement pastes*. Cement and Concrete Research, vol. 27, no. 7, pages 995 – 1007, 1997. 70

REFERENCES

- [Paciok 11] E. Paciok & B. Blümich. *Ultrafast Microscopy of Microfluidics: Compressed Sensing and Remote Detection*. *Ange wandte Chemie International Edition*, vol. 50, no. 23, pages 5258–5260, 2011. 17
- [Perlo 05] J. Perlo, F. Casanova & B. Blümich. *Profiles with microscopic resolution by single-sided NMR*. *Journal of Magnetic Resonance*, vol. 176, no. 1, pages 64 – 70, 2005. 8
- [Pilz 07] K. Pilz, C. von Saint-George & I. Schaefer. *Rembrandt Selbstbildnis untersuchungsbericht*. Rapport technique, Wallraf-Richartz-Museum and Fondation Corboud, 2007. 101
- [Pohlmeier 09] A. Pohlmeier, S. Haber-Pohlmeier & S. Stapf. *A fast field cycling nuclear magnetic resonance relaxometry study of natural soils*. *Vadose Zone Journal*, vol. 8, no. 3, pages 735–742, 2009. 43, 53, 55
- [Pollio 99] M. Vitruvius Pollio & I.D. Rowland. *Ten books on architecture*, volume 38. Cambridge Univ. Press, 1999. 90, 93
- [Presciutti 08] F. Presciutti, J. Perlo, F. Casanova, S. Glögler, C. Miliani, B. Blümich, B.G. Brunetti & A. Sgamellotti. *Noninvasive nuclear magnetic resonance profiling of painting layers*. *Appl. Phys. Lett.*, vol. 93, no. 3, page 033505, 2008. 104
- [Procacii 69] U. Procacii. *Frescoes from florence*. Arts Council, London, 1969. 82
- [Proietti 04] N. Proietti, D. Capitani, E. Pedemonte, B. Blümich & A.L. Segre. *Monitoring degradation in paper: non-invasive analysis by unilateral NMR. Part II*. *Journal of Magnetic Resonance*, vol. 170, no. 1, pages 113 – 120, 2004. 81

REFERENCES

- [Proietti 05] N. Proietti, D. Capitani, R. Lamanna, F. Presciutti, E. Rossi & A.L. Segre. *Fresco paintings studied by unilateral NMR*. Journal of Magnetic Resonance, vol. 177, no. 1, pages 111 – 117, 2005. 81
- [Proietti 06] N. Proietti, D. Capitani, S. Cozzolino, M. Valentini, E. Pedemonte, E. Princi, S. Vicini & A.L. Segre. *In situ and frontal polymerization for the consolidation of porous stones: A unilateral NMR and magnetic resonance imaging study*. The Journal of Physical Chemistry B, vol. 110, no. 47, pages 23719–23728, 2006. 81
- [Proietti 07] N. Proietti, D. Capitani, E. Rossi, S. Cozzolino & A.L. Segre. *Unilateral NMR study of a XVI century wall painted*. Journal of Magnetic Resonance, vol. 186, no. 2, pages 311 – 318, 2007. 81
- [Project 07] Herculaneum Conservation Project. *Special edition on Herculaneum*. Conservation and management of archaeological sites., vol. 8.4, page all, 2007. 82
- [Provencher 82a] S. W. Provencher. *A constrained regularization method for inverting data represented by linear algebraic or integral equations*. Computer Physics Communications, vol. 27, no. 3, pages 213 – 227, 1982. 24
- [Provencher 82b] S.W. Provencher. *CONTIN: A general purpose constrained regularization program for inverting noisy linear algebraic and integral equations*. Computer Physics Communications, vol. 27, pages 229–242, 1982. 24
- [Raich 04] H. Raich & P. Blümli. *Design and construction of a dipolar Halbach array with a homogeneous field from identical bar magnets: NMR Mandhalas*. Concepts in Magnetic Resonance Part B: Magnetic Resonance Engineering, vol. 23B, no. 1, pages 16–25, 2004. 8, 10

REFERENCES

- [Rühli 07] F.J. Rühli, T. Böni, J. Perlo, F. Casanova, M. Baías, E. Egarter & B. Blümich. *Non-invasive spatial tissue discrimination in ancient mummies and bones in situ by portable nuclear magnetic resonance*. Journal of Cultural Heritage, vol. 8, no. 3, pages 257 – 263, 2007. 81
- [Rosi 09] F. Rosi, A. Daveri, C. Miliani, G. Verri, P. Benedetti, F. Pique, B. Brunetti & A. Sgamellotti. *Non-invasive identification of organic materials in wall paintings by fiber optic reflectance infrared spectroscopy: a statistical multivariate approach*. Analytical and Bioanalytical Chemistry, vol. 395, pages 2097–2106, 2009. 84
- [Schaefer 12] I. Schaefer, K. Pilz & C. von Saint-George. *Rembrandts Selbstbildnis als Zeuxis; Neues zum Original, zur Erhaltung und zur Frage der Restaurierung*. Journal of Art Technology and Conservation, vol. 2, pages 285–323, 2012. 99
- [Schaumann 05] G. Schaumann, E. Hobley, J. Hurraß & W. Rotard. *^1H -NMR relaxometry to monitor wetting and swelling kinetics in high-organic matter soils*. Plant and Soil, vol. 275, pages 1–20, 2005. 43
- [Sharma 03] S. Sharma, F. Casanova, W. Wache, A.L. Segre & B. Blümich. *Analysis of historical porous building materials by the NMR-MOUSE*. Magnetic Resonance Imaging, vol. 21, no. 3-4, pages 249 – 255, 2003. 81
- [Song 02a] Y.-Q. Song. *Categories of Coherence Pathways for the CPMG Sequence*. Journal of Magnetic Resonance, vol. 157, no. 1, pages 82 – 91, 2002. 12
- [Song 02b] Y.-Q. Song, L. Venkataramanan, M.D. Hürlimann, M. Flaum, P. Frulla & C. Straley. *T_1 - T_2 correlation spectra obtained using a fast two-dimensional Laplace inversion*

REFERENCES

- on. *Journal of Magnetic Resonance*, vol. 154, no. 2, pages 261 – 268, 2002. 23, 25, 44
- [Song 08] Yi-Qiao Song, H. Cho, Tim Hopper, Andrew E. Pomerantz & Phillip Zhe Sun. *Magnetic resonance in porous media: Recent progress*. *Journal of Chemical Physics*, vol. 128, no. 5, page 052212, 2008. 44
- [Song 10] Y.-Q. Song. *Recent progress of nuclear magnetic resonance applications in sandstones and carbonate rocks*. *Vandose Zone Journal*, vol. 9, pages 828–834, 2010. 23
- [Stallons 01] J.M. Stallons & E. Iglesia. *Simulation of the structure and properties of amorphous silica surfaces*. *Chemical Engineering Science*, vol. 56, pages 4205–4216, 2001. 35
- [Stingaciu 09] L.R. Stingaciu, A. Pohlmeier, P. Blümmler, L. Weihermüller, S. Stapf & H. Vereecken. *Characterization of unsaturated porous media by high-field and low-field NMR relaxometry*. *Water Resour. Res.*, vol. 45, page W08412, 2009. 43, 56
- [Stober 68] W. Stober, A. Fink & E. Bohn. *Controlled growth of monodisperse silica spheres in the micron size range*. *Journal of Colloid and Interface Science*, vol. 26, no. 1, pages 62 – 69, 1968. 27
- [Straley 97] C. Straley, D. Rossini, H. Vinegar, P. Tutunjian & C. Morris. *Core analysis by low-field NMR*. *Log Anal.*, vol. 38(2), page 84–91, 1997. 45
- [Todoruk 03] T.R. Todoruk, C.H. Langford & A. Kantzas. *Pore-scale redistribution of water during wetting of air-dried soils as studied by low-field NMR relaxometry*. *Environ. Sci. Technol.*, vol. 37, page 2707–2713, 2003. 43
- [Unesco 11] Unesco. *Cultural heritage definition*, October 2011. 81

REFERENCES

- [Venkataramanan 02] L.. Venkataramanan, Y-Q. Song & M.D. Hurlimann. *Solving Fredholm integrals of the first kind with tensor product structure in 2 and 2.5 dimensions*. IEEE Transactions on Signal Processing, vol. 50, no. 5, pages 1017 –1026, 2002. 18, 23, 24
- [Washburn 06] K.E. Washburn & P.T. Callaghan. *Tracking pore to pore exchange using relaxation exchange spectroscopy*. Phys. Rev. Lett., vol. 97, page 175502, Oct 2006. 14, 18, 44
- [Weise 10] F. Weise. *Forschungsbericht DAfStb V 463*. Rapport technique, BAM, 2010. 67, 68, 70, 76

REFERENCES

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- B. Blümich, J. Mauler, A. Haber, J. Perlo, E. Danieli, F. Casanova, Mobile NMR for geophysical analysis and materials testing, **Petroleum Science** 6, 1, 1-7, 2009.
- M. Van Landeghem, A. Haber, J-B. D'Espinose de Lacaillerie, B. Blümich, Analysis of multisite 2D relaxation exchange NMR, **Concept Magnetic Res.** 36A, 3, 153-169, 2010.
- B. Blümich, A. Haber, F. Casanova, E. Del Federico, V. Boardman, G. Wahl, A. Stelliano, L. Isolani, Noninvasive depth profiling of walls by portable nuclear magnetic resonance, **Anal. Bioanal. Chem.**, 397, 3117-3125, 2010.
- A. Haber, S. Haber-Pohlmeier, F. Casanova, B. Blümich, Relaxation-relaxation experiments in natural porous media with a portable Halbach magnet, **Vandose Zone Journal**, 9, 893-897, 2010.
- A. Haber, B. Blümich, D. Souvorova, E. Del Federico, Ancient roman wall paintings mapped nondestructively by portable NMR, **Anal. Bioanal. Chem.**, 401, 1441-1452, 2011.
- B. Blümich, F. Casanova, M. Dabrowski, E. Danieli, L. Evertz, A. Haber, M. Van Landeghem, S. Haber-Pohlmeier, A. Olaru, J. Perlo, O. Sucre, Small-scale instrumentation for nuclear magnetic resonance of porous media, **New Journal of Physics**, 13, 015003, 2011.
- E. Paciok*, A. Haber*, M. Van Landeghem, B. Blümich, Relaxation exchange in nanoporous silica by low-field NMR, **Z. Phys. Chem.**, submitted, 2012. * These authors contributed equally to this work.

